Analysis Topics

Problem

Let $X \subset \mathbb{R}^2$ be non-compact. Prove that there exists a continuous unbounded function $f : X \to \mathbb{R}$.

Theory / tools

- **Heine–Borel in $\mathbb{R}^2$.**
  A set is compact iff it is *closed and bounded.*
  Thus, if X is non-compact, then either
  (i) X is unbounded,
  or
  (ii) X is bounded but not closed (so $\overline{X} \setminus X \neq \varnothing$).
- **Distance maps are continuous.**
  For any fixed $p \in \mathbb{R}^2$, the maps $x \mapsto \|x\|$ and $x \mapsto \|x - p\|$ are continuous on $\mathbb{R}^2$.
  If $p \notin X$, then $x \mapsto 1/\|x - p\|$ is continuous on X.

Solution

Case 1: X unbounded.

Define

$$f(x) = \|x\|, \qquad x \in X.$$

Then f is continuous. Since X is unbounded, for every $M > 0$ there exists $x \in X$ with $\|x\| > M$; hence f is unbounded.

Case 2: X bounded but not closed.

Choose $p \in \overline{X} \setminus X$ and define

$$f(x) = \frac{1}{\|x - p\|}, \qquad x \in X.$$

The denominator never vanishes on X, so f is continuous there. Because p is a limit point of X, there exists a sequence $x_n \in X$ with $x_n \to p$; thus $\|x_n - p\| \to 0$ and $f(x_n) \to \infty$. Therefore f is unbounded.

In either case there exists a continuous unbounded $f : X \to \mathbb{R}$. $\square$

Examples

1. **Open unit disk.**
   $X = \{x \in \mathbb{R}^2 : \|x\| < 1\}$. Take $p = (1,0) \in \partial X$. Then $f(x) = 1/\|x - p\|$ is continuous on X and unbounded near p.
2. **Parabola (unbounded).**
   $X = \{(t, t^2) : t \in \mathbb{R}\}$. The map $f(x) = \|x\|$ (equivalently $f(t, t^2) = \sqrt{t^2 + t^4}$) is continuous and unbounded.

Problem

Consider the metric space $(\mathbb{Q}, d)$ where $\mathbb{Q}$ is the set of rational numbers and $d(x, y) = |x - y|$ is the usual Euclidean metric.

Let

$$P = \{\, p \in \mathbb{Q} : 2 < p^2 < 3 \,\}.$$

Prove that P is closed and bounded in $\mathbb{Q}$, but not compact.

Give an open cover of P which has no finite subcover.

Theory

Open (resp. closed) sets of $\mathbb{Q}$ are precisely intersections of open (resp. closed) sets of $\mathbb{R}$ with $\mathbb{Q}$.

For any $A \subset \mathbb{R}$,

$$\overline{A \cap \mathbb{Q}}^{\mathbb{Q}} \;=\; \overline{A}^{\mathbb{R}} \cap \mathbb{Q}.$$

Hence if the only new points in $\overline{A}^{\mathbb{R}}$ are irrational, they vanish after intersecting with $\mathbb{Q}$.

Boundedness in $\mathbb{Q}$ is the same as boundedness in $\mathbb{R}$ because the metric is the restriction of the Euclidean metric.

Compactness means every open cover admits a finite subcover. In metric spaces, compactness implies completeness and total boundedness; since $(\mathbb{Q}, d)$ is not complete, many closed-and-bounded subsets are not compact.

Solution

Closedness.

Note that

$$P = \mathbb{Q} \cap \Big((\sqrt{2}, \sqrt{3}) \cup (-\sqrt{3}, -\sqrt{2})\Big).$$

The real closure of $(\sqrt{2}, \sqrt{3}) \cup (-\sqrt{3}, -\sqrt{2})$ is

$$[\sqrt{2}, \sqrt{3}] \cup [-\sqrt{3}, -\sqrt{2}].$$

Intersecting with $\mathbb{Q}$ removes the irrational endpoints, so

$$\overline{P}^{\mathbb{Q}} \;=\; \Big([\sqrt{2}, \sqrt{3}] \cup [-\sqrt{3}, -\sqrt{2}]\Big) \cap \mathbb{Q} \;=\; P.$$

Therefore P is closed in $\mathbb{Q}$.

Boundedness.

For any $p \in P$, $2 < p^2 < 3$ implies $\sqrt{2} < |p| < \sqrt{3}$. Thus $P \subset (-\sqrt{3}, \sqrt{3})$ and is bounded.

Non-compactness via an open cover with no finite subcover.

For each $n \in \mathbb{N}$, define

$$U_n = \Big(\sqrt{2},\, \sqrt{3} - \tfrac{1}{n}\Big) \cap \mathbb{Q}, \qquad V_n = \Big(-\sqrt{3} + \tfrac{1}{n},\, -\sqrt{2}\Big) \cap \mathbb{Q}.$$

Each U_n and V_n is open in $\mathbb{Q}$ (intersection of a real open interval with $\mathbb{Q}$).

Moreover,

$$P \;=\; \Big(\bigcup_{n=1}^{\infty} U_n\Big) \;\cup\; \Big(\bigcup_{n=1}^{\infty} V_n\Big),$$

so the family $\mathcal{U} = \{U_n, V_n : n \in \mathbb{N}\}$ is an open cover of P.

Assume a finite subfamily covers P and let N be the largest index used. Then the rationals in $(\sqrt{3} - \frac{1}{N}, \sqrt{3})$ are not contained in any U_n with $n \leq N$, and similarly the rationals in $(-\sqrt{3}, -\sqrt{3} + \frac{1}{N})$ are not contained in any V_n with $n \leq N$. Hence that finite subfamily fails to cover P, a contradiction.

Therefore P is not compact.

Examples / Remarks

P is closed in $\mathbb{Q}$ but not closed in $\mathbb{R}$; its real closure adds the irrational endpoints $\pm\sqrt{2}, \pm\sqrt{3}$.

Any finite subset of $\mathbb{Q}$ is compact. By contrast, $[0,1] \cap \mathbb{Q}$ is closed and bounded in $\mathbb{Q}$ but not compact; one can build a similar open cover that leaves uncovered rationals arbitrarily close to 0 or 1.

Problem 3 — Positive separation

Let $K \subset \mathbb{R}^2$ be compact and let $F \subset \mathbb{R}^2$ be closed with $K \cap F = \varnothing$.

Show that there exists $\delta > 0$ such that $d(x, y) \geq \delta$ for every $x \in K$ and every $y \in F$.

Give two proofs: one from compactness of K, and one using sequential compactness.

Theory

For $A \subset \mathbb{R}^2$ and $x \in \mathbb{R}^2$,

$$\operatorname{dist}(x, A) = \inf\{\|x - a\| : a \in A\}.$$

If A is closed and $x \notin A$, then $\operatorname{dist}(x, A) > 0$.

Compactness: every open cover has a finite subcover.

Sequential compactness: every sequence in K has a convergent subsequence with limit in K.

Closed sets are sequentially closed.

Proof A (open-cover argument)

For each $x \in K$, since $x \notin F$ and F is closed, we have $\operatorname{dist}(x, F) > 0$. Let

$$r_x = \tfrac{1}{2}\operatorname{dist}(x, F).$$

Then $B(x, r_x) \cap F = \varnothing$. The family $\{B(x, r_x) : x \in K\}$ covers K.

By compactness of K, choose $x_1, \ldots, x_m \in K$ with

$$K \subset \bigcup_{i=1}^{m} B(x_i, r_{x_i}).$$

Set

$$\delta = \min_{1 \le i \le m} r_{x_i} > 0.$$

If $x \in K$ and $y \in F$, choose i so that $x \in B(x_i, r_{x_i})$. Because $B(x_i, r_{x_i}) \cap F = \varnothing$, we have $d(x, y) \ge r_{x_i} \ge \delta$. Thus $d(x, y) \ge \delta$ for all $x \in K,\ y \in F$.

Proof B (sequential compactness)

Assume no such $\delta > 0$ exists. Then for each $n \in \mathbb{N}$ there exist $x_n \in K$ and $y_n \in F$ with

$$d(x_n, y_n) < \tfrac{1}{n}.$$

By sequential compactness, there is a subsequence $x_{n_k} \to x \in K$. Since $d(x_{n_k}, y_{n_k}) \to 0$, it follows that $y_{n_k} \to x$. Because F is closed and all $y_{n_k} \in F$, we have $x \in F$. Hence $x \in K \cap F$, a contradiction.

Therefore some $\delta > 0$ exists.

Examples / Remarks

If $K = \{(x, y) : x^2 + y^2 = 1\}$ and $F = \{(3, t) : t \in \mathbb{R}\}$, then the minimal distance is 2.

If compactness of K is dropped, the claim may fail; there are disjoint closed sets with zero infimum distance.

Sequential compactness: examples and counterexamples

Definition

A topological space X is sequentially compact if every sequence in X has a convergent subsequence with limit in X.

Examples (are sequentially compact)

- Closed and bounded subsets of $\mathbb{R}^n$ (e.g. $[0,1]^n$, spheres, closed balls). By Bolzano–Weierstrass.
- Any compact metric space (in metric spaces, compact $\Leftrightarrow$ sequentially compact).
- Finite spaces (every sequence is eventually constant).

Non-examples in metric spaces (not sequentially compact)

- $(0,1) \subset \mathbb{R}$: the sequence $x_n = 1/n$ has no convergent subsequence in $(0,1)$.
- $[0,\infty)$: the sequence $x_n = n$ has no convergent subsequence.
- $[0,1] \cap \mathbb{Q}$: rationals $q_n \to \sqrt{2}$ (in $\mathbb{R}$) yield no convergent subsequence in the space.

Sequentially compact but not compact (non-metric)

- An uncountable set with the co-countable topology.

  Any sequence has a convergent subsequence (indeed, often converges to every point), so the space is sequentially compact. The open cover $\{ X \setminus \{x\} : x \in X \}$ has no finite subcover, so the space is not compact.

Compact but not sequentially compact (non-metric)

- The Cantor cube $2^{\mathbb{R}}$ (product of $\{0,1\}$ over an uncountable index set).

  Compact by Tychonoff, but not sequentially compact; there exist sequences with no convergent subsequence (product convergence is coordinatewise and can be made to oscillate on infinitely many coordinates).
- The Stone–Čech compactification $\beta\mathbb{N}$.

Compact Hausdorff, yet not sequentially compact.

Summary. In metric spaces, compactness and sequential compactness coincide. In general topological spaces, neither implies the other.

Definitions and comparisons

Compact. Every open cover has a finite subcover.

Paracompact. Every open cover has a locally finite open refinement (i.e., each point meets only finitely many sets of the refinement).

Precompact (totally bounded / relatively compact). In a metric (or uniform) space: for every $\varepsilon > 0$ there exist finitely many ε-balls covering the space. Equivalently, the closure in the completion is compact.

Sequentially compact. Every sequence has a convergent subsequence with limit in the space.

Lindelöf. Every open cover admits a countable subcover.

Implications (high level)

General spaces.

$$\text{Compact} \Rightarrow \text{Lindelöf} \quad \text{and} \quad \text{Compact} \Rightarrow \text{Paracompact}.$$

The converses need not hold.

Metric spaces.

$$\text{Compact} \iff \text{Sequentially compact} \iff \text{Complete} + \text{Totally bounded (Precompact)}.$$

Also, every metric space is paracompact, and

$$\text{Separable} \iff \text{Second countable} \iff \text{Lindelöf}.$$

Mini examples

Precompact but not compact: $(0, 1) \subset \mathbb{R}$.

Lindelöf but not compact: $\mathbb{R}$ with the usual topology.

Compact hence paracompact: any compact Hausdorff space.

Sequentially compact = compact (metric): closed intervals $[a, b] \subset \mathbb{R}$.

Problem 4 — Complete metric on $(0, 1]$ with the usual topology

Find a metric d on $X = (0, 1]$ such that (X, d) is complete and such that the d-open subsets are exactly the open subsets of X in the Euclidean subspace topology.

Theory

If $\phi : (X, \tau) \to (Y, \rho)$ is a homeomorphism and (Y, ρ) is complete, then $d(x, y) := \rho(\phi(x), \phi(y))$ makes (X, d) complete and induces τ.

For $X = (0, 1]$, the map $\phi(x) = 1/x$ is a homeomorphism onto $[1, \infty)$, which is complete under the Euclidean metric.

Solution

Define

$$d(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|, \qquad x, y \in (0, 1].$$

Same topology. The metric d is the pullback of the Euclidean metric on $[1, \infty)$ via the homeomorphism $\phi(x) = 1/x$. Hence the d-open sets coincide with the Euclidean-open sets of $(0, 1]$.

Completeness. Let (x_n) be d-Cauchy. Then $(\phi(x_n)) = (1/x_n)$ is Cauchy in $[1, \infty)$, so $\phi(x_n) \to L \in [1, \infty)$. Therefore $x_n = \phi^{-1}(\phi(x_n)) \to \phi^{-1}(L) = 1/L \in (0, 1]$. Thus every d-Cauchy sequence converges in X, i.e., (X, d) is complete.

Examples / Counterexamples

Euclidean metric not complete: $x_n = 1/n$ is Euclidean-Cauchy in $(0, 1]$ but converges to $0 \notin X$.

Under d: $\phi(x_n) = n$ is not Cauchy, so no issue with completeness.

A d-Cauchy sequence: $x_n = 1/(1 + 1/n)$ has $\phi(x_n) = 1 + 1/n \to 1$, hence $x_n \to 1 \in X$.

Alternative metric: $d'(x, y) = |x - y| + \left|1/x - 1/y\right|$ is also complete and induces the same topology.

Understanding d-open sets and the subspace topology

What "d-open" means

Given a metric d on a set X, we say a subset $U \subseteq X$ is *d-open* if for every point $x \in U$ there exists a radius $r > 0$ such that the *open d-ball*

$$B_d(x, r) = \{\, y \in X : \ d(x, y) < r \,\}$$

is contained in U.

Equivalently, U is a union of d-balls. The collection of all d-open sets forms the **topology induced by** d.

What the subspace topology is

Suppose Y is a topological space (for example, a metric space with its metric–induced open sets), and $X \subseteq Y$.

The **subspace topology** on X is defined as

$$\mathcal{T}_X = \{\, U \cap X : U \subseteq Y \text{ is open in } Y \,\}.$$

So a set $A \subseteq X$ is open in the subspace topology if and only if it is the intersection of X with some open subset of the ambient space Y.

How these relate (and to Problem 4)

- Start with $Y = \mathbb{R}$ equipped with the usual metric $|\cdot|$.
- Take $X = (0, 1] \subset Y$.
- The subspace topology on X consists of all sets of the form $(a, b) \cap (0, 1]$.
  For example:
  - The set $(0.2, 0.7]$ is *not* open in X because the point 0.7 would need a right-hand neighborhood inside X, which does not exist.
  - The set $(0.2, 0.7)$ *is* open in X.

In Problem 4 we defined a new metric on X by

$$d(x, y) = \left|\frac{1}{x} - \frac{1}{y}\right|.$$

The key property required was:

a subset of X is open with respect to d iff it is open in the usual Euclidean subspace topology.

That means:

$$d\text{-open sets on } X \;=\; \text{open sets of the subspace topology from } \mathbb{R}.$$

So the metric d does not change which sets are open; it only changes the distances so that completeness is recovered.

Quick examples

In $X = (0, 1]$:

- $U = (0.3, 0.8)$ is open in the subspace topology (it is $(0.3, 0.8) \cap (0, 1]$); hence it is d-open as well.
- $V = (0, 1]$ is open in the subspace topology (it is $\mathbb{R} \cap (0, 1]$); hence also d-open.
- $W = (0, 1] \setminus \{1\}$ is *not* open in the subspace topology because near the endpoint 1 one cannot fit a right-hand neighborhood contained in W; therefore W is not d-open either.

Problem 5 — Fixed points on the open disc

Let $B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. If $f : B \to B$ is continuous, must f have a fixed point? Does the answer change if f is surjective?

Theory

Brouwer's fixed point theorem guarantees a fixed point for continuous self-maps of the *closed* disc $\overline{B}$. It does not apply to the non-compact open disc B. Automorphisms of the unit disc $B \subset \mathbb{C}$ are Möbius maps

$$\phi_{a,\theta}(z) = e^{i\theta}\frac{z-a}{1-\bar{a}z}, \qquad |a| < 1,$$

many of which have no fixed point in B.

Solution

Claim. There exist continuous maps $f : B \to B$ with no fixed point. This remains true even if f is surjective.

Example (surjective, no fixed point). For $r \in (0, 1)$, define

$$f(z) = \frac{z+r}{1+rz}, \qquad |z| < 1.$$

Then $f : B \to B$ is a disc automorphism with inverse $f^{-1}(w) = \dfrac{w-r}{1-rw}$, hence bijective and in particular surjective.

To find fixed points, solve $f(z) = z$:

$$z = \frac{z+r}{1+rz} \iff z(1+rz) = z+r \iff rz^2 = r \iff z = \pm 1.$$

Thus all fixed points lie on the boundary ∂B, so f has no fixed point in B.

Therefore a continuous self-map of the *open* disc need not have a fixed point, and this remains true even under surjectivity. This does not contradict Brouwer, which concerns the *closed* disc.

Remarks

Simple affine contractions of B typically have a fixed point (by Banach's fixed point theorem). The Möbius automorphisms provide clean counterexamples on the open disc.

Problem 6 — Expanding maps on a compact space are onto

Let (X, d) be a compact metric space and $f : X \to X$ be continuous with

$$d(f(x), f(y)) \geq d(x, y) \qquad (\forall x, y \in X).$$

Prove that f is surjective.

Theory

Compact images are compact. If X is compact and $f : X \to X$ is continuous, then $f(X)$ is compact (hence closed) in X.

Positive distance from a compact set. If $C \subset X$ is compact and $x \notin C$, then $\operatorname{dist}(x, C) = \min_{z \in C} d(x, z) > 0$.

Sequential compactness. In metric spaces, compactness $\Leftrightarrow$ every sequence admits a convergent subsequence.

Monotone gaps under expansion. From $d(fu, fv) \geq d(u, v)$ it follows inductively that for the orbit $x_{n+1} = f(x_n)$, the differences satisfy $d(x_n, x_{n+1}) \geq d(x_{n-1}, x_n)$ for all n.

Injectivity (useful side fact). If $d(fx, fy) \geq d(x, y)$ and $f(x) = f(y)$, then $0 \geq d(x, y) \Rightarrow x = y$; hence f is injective.

Solution

Assume f is not surjective. Choose $x_0 \in X \setminus f(X)$.

Since $f(X)$ is compact, the distance

$$r = \operatorname{dist}(x_0, f(X)) = \min_{y \in f(X)} d(x_0, y)$$

satisfies $r > 0$. Hence $d(x_0, y) \geq r$ for all $y \in f(X)$.

Consider the sequence $x_{n+1} = f(x_n)$. Then $x_n \in f(X)$ for all $n \geq 1$, so $d(x_0, x_1) \geq r$. Using $d(fu, fv) \geq d(u, v)$ iteratively, we obtain

$$d(x_n, x_{n+1}) \ \geq \ d(x_{n-1}, x_n) \ \geq \ \cdots \ \geq \ d(x_0, x_1) \ \geq \ r \qquad (n \geq 1).$$

Therefore (x_n) is not Cauchy and hence has no convergent subsequence.

This contradicts the sequential compactness of X. Therefore f must be surjective. $\square$

Examples / Counterexamples

Why compactness matters (counterexample). On $X = (0,1]$ with $d(x,y) = \big|1/x - 1/y\big|$, the map $f(x) = x/2$ is continuous, not surjective (image $(0, \frac{1}{2}]$), and

$$d(f(x), f(y)) = \left|\tfrac{2}{x} - \tfrac{2}{y}\right| = 2\, d(x,y) \geq d(x,y).$$

So the conclusion can fail without compactness.

Typical positive example. Let $X = S^1$ with the arc-length metric and f be a rotation. Then $d(fx, fy) = d(x,y)$ (an isometry), and f is bijective in particular surjective, in line with the theorem.

No constant maps allowed. A constant map $f(x) \equiv p$ would force $0 = d(fx, fy) \geq d(x,y)$ for all x, y, impossible unless X is a singleton.

Problem 7 — Radial projection to ∂D and continuity in the base point

Let $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$ be the closed unit disc and $\partial D = \{z : \|z\| = 1\}$ its boundary.

(i) Fix $x \in D$. Let $K \subset D$ be compact with $x \notin K$. For $y \in K$, follow the ray

$$\gamma_{x,y}(t) = x + t(y - x), \qquad t \geq 0,$$

until it first meets ∂D; call this intersection $f_x(y) \in \partial D$. Show that $f_x : K \to \partial D$ is continuous.

(ii) Let $g : D \to D$ be continuous and let $x_0 \in D$ with $x_0 \neq g(x_0)$. Show that there is $r > 0$ with $x \neq g(x)$ for all $x \in B_r(x_0)$. For $x \in B_r(x_0)$ define $h(x) = f_x\big(g(x)\big)$ (using f_x from (i)). Prove that h is continuous at x_0.

Theory you'll use

Ray/affine parametrization. With $v = y - x \neq 0$, the first hit of the ray $x + t(y - x)$ with ∂D occurs at the unique $t = \tau(x,y) \geq 1$ solving $\|x + t(y - x)\| = 1$. Equivalently,

$$\|v\|^2\, \tau^2 + 2\langle x, v\rangle \tau + (\|x\|^2 - 1) = 0,$$

so the relevant root is

$$\tau(x,y) = \frac{-\langle x, v\rangle + \sqrt{\langle x, v\rangle^2 + (1 - \|x\|^2)\,\|v\|^2}}{\|v\|^2}, \qquad v = y - x \neq 0,$$

and

$$f_x(y) = x + \tau(x,y)\,(y-x).$$

This formula is continuous in (x,y) on $U = \{(x,y) \in D \times D : \ y \neq x\}$.

Uniform separation on compact K. If $K \subset D$ is compact and $x \notin K$, then $\delta := \inf_{y\in K} \|y-x\| > 0$, so $v = y-x$ never vanishes on K.

Composition principle. If $F(x,y)$ is continuous on a set and g is continuous with $(x, g(x))$ remaining in that set near x_0, then $x \mapsto F(x, g(x))$ is continuous at x_0.

Positivity near x_0. The map $x \mapsto \|x - g(x)\|$ is continuous; if it is > 0 at x_0, it stays > 0 on some ball $B_r(x_0)$.

(i) Continuity of f_x for fixed x and compact $K \subset D \setminus \{x\}$

For $y \in K$ put $v(y) = y - x \neq 0$ and define $\tau(x,y)$ as above. Since K is compact and $x \notin K$, we have $\inf_{y\in K} \|v(y)\| > 0$; hence the denominator $\|v(y)\|^2$ stays away from 0. The map

$$y \longmapsto \tau(x,y) = \frac{-\langle x, v(y)\rangle \ + \ \sqrt{\langle x, v(y)\rangle^2 + (1-\|x\|^2)\,\|v(y)\|^2}}{\|v(y)\|^2}$$

is continuous on K. Therefore

$$f_x(y) = x + \tau(x,y)\,v(y)$$

is a continuous function $K \to \partial D$.

(Geometric remark.) One may also argue by the implicit function theorem applied to $F(t,y) = \|x + t(y-x)\|^2 - 1$: at the first hit, $\partial_t F = 2\langle x + t(y-x),\, y-x\rangle > 0$, so $t = \tau(x,y)$ depends continuously on y.

Solution to (ii)

Define $\phi(x) = \|x - g(x)\|$. Then ϕ is continuous and $\phi(x_0) =: a > 0$. Choose $r > 0$ such that $\phi(x) \geq a/2 > 0$ for all $x \in B_r(x_0)$. Thus $g(x) \neq x$ on $B_r(x_0)$, so $f_x(g(x))$ is well-defined there.

Let

$$F : U \to \partial D, \qquad F(x,y) = x + \tau(x,y)\,(y-x), \qquad U = \{(x,y) \in D \times D : \ y \neq x\}.$$

We know F is continuous on U. Since $(x, g(x)) \in U$ for all $x \in B_r(x_0)$ and g is continuous, the composition

$$h(x) = F\Big(x, g(x)\Big) = f_x\Big(g(x)\Big)$$

is continuous at x_0. $\square$

Examples / remarks

Special case $x = 0$. Then $f_0(y) = y/\|y\|$ (radial projection to ∂D), continuous on any compact $K \subset D \setminus \{0\}$. If $g(0) \neq 0$, part (ii) gives continuity at x_0 of $h(x) = \dfrac{g(x) - x}{\|g(x) - x\|}$ (up to the affine rescaling along the ray).

No extension through $y = x$. In general f_x cannot be extended continuously to $y = x$: approaching x along different directions sends $f_x(y)$ to different points of ∂D.

Deriving and explaining the formula for $\tau(x, y)$

Fix $x \in D$ and $y \in D \setminus \{x\}$, and consider the ray

$$\gamma_{x,y}(t) = x + t(y - x), \qquad t \geq 0.$$

We want the *first* parameter $t = \tau(x, y)$ for which $\gamma_{x,y}(t) \in \partial D$, i.e.

$$\|\gamma_{x,y}(t)\| = \|x + t(y - x)\| = 1.$$

Quadratic equation. Set $v := y - x \neq 0$. Then

$$\|x + tv\|^2 = \|x\|^2 + 2t \langle x, v \rangle + t^2 \|v\|^2.$$

The boundary condition $\|x + tv\|^2 = 1$ becomes

$$\|v\|^2 t^2 + 2\langle x, v \rangle t + (\|x\|^2 - 1) = 0.$$

Let $a = \|v\|^2 > 0$, $b = 2\langle x, v \rangle$, and $c = \|x\|^2 - 1 \leq 0$ (since $x \in D$). By the quadratic formula,

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-\langle x, v \rangle \pm \sqrt{\langle x, v \rangle^2 + (1 - \|x\|^2)\, \|v\|^2}}{\|v\|^2}.$$

Choosing the correct root (first hit for $t \geq 0$). The discriminant is

$$\Delta = \langle x, v \rangle^2 + (1 - \|x\|^2)\, \|v\|^2 \;\geq\; 0.$$

Moreover $\sqrt{\Delta} \geq |\langle x, v\rangle|$, hence the *plus* root

$$\boxed{\tau(x,y) = \frac{-\langle x, v\rangle + \sqrt{\langle x, v\rangle^2 + (1 - \|x\|^2)\, \|v\|^2}}{\|v\|^2}}$$

is nonnegative (its numerator is ≥ 0).

The *minus* root has numerator $-\langle x, v\rangle - \sqrt{\Delta} \leq 0$, so it is ≤ 0 and corresponds to moving *backwards* along the line (negative t), not along our ray $t \geq 0$.

Uniqueness of the positive intersection. Let $s(t) = \|x + tv\|^2$. Then s is a strictly convex quadratic in t with $s(0) = \|x\|^2 \leq 1$ and $s(t) \to \infty$ as $t \to \infty$. If $x \in \operatorname{int} D$, then $s(0) < 1$, so the equation $s(t) = 1$ has exactly one solution with $t > 0$ (the other solution is negative). If $x \in \partial D$ and $y \neq x$, then $t = 0$ is one solution and the other solution is positive; this positive one is the "first hit after leaving x." In either case, the ray meets ∂D at exactly one parameter $t > 0$, namely $t = \tau(x, y)$ above.

Continuity of τ and of the projection f_x. On the set

$$U = \{(x, y) \in D \times D :\ y \neq x\}$$

all operations in the displayed formula are continuous: $(x, y) \mapsto \langle x, y - x\rangle$, $(x, y) \mapsto \|x\|^2$, $(x, y) \mapsto \|y - x\|^2$, the square and square-root maps, and division by $\|y - x\|^2 > 0$. Hence $(x, y) \mapsto \tau(x, y)$ is continuous on U. Consequently,

$$f_x(y) = x + \tau(x, y)\,(y - x)$$

depends continuously on (x, y) on U, and in particular $y \mapsto f_x(y)$ is continuous on any compact $K \subset D \setminus \{x\}$ (where $\|y - x\|$ is bounded away from 0).

Geometric remark (alternative justification). Define $F(t, y) = \|x + t(y - x)\|^2 - 1$. At the first hit, we have $\partial_t F = 2\langle x + t(y - x),\, y - x\rangle > 0$, so by the implicit function theorem the solution $t = \tau(x, y)$ depends continuously on y, giving the same conclusion.

Problem 8 — Brouwer $\Rightarrow$ a positive eigenvector

Let A be a 3×3 matrix with positive entries. Use Brouwer's fixed-point theorem to prove that A has an eigenvector with positive entries.

Hint. Use A to define a self-map of the triangle

$$T = \operatorname{conv}\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} = \{x \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 = 1\}.$$

Theory

T is compact and convex. Brouwer's theorem: every continuous self-map of a compact convex subset of $\mathbb{R}^n$ has a fixed point. If $A > 0$ (entrywise) and $x \in T$, then Ax has strictly positive coordinates, and normalizing by the sum of its coordinates keeps the result in T. If some coordinate of x is 0, the image under A has all coordinates > 0, so the normalization lands in the interior of T.

Solution

Define

$$F : T \to T, \qquad F(x) = \frac{Ax}{\mathbf{1}^\top Ax}, \qquad \mathbf{1} = (1, 1, 1)^\top.$$

Because $A > 0$, we have $Ax \in (0, \infty)^3$ for every $x \in T$, hence $\mathbf{1}^\top Ax > 0$ and F is well-defined and continuous. Also, $F(T) \subseteq T$ since the coordinates are nonnegative and sum to 1.

By Brouwer's fixed-point theorem there exists $x^* \in T$ with $F(x^*) = x^*$. Let $\lambda = \mathbf{1}^\top Ax^* > 0$. Then

$$Ax^* = \lambda x^*,$$

so x^* is an eigenvector with eigenvalue $\lambda > 0$.

We claim x^* has strictly positive entries. If some coordinate of x^* were 0, then Ax^* would have all coordinates > 0 (since $A > 0$), and so $F(x^*) = \dfrac{Ax^*}{\mathbf{1}^\top Ax^*}$ would lie in the interior of T, contradicting $x^* = F(x^*)$. Hence x^* has all coordinates > 0.

$\square$

Example

For

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad F(x) = \frac{Ax}{\mathbf{1}^\top Ax}.$$

We have $A(1, 1, 1)^\top = 4(1, 1, 1)^\top$, so the fixed point in T is $x^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ with eigenvalue $\lambda = 4$, and all entries are positive.

Theory of Matrix Eigenvectors

Definitions

Let $\mathbb{F}$ be a field (typically $\mathbb{R}$ or $\mathbb{C}$) and let $A \in \mathbb{F}^{n\times n}$ represent a linear operator $T : \mathbb{F}^n \to \mathbb{F}^n$. A nonzero vector $v \in \mathbb{F}^n$ is an *eigenvector* of A with *eigenvalue* $\lambda \in \mathbb{F}$ if

$$Av = \lambda v.$$

Equivalently, $(A - \lambda I)v = 0$, so $v \in \ker(A - \lambda I)$ and λ is an eigenvalue if and only if

$$\det(A - \lambda I) = 0.$$

The *eigenspace* associated with λ is $E_\lambda = \ker(A - \lambda I)$.

Characteristic polynomial and spectrum

The *characteristic polynomial* of A is

$$p_A(\lambda) = \det(\lambda I - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0.$$

The *spectrum* (set of eigenvalues) is $\sigma(A) = \{\lambda \in \mathbb{F} : p_A(\lambda) = 0\}$.

Algebraic and geometric multiplicities

For $\lambda \in \sigma(A)$:

- The *algebraic multiplicity* $m_a(\lambda)$ is the multiplicity of λ as a root of p_A.
- The *geometric multiplicity* $m_g(\lambda)$ is the dimension of the eigenspace $E_\lambda = \ker(A - \lambda I)$.

These satisfy

$$1 \le m_g(\lambda) \le m_a(\lambda).$$

Eigenspaces, independence, invariant subspaces

Each eigenspace E_λ is an A-invariant subspace: if $x \in E_\lambda$, then $Ax = \lambda x \in E_\lambda$. If $\lambda_1, \ldots, \lambda_k$ are pairwise distinct eigenvalues, then $E_{\lambda_1}, \ldots, E_{\lambda_k}$ are linearly independent subspaces, and any collection of eigenvectors from distinct eigenvalues is linearly independent.

Diagonalizability and Jordan form

The matrix A is *diagonalizable* over $\mathbb{F}$ if there exists a basis of $\mathbb{F}^n$ consisting of eigenvectors of A. Equivalently, there exists an invertible P with

$$A = PDP^{-1}, \qquad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

This holds if and only if $\sum_{\lambda \in \sigma(A)} m_g(\lambda) = n$, which in turn implies $m_g(\lambda) = m_a(\lambda)$ for each eigenvalue λ. If A is not diagonalizable, there exists an invertible P with

$$A = PJP^{-1},$$

where J is the *Jordan normal form*, a block-diagonal matrix of Jordan blocks $J_k(\lambda) = \lambda I_k + N_k$ with N_k nilpotent, $(N_k)_{i,i+1} = 1$, zeros elsewhere.

Real vs. complex eigenstructure

Over $\mathbb{C}$, the characteristic polynomial splits into linear factors, so A has n complex eigenvalues counting algebraic multiplicity. Over $\mathbb{R}$, eigenvalues may be complex or may fail to be real (e.g. a planar rotation without reflection has no real eigenvalues).

Special matrix classes

- **Normal matrices** ($AA^* = A^*A$): unitarily diagonalizable, i.e. $A = UDU^*$ with U unitary and D diagonal.
- **Hermitian / symmetric** ($A^* = A$ or $A^\top = A$): all eigenvalues are real; there exists an orthonormal eigenbasis (Spectral Theorem).
- **Positive (semi)definite**: $x^*Ax > 0$ (or ≥ 0) for $x \neq 0$; eigenvalues are positive (or nonnegative).
- **Unitary / orthogonal**: $A^*A = I$ (or $A^\top A = I$); eigenvalues have modulus 1.
- **Nonnegative / stochastic**: For irreducible nonnegative A, Perron–Frobenius guarantees a largest eigenvalue $\rho(A) > 0$ with a strictly positive eigenvector.

Similarity invariants and the minimal polynomial

Matrices A and B are similar if $B = P^{-1}AP$ for some invertible P. Similar matrices share spectrum (with algebraic multiplicities), the characteristic polynomial, and the *minimal polynomial* μ_A, the monic polynomial of least degree with $\mu_A(A) = 0$. One has $\mu_A \mid p_A$. Moreover, A is diagonalizable if and only if μ_A splits over $\mathbb{F}$ into pairwise distinct linear factors.

Functional calculus for polynomials

If q is a polynomial, then the eigenvalues of $q(A)$ are $\{q(\lambda) : \lambda \in \sigma(A)\}$, with eigenvectors preserved: if $Av = \lambda v$, then $q(A)v = q(\lambda)v$.

Basic stability properties

If $\lambda \in \sigma(A)$ and $k \in \mathbb{N}$, then $\lambda^k \in \sigma(A^k)$. If A is invertible, then $\lambda \neq 0$ and $\lambda^{-1} \in \sigma(A^{-1})$. The spectra of A, $A^\top$, and A^* coincide (over $\mathbb{C}$, counting multiplicity).

Computation of eigenpairs

- *Exact (small n)*: compute p_A, find its roots, then for each root λ, solve $(A-\lambda I)v = 0$.
- *Numerical*: QR algorithm (all eigenvalues for dense A), power iteration (dominant eigenpair), inverse iteration / Rayleigh quotient iteration (targeted eigenpairs), Arnoldi/Lanczos (large sparse).

Examples

Example 1 (Diagonal). $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ has $p_A(\lambda) = (\lambda - 3)(\lambda - 1)$, eigenvectors e_1 for $\lambda = 3$ and e_2 for $\lambda = 1$. Here $m_a = m_g = 1$ for each eigenvalue, so A is diagonalizable.

Example 2 (Defective). $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $p_A(\lambda) = (\lambda - 1)^2$. The eigenspace is $\ker(A - I) = \operatorname{span}\{(1, 0)^\top\}$, so $m_a(1) = 2$ but $m_g(1) = 1$; A is not diagonalizable and is already a Jordan block.

Theory of matrix eigenvectors

Definitions

Let $A \in \mathbb{F}^{n\times n}$ with $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A nonzero vector $v \in \mathbb{F}^n$ is an *eigenvector* of A with *eigenvalue* $\lambda \in \mathbb{F}$ if

$$Av = \lambda v, \qquad v \neq 0.$$

The *eigenspace* for λ is $E_\lambda = \ker(A - \lambda I)$ and has dimension $g_\lambda = \dim E_\lambda$ (the *geometric multiplicity*). The *characteristic polynomial* is $\chi_A(t) = \det(tI - A)$. If λ is a root of χ_A, its multiplicity as a root is the *algebraic multiplicity* a_λ.

Basic properties

Over $\mathbb{C}$, χ_A splits and has degree n, so A has n eigenvalues counted with algebraic multiplicity. Always $1 \le g_\lambda \le a_\lambda$ and $\sum_\lambda a_\lambda = n$. Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Diagonalization

A is *diagonalizable* over $\mathbb{F}$ if there exists a basis of eigenvectors; equivalently

$$A = PDP^{-1}, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

or $\sum_\lambda g_\lambda = n$, or $g_\lambda = a_\lambda$ for all λ. Diagonalizable matrices admit spectral decompositions and satisfy $p(A) = P\,p(D)\,P^{-1}$ for polynomials p.

Left eigenvectors

A nonzero row vector $w^\top$ with $w^\top A = \lambda w^\top$ is a *left eigenvector*. Equivalently, w is a right eigenvector of $A^\top$ with the same eigenvalue.

Spectral radius and norms

The *spectral radius* is $\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}$. For any operator norm $\|\cdot\|$, $\rho(A) \le \|A\|$, with equality $\|A\|_2 = \rho(A)$ when A is normal ($A^*A = AA^*$), e.g. symmetric or unitary.

Cayley–Hamilton and minimal polynomial

Cayley–Hamilton: $\chi_A(A) = 0$. The *minimal polynomial* m_A is the unique monic polynomial of least degree with $m_A(A) = 0$. Its roots are the distinct eigenvalues of A. A is diagonalizable over $\mathbb{F}$ iff m_A splits over $\mathbb{F}$ and has no repeated roots.

Schur, spectral, and Jordan theorems

Schur (complex field): For $A \in \mathbb{C}^{n\times n}$ there exists unitary Q with

$$A = QTQ^*, \qquad T \text{ upper triangular, } \operatorname{diag}(T) = \{\text{eigenvalues of } A\}.$$

Spectral theorem (real symmetric / complex Hermitian): If $A = A^\top$ (or $A = A^*$), then

$$A = Q\Lambda Q^\top \quad (\text{or } Q\Lambda Q^*),$$

with Q orthogonal/unitary and Λ real diagonal. Hence A has an orthonormal eigenbasis.

Jordan normal form (over $\mathbb{C}$): There exists invertible S such that $A = SJS^{-1}$ with J block diagonal, each block $J_k(\lambda)$ having λ on the diagonal and ones on the superdiagonal. Nontrivial blocks (size > 1) occur exactly when $g_\lambda < a_\lambda$.

Invariant subspaces and polynomials in A

If $Av = \lambda v$, then for any polynomial p we have $p(A)v = p(\lambda)v$ and $A^k v = \lambda^k v$. If $|\lambda| < 1$, then $A^k v \to 0$; if $|\lambda| > 1$, then $\|A^k v\| \to \infty$. Eigenspaces are invariant subspaces; direct sums of eigenspaces are invariant and (when summed to full dimension) yield diagonalization.

Real vs. complex eigenstructure

Real matrices may lack real eigenvalues; complexification guarantees n eigenvalues counting multiplicity. Over $\mathbb{R}$, real canonical forms (e.g., real Jordan or real Schur) group complex conjugate pairs into 2×2 blocks.

Perron–Frobenius (positive / nonnegative matrices)

If $A > 0$ (all entries strictly positive), then:

- There exists a simple, real eigenvalue $\lambda_{\max} > 0$ with eigenvector $v > 0$ (all entries positive).
- $\lambda_{\max} = \rho(A)$ and, in the positive (or primitive) case, $|\lambda| < \lambda_{\max}$ for every other eigenvalue.
- The Perron eigenvector is unique up to positive scaling.

Examples

Diagonalizable.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \quad \chi_A(t) = (t-2)(t-3).$$

Eigenvectors for 2 and 3 span $\mathbb{R}^2$; $A = PDP^{-1}$ with $D = \operatorname{diag}(2,3)$.

Defective (not diagonalizable).

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \chi_A(t) = (t-1)^2,\ g_1 = 1 < a_1 = 2.$$

Jordan form has a single 2×2 block $J_2(1)$.

Symmetric (orthogonal eigenbasis).

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Eigenvalues 1 and 3; eigenvectors can be chosen orthonormal.

Perron vector.

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad A(1,1,1)^\top = 4(1,1,1)^\top.$$

Perron eigenpair $(\lambda_{\max}, v) = (4, (1,1,1)^\top)$.

Problem 9 — Brouwer gives a root in the unit disk

Use the Brouwer fixed point theorem to prove that there exists a complex number z with $|z| \le 1$ such that

$$z^4 - z^3 + 8z^2 + 11z + 1 = 0.$$

Theory

Identify $\mathbb{C}$ with $\mathbb{R}^2$. The closed unit disk

$$\overline{\mathbb{D}} = \{\, z \in \mathbb{C} : |z| \le 1 \,\}$$

is compact and convex. *Brouwer fixed point theorem:* every continuous map from a compact, convex subset of $\mathbb{R}^n$ to itself has a fixed point.

Solution

Rewrite the equation as a fixed point problem:

$$z^4 - z^3 + 8z^2 + 11z + 1 = 0 \quad \Longleftrightarrow \quad z \;=\; -\,\frac{z^4 - z^3 + 8z^2 + 1}{11} \;=:\; \Phi(z).$$

For any z with $|z| \le 1$ we have

$$|\Phi(z)| \;\le\; \frac{|z|^4 + |z|^3 + 8|z|^2 + 1}{11} \;\le\; \frac{1+1+8+1}{11} \;=\; 1.$$

Hence Φ is a continuous self-map of the closed unit disk $\overline{\mathbb{D}}$.

By the Brouwer fixed point theorem there exists $z^* \in \overline{\mathbb{D}}$ such that $\Phi(z^*) = z^*$. Substituting back gives

$$(z^*)^4 - (z^*)^3 + 8(z^*)^2 + 11z^* + 1 = 0,$$

so the polynomial has a root with $|z^*| \le 1$. $\square$

We define

$$\Phi(z) \;=\; -\,\frac{z^4 - z^3 + 8z^2 + 1}{11}.$$

Then

$$|\Phi(z)| \;=\; \frac{1}{11}\,\Big|\, z^4 - z^3 + 8z^2 + 1 \,\Big| \;\le\; \frac{1}{11}\,\Big(|z|^4 + |z|^3 + 8|z|^2 + |1|\Big),$$

where we used the triangle inequality $|a+b| \le |a| + |b|$ repeatedly on the four-term sum. If $|z| \le 1$, then $|z|^k \le 1$ for $k = 2, 3, 4$ and $|1| = 1$, hence

$$|\Phi(z)| \;\le\; \frac{1+1+8+1}{11} \;=\; 1.$$

Therefore Φ maps the closed unit disk into itself.

General trick. If you can write a polynomial equation in the form

$$z \;=\; \frac{q(z)}{c},$$

with $c > 0$ and $|q(z)| \le c$ whenever $|z| \le 1$, then

$$\Phi(z) \;=\; \frac{q(z)}{c}$$

is a continuous self-map of the closed unit disk $\overline{\mathbb{D}}$. By Brouwer's fixed point theorem, there exists a fixed point z with $|z| \le 1$.

In our case we chose $c = 11$, the coefficient of z, and put all the other terms into

$$q(z) = -\Big(z^4 - z^3 + 8z^2 + 1\Big),$$

whose modulus satisfies $|q(z)| \le 11$ on $|z| \le 1$. Therefore $\Phi(\overline{\mathbb{D}}) \subseteq \overline{\mathbb{D}}$, and a fixed point of Φ yields a root of the polynomial within the unit disk.

Problem 10 — Compactness and sequential compactness in $C[0,1]$

Let $C[0,1]$ be the metric space consisting of all continuous functions $f : [0,1] \to \mathbb{R}$, with distance

$$d(f,g) = \sup_{x\in[0,1]} |f(x) - g(x)|.$$

(i) Supremum is a maximum

Let $h(x) = |f(x)-g(x)|$. Since h is continuous on the compact interval $[0,1]$, the Extreme Value Theorem ensures that h attains its maximum and minimum values. Therefore

$$d(f,g) = \sup_{x\in[0,1]} |f(x) - g(x)| = \max_{x\in[0,1]} |f(x) - g(x)|.$$

(ii) X is not sequentially compact

Let

$$X = \{\, f \in C[0,1] : f([0,1]) \subseteq [0,1] \,\}.$$

Consider the sequence $f_n(x) = x^n$ for $x \in [0,1]$. Each f_n lies in X because $x^n \in [0,1]$ for $x \in [0,1]$.

For any fixed $x \in [0,1)$, $f_n(x) = x^n \to 0$, while $f_n(1) = 1$. Hence f_n converges pointwise to

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1, \end{cases}$$

which is discontinuous at $x = 1$.

Since the limit function is not continuous, the convergence cannot be uniform. No subsequence can converge uniformly either (every subsequence has the same pointwise limit). Thus (f_n) has no convergent subsequence in $C[0,1]$ under $d(f,g) = \|f - g\|_\infty$.

Therefore X is not sequentially compact.

(iii) X is not compact (open cover argument)

In metric spaces, compactness $\Leftrightarrow$ sequential compactness, so X is not compact. Nevertheless, we can give an explicit open cover with no finite subcover.

For each $n \in \mathbb{N}$, define

$$U_n = \{\, f \in X : f(1) > 1 - \tfrac{1}{n} \,\}.$$

Each U_n is open in the sup-norm topology. The collection $\{U_n : n \in \mathbb{N}\}$ covers X, since for every $f \in X$, $f(1) \in [0,1]$ so $f(1) > 1 - 1/n$ for some n.

However, no finite subcollection covers X. If $U_{n_1}, \ldots, U_{n_k}$ are finitely many, let $N = \max\{n_i\}$. Then the constant function $f(x) \equiv 1 - \frac{1}{2N}$ lies in X but satisfies $f(1) = 1 - \frac{1}{2N} < 1 - \frac{1}{N}$, hence $f \notin U_N$. Thus this cover has no finite subcover.

Remarks and examples

- The sequence $f_n(x) = x^n$ is a classical example in $C[0,1]$ showing failure of compactness.
- The family X is bounded and closed but not equicontinuous; by the Arzelà–Ascoli theorem it is not relatively compact.
- $C[0,1]$ with $\|\cdot\|_\infty$ is complete but infinite-dimensional, hence non-compact.

Supremum vs. Maximum

Definitions. For $S \subset \mathbb{R}$:

- The *maximum* $\max S$ is a number $m \in S$ with $s \leq m$ for all $s \in S$.
- The *supremum* $\sup S$ is the least upper bound: the smallest $\alpha \in \mathbb{R}$ such that $s \leq \alpha$ for all $s \in S$.

If $\max S$ exists, then $\sup S = \max S$. The supremum need not belong to S.

Examples.

- $S = (0,1)$: $\sup S = 1$ but $\max S$ does not exist (1 is not in S).
- $S = [0,1]$: $\sup S = \max S = 1$ (attained).
- $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$: $\sup S = 1$, no maximum (1 is only a limit point).
- $S = \{0, 1, 2, \ldots, 100\}$: $\sup S = \max S = 100$.
- $S = (-\infty, 0)$: $\sup S = 0$, no maximum.

For functions. If $f : [a, b] \to \mathbb{R}$ is continuous and $[a, b]$ is compact, then by the Extreme Value Theorem

$$\sup_{x\in[a,b]} f(x) \;=\; \max_{x\in[a,b]} f(x).$$

If the domain is not compact, or f is not continuous, the supremum may fail to be attained:

- $f(x) = x$ on $(0, 1)$: $\sup f = 1$ but f has no maximum.
- $f(x) = \sin x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: $\sup f = 1$ but no maximum (endpoints excluded).

Application to $C[0, 1]$. For $f, g \in C[0, 1]$, the function $h(x) = |f(x)-g(x)|$ is continuous on the compact interval $[0, 1]$. Hence

$$\sup_{x\in[0,1]} |f(x) - g(x)| \;=\; \max_{x\in[0,1]} |f(x) - g(x)|.$$

Extreme Value Theorem (EVT)

Statement. Let $K \subset \mathbb{R}^n$ be compact and let $f : K \to \mathbb{R}$ be continuous. Then there exist points $x_{\min}, x_{\max} \in K$ such that

$$f(x_{\min}) = \min_{x\in K} f(x), \qquad f(x_{\max}) = \max_{x\in K} f(x).$$

Equivalently,

$$\sup_{x\in K} f(x) = \max_{x\in K} f(x), \qquad \inf_{x\in K} f(x) = \min_{x\in K} f(x).$$

Proof sketch. Let $M = \sup_{x\in K} f(x)$. Choose $x_n \in K$ with $f(x_n) \to M$. By compactness, some subsequence $x_{n_k} \to x^* \in K$. By continuity, $f(x^*) = \lim_{k\to\infty} f(x_{n_k}) = M$, so the supremum is attained. Apply the same argument to $-f$ to get a minimum.

Why the hypotheses are needed.

- If the domain is not compact, even continuous functions may fail to attain extrema: $f(x) = x$ on $(0, 1)$ has $\sup = 1$ but no maximum.
- If f is not continuous, attainment can fail even on compact sets (pathologies exist).

Application to Problem 10

For $f, g \in C[0, 1]$,

set $h(x) = |f(x) - g(x)|$.

Then h is continuous on the compact interval $[0, 1]$. By the EVT,

$$\sup_{x\in[0,1]} |f(x) - g(x)| = \max_{x\in[0,1]} |f(x) - g(x)|.$$

Hence the metric $d(f, g) = \sup_{x\in[0,1]} |f(x) - g(x)|$ is the maximum value of $|f - g|$ on $[0, 1]$.

Open cover argument for non-compactness of X

Let $X = \{f \in C[0, 1] : 0 \le f(x) \le 1 \text{ for all } x \in [0, 1]\}$ with the sup norm.

Define the cover. For each $n \in \mathbb{N}$ set

$$U_n = \Big\{ f \in X : \ f(1) > 1 - \tfrac{1}{n} \Big\}.$$

U_n is open. The evaluation functional $E_1 : C[0, 1] \to \mathbb{R}$, $E_1(f) = f(1)$, is continuous since $|E_1(f) - E_1(g)| = |f(1) - g(1)| \le \|f - g\|_\infty$. Thus $U_n = E_1^{-1}\big((1 - \tfrac{1}{n}, \infty)\big)$ is open.

$\{U_n\}_{n\in\mathbb{N}}$ covers X. If $f \in X$ then $f(1) \in [0, 1]$, hence there exists n with $1 - \frac{1}{n} < f(1)$, i.e. $f \in U_n$.

No finite subcover. Given finitely many $U_{n_1}, \dots, U_{n_k}$, let $N = \max\{n_1, \dots, n_k\}$ and consider the constant function $f_N(x) \equiv 1 - \frac{1}{2N}$. Then $f_N \in X$ but

$$f_N(1) = 1 - \tfrac{1}{2N} \le 1 - \tfrac{1}{N},$$

so $f_N \notin U_N$, hence f_N is not contained in the finite union $U_{n_1} \cup \cdots \cup U_{n_k}$. Therefore no finite subcollection covers X.

Consequently, X is not compact.

Rational fibers and their role

For $f : \mathbb{R} \to \mathbb{R}$ and $q \in \mathbb{R}$, the *fiber* (level set) at q is

$$f^{-1}(q) = \{x \in \mathbb{R} : \ f(x) = q\}.$$

A *rational fiber* means $q \in \mathbb{Q}$.

Key mechanism (with IVP). Assume f has the intermediate value property and that each rational fiber $f^{-1}(q)$ is closed. Fix $q \in \mathbb{Q}$. Then the sublevel and superlevel sets

$$U_q = f^{-1}((-\infty, q)), \qquad V_q = f^{-1}((q, \infty))$$

are open. Indeed, if $x_0 \in U_q$ but every neighborhood of x_0 contains points with values $\geq q$, the IVP forces points with value exactly q arbitrarily close to x_0, hence $x_0 \in \overline{f^{-1}(q)}$. Since $f^{-1}(q)$ is closed, $x_0 \in f^{-1}(q)$, contradicting $f(x_0) < q$. Thus U_q is open; similarly V_q is open.

For rationals $r < s$,

$$f^{-1}((r,s)) = V_r \cap U_s$$

is open. Because intervals with rational endpoints form a base of $\mathbb{R}$, the preimage of every open set is open; hence f is continuous.

Examples.

- *IVP but some rational fiber not closed (discontinuous):* $f(x) = \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. For $q = \frac{1}{2}$, $f^{-1}(q)$ accumulates at 0 but does not contain 0.
- *Closed rational fibers but no IVP (discontinuous):* $f(x) = \mathbf{1}_{[0,\infty)}(x)$. Fibers at $q = 0, 1$ are closed; others are empty (closed). Yet f is discontinuous at 0.
- *Continuous:* any continuous f has all fibers $f^{-1}(q)$ closed for every $q \in \mathbb{R}$ and satisfies IVP.

Problem 12 — Minimal-matching distance on unordered q-tuples

Let $q \geq 2$ and $n \geq 1$ be integers. Denote by $\mathcal{Q}$ the set of all unordered q-tuples of points in $\mathbb{R}^n$ (repetitions allowed):

$$\mathcal{Q} = \Big\{\{x_1, x_2, \ldots, x_q\} : x_j \in \mathbb{R}^n\Big\}.$$

Define $\mathcal{G} : \mathcal{Q} \times \mathcal{Q} \to \mathbb{R}$ by

$$\mathcal{G}(\{x_1, \ldots, x_q\}, \{y_1, \ldots, y_q\}) = \inf_{\sigma \in S_q} \left(\sum_{j=1}^{q} \| \, y_j - x_{\sigma(j)} \, \|^2 \right)^{1/2}.$$

(i) Show that $\mathcal{G}$ is a metric on $\mathcal{Q}$.

(ii) Assume $n = 1$. In this case, for any point $x = \{x_1, \ldots, x_q\} \in \mathcal{Q}$, relabel so that $x_1 \leq \cdots \leq x_q$, and do likewise for $y = \{y_1, \ldots, y_q\}$. Is it true that

$$\mathcal{G}(\{x_1, \ldots, x_q\}, \{y_1, \ldots, y_q\}) = \left(\sum_{j=1}^{q} (x_j - y_j)^2 \right)^{1/2} ?$$

Fix $q \geq 2$ and $n \geq 1$. Let $\mathcal{Q}$ be the set of unordered q-tuples (multisets) of points in $\mathbb{R}^n$. For $x = \{x_1, \dots, x_q\}$ and $y = \{y_1, \dots, y_q\}$ define

$$\mathcal{G}(x, y) = \inf_{\sigma \in S_q} \left(\sum_{j=1}^{q} \|y_j - x_{\sigma(j)}\|^2 \right)^{1/2}.$$

(i) $\mathcal{G}$ is a metric on $\mathcal{Q}$

Nonnegativity and symmetry are immediate.

Identity of indiscernibles. If $\mathcal{G}(x, y) = 0$, there exist permutations σ_k with $\sum_j \|y_j - x_{\sigma_k(j)}\|^2 \to 0$, hence (passing to a subsequence of permutations if needed) $y_j = x_{\sigma(j)}$ for all j, i.e. the multisets coincide. Conversely, equal multisets give zero cost.

Triangle inequality. Let $\alpha, \beta \in S_q$. For each i, $\|x_i - z_{\beta(i)}\| \leq \|x_i - y_{\alpha(i)}\| + \|y_{\alpha(i)} - z_{\beta(i)}\|$. Applying Minkowski's inequality in $\mathbb{R}^{nq}$,

$$\left(\sum_{i=1}^{q} \|x_i - z_{\beta(i)}\|^2 \right)^{1/2} \leq \left(\sum_{i=1}^{q} \|x_i - y_{\alpha(i)}\|^2 \right)^{1/2} + \left(\sum_{i=1}^{q} \|y_{\alpha(i)} - z_{\beta(i)}\|^2 \right)^{1/2}.$$

Taking the infimum over α on the right and over β on the left yields $\mathcal{G}(x, z) \leq \mathcal{G}(x, y) + \mathcal{G}(y, z)$. Thus $\mathcal{G}$ is a metric.

(ii) The one-dimensional case

Assume $n = 1$ and label so that $x_1 \leq \cdots \leq x_q$ and $y_1 \leq \cdots \leq y_q$. By the rearrangement inequality,

$$\sum_{j=1}^{q} (x_j - y_j)^2 \;\leq\; \sum_{j=1}^{q} (x_j - y_{\sigma(j)})^2 \qquad \text{for all } \sigma \in S_q.$$

Hence the infimum in the definition of $\mathcal{G}$ is attained at $\sigma = \mathrm{id}$, and

$$\mathcal{G}\big(\{x_1, \dots, x_q\}, \{y_1, \dots, y_q\}\big) = \left(\sum_{j=1}^{q} (x_j - y_j)^2 \right)^{1/2}.$$

Problem 1 — Winding number of a polynomial image

Let $p(z) = z^2 - 4z + 3$ and define $\gamma : [0, 1] \to \mathbb{C}$ by $\gamma(t) = p(e^{2\pi i t})$.

(a) Show that the closed path associated with γ does not pass through 0.

(b) Compute $w(\gamma, 0)$: (i) directly from the definition; (ii) by factoring; (iii) by the dog–walking lemma.

Solution

(a) Put $\theta = 2\pi t$. Then

$$\Re\,\gamma(t) = \cos(2\theta) - 4\cos\theta + 3 = 2(\cos\theta - 1)^2 \geq 0,$$

with equality only at $\theta = 0, 2\pi$. Thus for $t \in (0,1)$ we have $\Re\,\gamma(t) > 0$, so $\gamma(t) \neq 0$. The associated closed loop (identify $t = 0$ and $t = 1$) avoids 0.

(b)(i) Direct computation. With $\theta = 2\pi t$,

$$\Im\,\gamma(t) = \sin(2\theta) - 4\sin\theta = 2\sin\theta(\cos\theta - 2),$$

which is negative on $(0, \pi)$, zero at $\theta = \pi$ (where $\gamma = p(-1) = 8 > 0$), and positive on $(\pi, 2\pi)$. Since $\Re\,\gamma(t) > 0$ on $(0,1)$, the loop lies in the right half–plane, crossing the real axis once at a positive point. The argument increases from just below $-\frac{\pi}{2}$ to just above $+\frac{\pi}{2}$ (change $+\pi$), and the closing jump at the basepoint contributes $-\pi$. Total change 0, hence $w(\gamma, 0) = 0$.

(b)(ii) By factoring. $p(z) = (z-1)(z-3)$, so

$$\gamma(t) = \left(e^{2\pi i t} - 1\right)\left(e^{2\pi i t} - 3\right).$$

The loop $e^{2\pi i t} - 3$ is the circle centered at -3 of radius 1 and does not encircle 0, so its winding number is 0. The loop $e^{2\pi i t} - 1$ is the circle centered at -1 of radius 1, touching 0 only at the basepoint; its winding is 0. Therefore $w(\gamma, 0) = 0$.

(b)(iii) By the dog–walking lemma. Let $a(t) = e^{2\pi i t} - 1$ and $b(t) = e^{2\pi i t} - 3$. Then $w(\gamma, 0) = w(a \cdot b, 0) = w(a, 0) + w(b, 0) = 0 + 0 = 0$. □

Winding Number

Given a continuous closed path (loop)

$$\gamma : [0,1] \to \mathbb{C}$$

and a point $a \in \mathbb{C}$ with $\gamma(t) \neq a$ for all t, the *winding number* (or *index*) of γ about a is defined by

$$w(\gamma, a) = \frac{1}{2\pi}\Delta_{t\in[0,1]} \arg\left(\gamma(t) - a\right) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a},$$

i.e. the total change of the argument of $\gamma(t) - a$ divided by 2π, equivalently the contour integral.

Intuitively: $w(\gamma, a)$ counts how many times (with orientation) the curve γ goes around the point a.

Key properties

- **Integer-valued:** $w(\gamma, a) \in \mathbb{Z}$ (it is always an integer).
- **Homotopy invariance:** If two loops avoiding a can be continuously deformed into one another without crossing a, then they have the same winding number about a.
- **Additivity:** For loop concatenation $\gamma_1 * \gamma_2$,
$$w(\gamma_1 * \gamma_2, a) = w(\gamma_1, a) + w(\gamma_2, a).$$
- **Orientation:** Reversing direction flips the sign:
$$w(\gamma^{-1}, a) = -\,w(\gamma, a).$$
- **Locality:** If a lies in a component of $\mathbb{C} \setminus \gamma([0,1])$ that is "outside," then $w(\gamma, a) = 0$. For a positively oriented simple closed curve, $w(\gamma, a) = 1$ for points inside and $w(\gamma, a) = 0$ for points outside (Jordan curve theorem).
- **Multiplicativity for products ("dog-walking lemma"):**
$$w(\gamma_1 \cdot \gamma_2, 0) = w(\gamma_1, 0) + w(\gamma_2, 0),$$
for loops avoiding 0, where $(\gamma_1 \cdot \gamma_2)(t) = \gamma_1(t)\gamma_2(t)$.

How to compute in practice

1. **Total argument change.** Track $\arg(\gamma(t) - a)$ continuously and add up its net change over one loop; divide by 2π.
2. **Integral formula.**
$$w(\gamma, a) = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z - a}.$$
For piecewise C^1 curves this is straightforward; residues help if the integrand is more general.
3. **Counting signed crossings (real-axis trick).** When γ never hits a and you can choose a ray from a that the curve crosses transversely, $w(\gamma, a)$ equals the number of crossings from below to above *minus* the number from above to below (signs determined by orientation).

Examples

- $\gamma(t) = e^{2\pi it}$: unit circle around 0; $w(\gamma, 0) = 1$.
- $\gamma(t) = e^{2\pi ikt}$: k-fold loop; $w(\gamma, 0) = k$.
- $\gamma(t) = e^{2\pi it} - 3$: shifted circle not enclosing 0; $w(\gamma, 0) = 0$.
- $\gamma(t) = (e^{2\pi it}-1)(e^{2\pi it}-3)$: product path; $w(\gamma, 0) = w(e^{2\pi it}-1, 0)+w(e^{2\pi it}-3, 0) = 0$.

Problem 2 — Fixed and antipodal points for a disk–extendable circle map

Let $g : S^1 \to S^1$ be continuous, where $S^1 = \{z \in \mathbb{C} : \ |z| = 1\}$. Suppose g admits a continuous extension $G : D \to S^1$ to the closed unit disk $D = \{z \in \mathbb{C} : \ |z| \le 1\}$ (so $G|_{S^1} = g$). Prove that:

(a) there exists $z \in S^1$ with $g(z) = z$;
(b) there exists $z \in S^1$ with $g(z) = -z$.

Theory you'll use

- **Degree on S^1.** For $g : S^1 \to S^1$, there is an integer $\deg g$ such that, if

$$g(e^{i\theta}) \ = \ e^{if(\theta)}$$

for a continuous lift $f : \mathbb{R} \to \mathbb{R}$ (possible since $\exp(i\cdot) : \mathbb{R} \to S^1$ is a covering), then

$$\deg g \ = \ \frac{f(\theta + 2\pi) - f(\theta)}{2\pi} \ \in \ \mathbb{Z}.$$

If g is *null–homotopic* (homotopic to a constant), then $\deg g = 0$.

- **Why $\deg g = 0$ here.** The extension $G : D \to S^1$ gives a homotopy on the boundary:

$$H(r, e^{i\theta}) \ = \ G\big(re^{i\theta}\big), \qquad r \in [0, 1].$$

Thus g is homotopic on S^1 to the constant map $z \mapsto G(0)$. Hence g is null–homotopic and $\deg g = 0$.

- **Intermediate value on a lift.** With $\deg g = 0$ we may choose a lift f with $f(\theta + 2\pi) = f(\theta)$. Put $h(\theta) = f(\theta) - \theta$. Then

$$h(\theta + 2\pi) = f(\theta + 2\pi) - (\theta + 2\pi) = h(\theta) - 2\pi,$$

so as θ runs from 0 to 2π, the continuous function h drops by 2π. Therefore it attains every value in an interval of length 2π; in particular some $2k\pi$ and some $\pi + 2k\pi$.

- **Translations to fixed/antipodal points.**

$$g(e^{i\theta}) = e^{i\theta} \iff f(\theta) \equiv \theta \pmod{2\pi} \iff h(\theta) \in 2\pi\mathbb{Z}.$$

$$g(e^{i\theta}) = -\,e^{i\theta} \iff f(\theta) \equiv \theta + \pi \pmod{2\pi} \iff h(\theta) \in \pi + 2\pi\mathbb{Z}.$$

Theory

- **Degree and lifts.** For $g : S^1 \to S^1$ there is an integer $\deg g$ with the property: there exists a continuous lift $f : \mathbb{R} \to \mathbb{R}$ such that $g(e^{i\theta}) = e^{if(\theta)}$ and

$$\deg g = \frac{f(\theta + 2\pi) - f(\theta)}{2\pi}.$$

  If g is null–homotopic, then $\deg g = 0$.

- **Extension ⇒ degree 0.** Setting $H(r, e^{i\theta}) = G(re^{i\theta})$ for $r \in [0, 1]$ gives a homotopy on S^1 from g to the constant map $z \mapsto G(0)$. Hence g is null–homotopic and $\deg g = 0$.
- **Intermediate value on $h(\theta) = f(\theta) - \theta$.** When $\deg g = 0$ we can choose f with $f(\theta + 2\pi) = f(\theta)$. Then $h(\theta + 2\pi) = h(\theta) - 2\pi$, so h takes every value in an interval of length 2π.

Solution

Since $\deg g = 0$, choose a lift f with $f(\theta + 2\pi) = f(\theta)$ and write $h(\theta) = f(\theta) - \theta$. Then h is continuous and $h(\theta + 2\pi) = h(\theta) - 2\pi$; hence as θ runs from 0 to 2π, h takes every value in an interval of length 2π.

(a) Fixed point. There exists θ_0 with $h(\theta_0) = 2k\pi$ for some $k \in \mathbb{Z}$, i.e. $f(\theta_0) = \theta_0 + 2k\pi$. Therefore

$$g(e^{i\theta_0}) = e^{if(\theta_0)} = e^{i(\theta_0 + 2k\pi)} = e^{i\theta_0},$$

so $z = e^{i\theta_0}$ satisfies $g(z) = z$.

(b) Antipodal point. Likewise there exists θ_1 with $h(\theta_1) = \pi + 2k\pi$. Then

$$g(e^{i\theta_1}) = e^{if(\theta_1)} = e^{i(\theta_1 + \pi)} = -\,e^{i\theta_1},$$

so $z = e^{i\theta_1}$ satisfies $g(z) = -z$. □

Remarks / Examples

If g is constant, both conclusions are immediate. If $g(z) = \overline{z}$, then g extends to $G(re^{i\theta}) = e^{-i\theta}$ and one finds fixed points at $z = \pm 1$ and antipodal points at $z = \pm i$. Maps like $g(z) = z^k$ with $|k| \geq 2$ have $\deg g = k \neq 0$ and do not extend to D, so (a)–(b) need not hold for them.

Degree of a map $g : S^1 \to S^1$

Intuition. The degree $\deg g \in \mathbb{Z}$ tells how many times (with orientation) the image loop $g \circ \gamma$ winds around the circle while the domain circle is traversed once. Positive degree means counterclockwise wrapping; negative degree means clockwise.

Definition via a lift. Parametrize S^1 by $\theta \mapsto e^{i\theta}$. Since $\exp(i\,\cdot) : \mathbb{R} \to S^1$ is a covering map, there exists a continuous *lift* $f : \mathbb{R} \to \mathbb{R}$ with

$$g\left(e^{i\theta}\right) = e^{if(\theta)} \qquad (\theta \in \mathbb{R}).$$

Then

$$\deg g \;=\; \frac{f(\theta + 2\pi) - f(\theta)}{2\pi} \;\in\; \mathbb{Z}.$$

Equivalently, $\deg g$ is the total change of the argument of $g(e^{i\theta})$ over one lap, divided by 2π.

Equivalent description (winding number). Let $\gamma(\theta) = g(e^{i\theta})$. Then

$$\deg g \;=\; \frac{1}{2\pi}\,\Delta_{\theta\in[0,2\pi]} \arg \gamma(\theta) \;=\; \frac{1}{2\pi i}\int_\gamma \frac{dz}{z}.$$

Basic facts.

- $\deg(g_1 \circ g_2) = \deg(g_1)\,\deg(g_2)$.
- g is null–homotopic (homotopic to a constant) $\iff$ $\deg g = 0$.
- Two maps $S^1 \to S^1$ are homotopic $\iff$ they have the same degree.

Examples.

$$g(z) = z^k \;\Rightarrow\; \deg g = k; \qquad g(z) = \overline{z} \;\Rightarrow\; \deg g = -1; \qquad g(z) \equiv 1 \;\Rightarrow\; \deg g = 0.$$

Problem 3 — Can a discontinuous function be uniformly approximated by polynomials?

Question. Does there exist a function $f : [0,1] \to \mathbb{R}$ with a discontinuity which can be approximated *uniformly on* $[0,1]$ by polynomials?

Theory

- **Weierstrass theorem.** Every continuous $f : [0,1] \to \mathbb{R}$ can be uniformly approximated by polynomials.
- **Uniform limits preserve continuity.** If p_n are continuous and $\|p_n - f\|_\infty \to 0$, then f is continuous.
- **Jump lower bound.** If f has a jump at x_0 with left/right limits $a < b$, then for any continuous g
$$\|f - g\|_\infty \;\geq\; \frac{b-a}{2}.$$
In particular, the step function $f = \mathbf{1}_{[1/2,1]}$ satisfies $\inf_{g \in C[0,1]} \|f - g\|_\infty \geq \frac{1}{2}$.

Answer and proof

Answer: No. Suppose there were polynomials p_n with $p_n \to f$ uniformly on $[0,1]$. Since each p_n is continuous and uniform limits of continuous functions are continuous, f must be continuous, a contradiction. Hence a discontinuous function cannot be uniformly approximated on $[0,1]$ by polynomials.

Examples / remarks

- *Pointwise vs. uniform:* One can build polynomials that converge pointwise to a step function away from the jump, but the uniform error on $[0,1]$ never goes below $1/2$.
- *Uniform on punctured sets:* For any $\varepsilon > 0$, the step function can be uniformly approximated on $[0,1] \setminus (1/2 - \varepsilon,\, 1/2 + \varepsilon)$.
- *L^p approximation:* For $1 \leq p < \infty$, polynomials are dense in $L^p[0,1]$; discontinuous functions can be approximated in L^p (but not uniformly).

Problem 4 — Degrees must blow up in uniform polynomial approximation

Let $f : [0, 1] \to \mathbb{R}$ be continuous and not a polynomial. Suppose (p_n) are polynomials with $p_n \to f$ uniformly on $[0, 1]$, and set $d_n = \deg p_n$. Prove that $d_n \to \infty$.

Theory

- **Finite-dimensional subspaces are closed.** In any normed linear space X, every finite-dimensional subspace V is closed in X.
- **Bounded degree subspace.** $\mathcal{P}_{\leq m} := \{p : \deg p \leq m\} \subset C[0, 1]$ has dimension $m + 1$ (basis $1, x, \ldots, x^m$), hence is a closed subspace of $(C[0, 1], \|\cdot\|_\infty)$.
- **Uniform limit in a closed set.** If $E \subset X$ is closed and $x_n \in E$ with $x_n \to x$, then $x \in E$.

Solution

Assume $\{d_n\}$ is not unbounded. Then there exists m and an infinite subsequence (p_{n_k}) with $\deg p_{n_k} \leq m$ for all k, i.e. $p_{n_k} \in \mathcal{P}_{\leq m}$. Since $\mathcal{P}_{\leq m}$ is closed in $C[0, 1]$ and $p_{n_k} \to f$ uniformly, we must have $f \in \mathcal{P}_{\leq m}$; thus f is a polynomial of degree $\leq m$, contradicting the assumption that f is not a polynomial. Hence $d_n \to \infty$. $\square$

Remarks

Examples such as $f(x) = |x|$, $f(x) = \sqrt{x}$, or $f(x) = e^x$ on $[0, 1]$ show that any uniformly convergent polynomial approximation requires degrees tending to infinity; a bounded-degree family cannot approximate a non-polynomial uniformly.

Practical meaning and approximation error

The theorem implies that if f is continuous but not a polynomial, then any sequence of polynomials (p_n) converging uniformly to f must have degrees $\deg p_n \to \infty$. In practice, this means:

- We can only use a *finite-degree* polynomial p_m at a time, which yields an approximation error that depends on m.
- Increasing the degree m decreases the approximation error; we stop once the error is below the desired tolerance.

Define the *best uniform (Chebyshev) approximation error* by

$$E_m(f) \ := \inf_{\deg p \le m} \|f - p\|_\infty.$$

Then $E_m(f) \to 0$ as $m \to \infty$ (by Weierstrass), but the rate of decay depends on the smoothness of f:

- If f is **analytic** in a neighborhood of $[0, 1]$, then $E_m(f)$ decays **exponentially fast** ("spectral accuracy").
- If $f \in C^k[0, 1]$, then $E_m(f) \lesssim C\, m^{-k}$ (a Jackson-type estimate).
- If f is merely continuous, $E_m(f) \to 0$ but may do so **arbitrarily slowly**.

Practical approximation strategies.

- Use **Chebyshev polynomials** or the **Remez algorithm** to find near-optimal or optimal polynomial approximations in the uniform norm.
- Choose the degree m so that the maximum (or sampled) error is below the target tolerance ε.
- For interpolation, prefer **Chebyshev nodes** rather than equispaced points, to avoid large oscillations (Runge phenomenon).

Thus, polynomial approximation of a non-polynomial function is always an *infinite-degree process* in theory, but a *finite-degree compromise* in practice: we stop at the degree m that yields the desired accuracy.

Approximation without uniformity

Pointwise at continuity points (Bernstein). For a bounded function $f : [0, 1] \to \mathbb{R}$, the Bernstein polynomials

$$B_n[f](x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}$$

satisfy $B_n[f](x) \to f(x)$ at every continuity point x of f. If f is continuous, the convergence is uniform (Weierstrass); if f is discontinuous, the convergence is typically pointwise but not uniform.

L^p-approximation ($1 \leq p < \infty$). Polynomials are dense in $L^p([0,1])$; i.e., for any $f \in L^p$ and $\varepsilon > 0$ there exists a polynomial p with $\|f - p\|_{L^p} < \varepsilon$. Thus discontinuous functions can be approximated by polynomials in L^p, even though uniform approximation may be impossible.

Example: a step function

Let $H(x) = \mathbf{1}_{[1/2,\,1]}(x)$. Then

$$B_n[H](x) = \sum_{k=\lceil n/2 \rceil}^{n} \binom{n}{k} x^k (1-x)^{n-k}.$$

One has

$$B_n[H](x) \to \begin{cases} 0, & x < \frac{1}{2}, \\ \frac{1}{2}, & x = \frac{1}{2}, \\ 1, & x > \frac{1}{2}, \end{cases} \qquad (n \to \infty).$$

Hence $B_n[H] \to H$ pointwise at all continuity points of H, but not uniformly. Moreover, for each $1 \leq p < \infty$,

$$\|B_n[H] - H\|_{L^p([0,1])} \longrightarrow 0,$$

so H is approximable by polynomials in L^p.

Summary

Dropping uniform convergence allows polynomial approximation of discontinuous functions:

- pointwise convergence at continuity points (e.g. Bernstein polynomials),
- L^p-convergence for every $1 \leq p < \infty$ (polynomials are dense in L^p),
- often almost-everywhere convergence for suitable subsequences.

Uniform convergence on the whole interval still fails for discontinuous f.

Degrees still blow up without uniformity

Let (p_n) be polynomials on $[0,1]$ converging to f (pointwise, or in L^p, $1 \le p < \infty$). If f is not a polynomial, then necessarily $\deg p_n \to \infty$.

Proof (pointwise case). Assume $\deg p_n \le m$ for all n. Let $\mathcal{P}_{\le m}$ be the space of polynomials of degree $\le m$. Choose distinct nodes $x_0, \ldots, x_m \in [0,1]$. The evaluation map

$$T : \mathcal{P}_{\le m} \to \mathbb{R}^{m+1}, \qquad T(p) = (p(x_0), \ldots, p(x_m))$$

is a linear isomorphism. Since $p_n(x_j) \to f(x_j)$ for each j, we have $T(p_n) \to v := (f(x_0), \ldots, f(x_m))$. Let $p := T^{-1}(v) \in \mathcal{P}_{\le m}$. By Lagrange interpolation, for each x there exists a linear functional $L_x : \mathbb{R}^{m+1} \to \mathbb{R}$ with $p_n(x) = L_x\big(T(p_n)\big)$; hence $p_n(x) \to L_x(v) = p(x)$. Therefore the pointwise limit f equals p, a polynomial of degree $\le m$. Contradiction. Thus $\deg p_n$ cannot be bounded; hence $\deg p_n \to \infty$. $\square$

Remark (normed convergences). For convergence in any norm (e.g. L^p), the same conclusion holds since $\mathcal{P}_{\le m}$ is finite-dimensional and therefore closed; a limit of (p_n) with $\deg p_n \le m$ must lie in $\mathcal{P}_{\le m}$.

Problem 5 — Interpolation remainder and uniform convergence

Let $f : [-1,1] \to \mathbb{R}$ be $(n+1)$-times continuously differentiable and $J_n = \{x_0, \ldots, x_n\} \subset [-1,1]$ distinct. Let $P_{J_n} \in \Pi_n$ be the unique polynomial with $P_{J_n}(x_j) = f(x_j)$. Define

$$\beta_{J_n}(x) = (x - x_0) \cdots (x - x_n).$$

Remainder formula. For every $x \in [-1,1]$ there exists $\zeta \in (-1,1)$ such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!}\, \beta_{J_n}(x).$$

Proof. If $x = x_j$ this is trivial. Otherwise choose λ so that $g(y) = f(y) - P_{J_n}(y) - \lambda\, \beta_{J_n}(y)$ satisfies $g(x) = 0$. Since $g(x_j) = 0$ for $j = 0, \ldots, n$, g has $n+2$ distinct zeros. By repeated Rolle, there exists ζ with $g^{(n+1)}(\zeta) = 0$. Because $\deg P_{J_n} \le n$, $g^{(n+1)}(y) = f^{(n+1)}(y) - \lambda(n+1)!$, hence $\lambda = f^{(n+1)}(\zeta)/(n+1)!$. Evaluating at x yields the formula. $\square$

Uniform convergence under factorial control. Assume $f \in C^\infty[-1,1]$ and $\sup_x |f^{(n)}(x)| \le M^n$ for all $n \ge 1$. From the remainder formula and the bound

$$|\beta_{J_n}(x)| = \prod_{j=0}^{n} |x - x_j| \le 2^{n+1} \qquad (x, x_j \in [-1,1]),$$

we obtain, uniformly in x and in the choice of nodes J_n,

$$\|f - P_{J_n}\|_\infty \le \frac{\sup_y |f^{(n+1)}(y)|}{(n+1)!} \sup_x |\beta_{J_n}(x)| \le \frac{M^{n+1}\, 2^{n+1}}{(n+1)!} \xrightarrow[n\to\infty]{} 0.$$

Hence $P_{J_n} \to f$ uniformly on $[-1, 1]$. $\square$

Remarks

The crude bound $\sup |\beta_{J_n}| \le 2^{n+1}$ is node-independent. Sharper bounds are possible for specific choices, e.g. Chebyshev nodes minimize $\|\beta_{J_n}\|_\infty$ among monic polynomials, yielding $\|\beta_{J_n}\|_\infty = 2^{-n}$. Without derivative growth control, poor nodes can cause divergence (Runge phenomenon).

Approximation and interpolation: definitions

Approximation (by polynomials). Let $X \subset \mathbb{R}$ and $f : X \to \mathbb{R}$. A sequence of polynomials (p_n) *approximates* f on X if $p_n \to f$ in a prescribed mode:

- *Uniform approximation:* $\|f - p_n\|_\infty := \sup_{x\in X} |f(x) - p_n(x)| \to 0$.
- *Pointwise approximation:* for every $x \in X$, $p_n(x) \to f(x)$.
- L^p *approximation* ($1 \le p < \infty$)*:* $\|f - p_n\|_{L^p(X)} \to 0$.

For $m \in \mathbb{N}$, a *best degree-m approximation* is a polynomial $p_m^* \in \Pi_m$ such that

$$\|f - p_m^*\| = \inf_{q\in\Pi_m} \|f - q\|,$$

where $\|\cdot\|$ is the chosen norm (e.g., $\|\cdot\|_\infty$ or $\|\cdot\|_{L^p}$).

Interpolation (by polynomials). Given distinct nodes $x_0, \dots, x_n \in X$ and data $y_j = f(x_j)$, a polynomial $P \in \Pi_n$ *interpolates* f at the nodes if

$$P(x_j) = f(x_j) \qquad (j = 0, \dots, n).$$

For distinct nodes, such P exists and is unique; it can be written in *Lagrange form*

$$P(x) = \sum_{j=0}^{n} f(x_j)\, \ell_j(x), \qquad \ell_j(x) = \prod_{\substack{k=0\\k\neq j}}^{n} \frac{x - x_k}{x_j - x_k},$$

or in *Newton form* using divided differences.

Approximation vs. interpolation. Interpolation enforces exact matching at finitely many points; approximation aims to minimize a global error without necessarily matching at any specific point. An interpolant may approximate well (e.g. at Chebyshev nodes), but can also behave poorly (Runge phenomenon with equispaced nodes).

Problem 6 — Chebyshev nodes minimize the sup norm

Fix $n \geq 1$ and let $J = \{x_1, \dots, x_n\} \subset [-1, 1]$ be distinct points. Set $\beta_J(x) = (x - x_1) \cdots (x - x_n)$ and

$$F(x_1, \dots, x_n) = \sup_{x \in [-1,1]} |\beta_J(x)|.$$

Prove that F is minimized when $x_k = \cos\!\left(\frac{(2k-1)\pi}{2n}\right)$, $k = 1, \dots, n$.

Theory

The Chebyshev polynomials $T_n(x) = \cos(n \arccos x)$ satisfy $|T_n(x)| \leq 1$ on $[-1, 1]$ and $T_n(\cos(\frac{k\pi}{n})) = (-1)^k$ for $k = 0, \dots, n$. For $n \geq 1$, the leading coefficient of T_n is 2^{n-1}, so $q_n(x) := 2^{1-n} T_n(x)$ is monic and $\|q_n\|_\infty = 2^{1-n}$. Its zeros are $\cos(\frac{(2k-1)\pi}{2n})$, $k = 1, \dots, n$.

Solution

Let $p(x) = (x - x_1) \cdots (x - x_n)$ be any monic degree-n polynomial with nodes in $[-1, 1]$. Let $q_n(x) = 2^{1-n} T_n(x)$ be the monic Chebyshev polynomial.

Consider the $n + 1$ points

$$\xi_k := \cos\!\left(\frac{k\pi}{n}\right), \qquad k = 0, 1, \dots, n,$$

at which $T_n(\xi_k) = (-1)^k$ and thus $q_n(\xi_k) = 2^{1-n}(-1)^k$.

If $\|p\|_\infty < \|q_n\|_\infty = 2^{1-n}$, then

$$|p(\xi_k)| < |q_n(\xi_k)| = 2^{1-n} \qquad \text{for all } k,$$

so the values $p(\xi_k)$ and $q_n(\xi_k)$ have the *same sign* $(-1)^k$.

Hence the function $r := p - q_n$ satisfies

$$r(\xi_k) = p(\xi_k) - q_n(\xi_k)$$

and *alternates in sign* at the points $\xi_0, \xi_1, \dots, \xi_n$. By the intermediate value theorem, r has at least one zero in each open interval (ξ_{k+1}, ξ_k), i.e. at least n distinct zeros in $(-1, 1)$.

But $\deg r \le n-1$, a contradiction unless $r \equiv 0$. Therefore no monic polynomial can have smaller sup-norm than q_n, and the minimizer is unique:

$$\min_{p \in \Pi_n^{\mathrm{mon}}} \|p\|_\infty \;=\; \|q_n\|_\infty \;=\; 2^{1-n}.$$

Consequently, the product β_J attains the minimal possible $\sup_{[-1,1]} |\cdot|$ exactly when it equals q_n (hence has the same zeros), i.e. when

$$x_k \;=\; \cos\!\Bigl(\frac{(2k-1)\pi}{2n}\Bigr), \qquad k = 1, \dots, n.$$

Thus F is minimized at the **Chebyshev nodes**.

Examples

For $n = 1$, $q_1(x) = x$ and $\min \|p\|_\infty = 1$ (zero at 0). For $n = 2$, $q_2(x) = x^2 - \frac{1}{2}$ with zeros $\pm\frac{\sqrt{2}}{2}$ and $\|q_2\|_\infty = \frac{1}{2}$.

Theory you'll use

Chebyshev polynomials

The Chebyshev polynomials $T_n(x) = \cos(n \arccos x)$ satisfy

$$|T_n(x)| \le 1 \quad (x \in [-1, 1]), \qquad T_n\left(\cos\frac{k\pi}{n}\right) = \cos(k\pi) = (-1)^k \quad (k = 0, \dots, n).$$

For $n \ge 1$, the leading coefficient of T_n is 2^{n-1}. Hence

$$q_n(x) := 2^{1-n}\, T_n(x)$$

is *monic* of degree n and

$$\|q_n\|_\infty = \sup_{x \in [-1,1]} |q_n(x)| = \; 2^{1-n}.$$

Its zeros are precisely

$$x_k = \cos\!\Bigl(\frac{(2k-1)\pi}{2n}\Bigr), \qquad k = 1, \dots, n.$$

Extremal (alternation) property

Among all monic polynomials Π_n^{mon} of degree n,

$$\inf_{p\in\Pi_n^{\mathrm{mon}}} \|p\|_\infty \;=\; \|q_n\|_\infty \;=\; 2^{\,1-n},$$

and the minimizer is unique (up to equality) and equal to q_n. A concise proof uses the *alternation* of T_n at the points $\cos(k\pi/n)$.

Chebyshev polynomials T_n: core formulas

Definitions

$$T_n(x) = \cos\Big(n \arccos x\Big), \qquad x \in [-1,1],\; n \ge 0.$$

Equivalently, the three–term recurrence

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2x\, T_n(x) - T_{n-1}(x) \quad (n \ge 1).$$

Basic properties

$$|T_n(x)| \le 1 \;\; (x \in [-1,1]), \qquad T_n(1) = 1, \;\; T_n(-1) = (-1)^n.$$

Zeros:

$$T_n\Big(\cos\theta\Big) = 0 \iff \theta = \frac{(2k-1)\pi}{2n}, \quad x_k = \cos\Big(\frac{(2k-1)\pi}{2n}\Big), \;\; k = 1, \dots, n.$$

Extrema:

$$x = \cos\Big(\frac{k\pi}{n}\Big), \;\; k = 0, \dots, n, \qquad T_n\left(\cos\frac{k\pi}{n}\right) = \cos(k\pi) = (-1)^k.$$

Leading coefficient and monic scaling

For $n \ge 1$, the leading coefficient of T_n is $2^{\,n-1}$, so the monic Chebyshev is

$$q_n(x) := 2^{1-n} T_n(x), \qquad \|q_n\|_\infty = 2^{1-n}.$$

Explicit expansion

For $n \ge 0$,

$$T_n(x) = \frac{n}{2} \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j}{n-j} \binom{n-j}{j} (2x)^{n-2j},$$

or equivalently

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \, \frac{n}{n-j} \binom{n-j}{j} \, x^{\,n-2j}.$$

Generating function

$$\sum_{n=0}^{\infty} T_n(x)\, t^n = \frac{1-xt}{1-2xt+t^2}, \qquad |t|<1.$$

Derivative / relation with U_n

Let U_n be Chebyshev polynomials of the second kind. Then

$$\frac{d}{dx} T_n(x) = n\, U_{n-1}(x), \qquad U_n(\cos\theta) = \frac{\sin\big((n+1)\theta\big)}{\sin\theta}.$$

Orthogonality (on $[-1,1]$)

$$\int_{-1}^{1} \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}}\, dx = \begin{cases} 0, & m \neq n, \\ \pi, & n=m=0, \\ \frac{\pi}{2}, & n=m \geq 1. \end{cases}$$

Examples

1. $n=2$. Monic Chebyshev:

$$q_2(x) = 2^{1-2} T_2(x) = x^2 - \tfrac{1}{2}, \qquad \|q_2\|_\infty = \tfrac{1}{2}.$$

   Zeros at $\pm 2^{-1/2}$. Any other two nodes yield a larger sup norm.

2. $n=3$ **: Chebyshev vs. equispaced.**

$$q_3(x) = 2^{1-3} T_3(x) = x^3 - \tfrac{3}{4}x, \qquad \|q_3\|_\infty = \tfrac{1}{4}.$$

   Equispaced nodes $\{-1,0,1\}$ give $p(x) = x^3 - x$. Max on $[-1,1]$ at $x = \pm 1/\sqrt{3}$:

$$\|p\|_\infty = \frac{2}{3\sqrt{3}} \approx 0.385 \; > \; \tfrac{1}{4}.$$

3. $n=4$ **: Chebyshev vs. equispaced.**

$$q_4(x) = 2^{1-4} T_4(x) = x^4 - x^2 + \tfrac{1}{8}, \qquad \|q_4\|_\infty = \tfrac{1}{8}.$$

Equispaced nodes $\{-1, -\frac{1}{3}, \frac{1}{3}, 1\}$ give

$$p(x) = (x^2 - 1)\left(x^2 - \tfrac{1}{9}\right) = x^4 - \tfrac{10}{9}x^2 + \tfrac{1}{9}.$$

Critical points $x = 0,\ \pm\sqrt{5}/3$ yield

$$\|p\|_\infty = \max\left\{\tfrac{1}{9},\ \tfrac{16}{81}\right\} = \tfrac{16}{81} \approx 0.198\ >\ \tfrac{1}{8}.$$

Problem 7 — Uniqueness of the minimax polynomial

Exact statement

Let $f \in C([0,1])$. Suppose p is a polynomial of degree $< n$ that minimizes the uniform error

$$\|f - q\|_\infty \ :=\ \sup_{x\in[0,1]} |\, f(x) - q(x)\,|$$

among all polynomials q with $\deg q < n$. The *converse equal–ripple criterion* (equioscillation theorem) says there exist $n+1$ points

$$0 \le a_0 < a_1 < \cdots < a_n \le 1$$

such that either

$$f(a_k) - p(a_k)\ =\ (-1)^k\, \|f - p\|_\infty \qquad \text{for } k = 0, \ldots, n,$$

or

$$f(a_k) - p(a_k)\ =\ (-1)^{k+1}\, \|f - p\|_\infty \qquad \text{for } k = 0, \ldots, n.$$

Task. Assuming this converse criterion, prove: for every $f \in C([0,1])$ and every positive integer n, the minimizer of $\|f - q\|_\infty$ over $\deg q < n$ is *unique.*

Theory you'll use

- **Convexity.** The set $\Pi_{<n}$ of polynomials of degree $< n$ is convex, and the map $q \mapsto \|f - q\|_\infty$ is convex. Hence, if p_1, p_2 are both minimizers with common error E, the midpoint $r = \frac{1}{2}(p_1 + p_2)$ satisfies $\|f - r\|_\infty \le E$, with *strict* inequality unless the error oscillations at alternation points cancel exactly.
- **Equal–ripple (converse form).** If p is a minimizer, there exist $n+1$ points where the error equals $\pm E$ with *alternating sign.* Any such ordered set is called an *alternant* for p.
- **Zeros from alternating signs.** If two continuous functions coincide in value at $n+1$ points with alternating signs, then their difference has at least n sign changes; hence at least n distinct zeros in $(0,1)$ (by the intermediate value theorem).

Solution

Assume, for contradiction, there are two distinct minimizers $p_1 \neq p_2$ with common error $E = \|f - p_1\|_\infty = \|f - p_2\|_\infty$. By the converse equal–ripple criterion, there is an alternant $0 \leq a_0 < \cdots < a_n \leq 1$ with

$$f(a_k) - p_1(a_k) = \sigma\,(-1)^k E \qquad (k = 0, \dots, n),$$

for some $\sigma \in \{\pm 1\}$. Consider the midpoint $r = \frac{1}{2}(p_1 + p_2)$. If for some k the signs of $f(a_k) - p_2(a_k)$ agree with those of $f(a_k) - p_1(a_k)$, then $|f(a_k) - r(a_k)| < E$. Since the signs at the $n+1$ points a_k alternate, this would force $\|f - r\|_\infty < E$ (by the standard alternation argument), contradicting minimality.

Hence for every k we must have

$$\operatorname{sgn}\Big(f(a_k) - p_2(a_k)\Big) = -\operatorname{sgn}\Big(f(a_k) - p_1(a_k)\Big), \qquad |f(a_k) - p_2(a_k)| = E.$$

Subtracting yields $(p_2 - p_1)(a_k) = 0$ for $k = 0, \dots, n$. The polynomial $p_2 - p_1$ has degree $< n$ and at least $n+1$ distinct zeros, hence is identically zero. Therefore $p_1 = p_2$, a contradiction. Thus the minimax polynomial of degree $< n$ is unique. $\square$

Remarks / examples

For $f(x) = x^2$ on $[-1, 1]$ and $n = 1$, the unique minimizer is the constant $p(x) = 1/3$, with alternation at $-1, 0, 1$. For $f(x) = |x|$ and $n = 2$, the unique minimax line $p(x) = ax$ equioscillates at four points $-1, -\xi, \xi, 1$, giving a unique slope a.

Equal-ripple (equioscillation) and the minimax polynomial

Setting. Let $f \in C([a, b])$ and let Π_n be the set of polynomials of degree $\leq n$. The *best uniform (minimax) error* is

$$E_n \;=\; \min_{p \in \Pi_n} \|f - p\|_\infty \qquad \text{with} \qquad \|g\|_\infty := \max_{x \in [a,b]} |g(x)|.$$

Any $p_n^* \in \Pi_n$ attaining this minimum is called a minimax (or best uniform) polynomial of degree $\leq n$ for f.

Chebyshev alternation theorem (equal-ripple). A polynomial $p_n^* \in \Pi_n$ is a minimax approximant if and only if there exist $n + 2$ points

$$a \leq x_0 < x_1 < \cdots < x_{n+1} \leq b$$

such that

$$f(x_k) - p_n^*(x_k) = (-1)^k E_n \quad \text{for } k = 0, 1, \dots, n+1, \qquad \text{and} \qquad |f(x) - p_n^*(x)| \le E_n \text{ for all } x \in [a, b].$$

In words: the error $f - p_n^*$ *equioscillates* between $\pm E_n$ at $n + 2$ alternating points and nowhere exceeds E_n in magnitude. Under this condition the minimax approximant is unique (unless $f \in \Pi_n$, where the exact fit can be non-unique in a trivial sense).

Existence and "is the minimizer a polynomial?" Yes. Parametrize $p(x) = \sum_{k=0}^{n} c_k x^k$. The map $c \mapsto \|f - p\|_\infty$ is continuous, and a minimizing sequence can be confined to a compact set in coefficient space; hence a minimizer exists. By construction it lies in Π_n (possibly of degree $< n$ if the optimal top coefficients vanish). Uniqueness holds under the alternation condition above.

Weighted version. For a continuous positive weight $w : [a, b] \to (0, \infty)$, the weighted minimax problem

$$E_{n,w} \;=\; \min_{p \in \Pi_n} \; \| \, w(\cdot) \, (f - p) \, \|_\infty$$

has the same characterization with $w(x_k)\Big(f(x_k) - p_n^*(x_k)\Big) = (-1)^k E_{n,w}$ at $n+2$ alternating points, and $|w(x)(f(x) - p_n^*(x))| \le E_{n,w}$ on $[a, b]$.

Classic example: monic minimax on $[-1, 1]$. Among *monic* polynomials of degree n,

$$\min_{\substack{p \in \Pi_n \\ \text{lead coeff}=1}} \; \max_{|x| \le 1} |p(x)|$$

the unique solution is the scaled Chebyshev polynomial

$$p_n^*(x) = 2^{1-n} \, T_n(x),$$

because T_n has leading coefficient 2^{n-1}. On $[-1, 1]$ we have $|T_n(x)| \le 1$ and T_n attains ± 1 at $n + 1$ alternating extrema, so

$$\max_{|x| \le 1} |p_n^*(x)| \;=\; 2^{1-n},$$

and the error exhibits an equal-ripple pattern. (Recall $T_n(\cos\theta) = \cos(n\theta)$, and T_n equioscillates between ± 1 on $[-1, 1]$.)

8. Pointwise polynomial approximation on $\mathbb{R}$

Claim. For any continuous $f : \mathbb{R} \to \mathbb{R}$ there exist polynomials p_n such that $p_n(x) \to f(x)$ for every $x \in \mathbb{R}$.

Proof. For each $n \in \mathbb{N}$, by Weierstrass on $[-n, n]$ there is a polynomial q_n with

$$\sup_{|x| \le n} |q_n(x) - f(x)| \le \frac{1}{n}.$$

Set $p_n := q_n$. Fix $x \in \mathbb{R}$ and choose $N > |x|$. For all $n \geq N$, we have $x \in [-n, n]$, hence $|p_n(x) - f(x)| \leq 1/n \to 0$. Therefore $p_n(x) \to f(x)$ for every $x \in \mathbb{R}$. $\square$

Problem 9. Bernstein operator on x^3

Problem. Let $B_n : C[0,1] \to C[0,1]$ be the Bernstein operator

$$(B_n f)(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Show directly that $B_n f \to f$ uniformly on $[0,1]$ for the function $f(x) = x^3$.

Solution. For $f(x) = x^3$ write $K \sim \mathrm{Bin}(n, x)$ so that

$$(B_n f)(x) = \mathbb{E}\left[\left(\frac{K}{n}\right)^3\right] = \frac{\mathbb{E}[K^3]}{n^3}.$$

Use the factorial-moment identities

$$\mathbb{E}[K] = nx, \qquad \mathbb{E}[K(K-1)] = n(n-1)x^2, \qquad \mathbb{E}[K(K-1)(K-2)] = n(n-1)(n-2)x^3,$$

and the decomposition $K^3 = K(K-1)(K-2) + 3K(K-1) + K$. Hence

$$\mathbb{E}[K^3] = n(n-1)(n-2)x^3 + 3n(n-1)x^2 + nx,$$

so

$$(B_n f)(x) = \frac{(n-1)(n-2)}{n^2}\, x^3 + \frac{3(n-1)}{n^2}\, x^2 + \frac{1}{n^2}\, x.$$

Therefore

$$(B_n f)(x) - x^3 = -\Big(\frac{3}{n} - \frac{2}{n^2}\Big)x^3 + \frac{3(n-1)}{n^2}x^2 + \frac{1}{n^2}x.$$

For $x \in [0,1]$ this yields the uniform bound

$$\|B_n f - f\|_\infty \leq \frac{3}{n} - \frac{2}{n^2} + \frac{3(n-1)}{n^2} + \frac{1}{n^2} = \frac{3}{n} + \frac{2}{n^2} \xrightarrow[n\to\infty]{} 0.$$

Thus $B_n f \to f$ uniformly on $[0,1]$ for $f(x) = x^3$. $\square$

10. First five Chebyshev polynomials

Define $T_0(x) = 1$, $T_1(x) = x$, and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$. Then

$$\begin{aligned}
T_0(x) &= 1, \\
T_1(x) &= x, \\
T_2(x) &= 2x^2 - 1, \\
T_3(x) &= 4x^3 - 3x, \\
T_4(x) &= 8x^4 - 8x^2 + 1.
\end{aligned}$$

Problem 11. Legendre polynomials via Gram–Schmidt and Rodrigues-type formula

Problem. (i) Use orthogonality (the Gram–Schmidt method) to compute the Legendre polynomials p_n for $n = 0, 1, 2, 3$.

(ii) Explain why $\frac{d^m}{dx^m}(1-x)^n(1+x)^n$ vanishes at $x = 1$ or $x = -1$ whenever $m < n$. Suppose that $P_n(x) = \frac{d^n}{dx^n}(1-x^2)^n$. Use integration by parts to show that

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = 0 \qquad (m \neq n).$$

Conclude that the P_n are scalar multiples of the Legendre polynomials p_n.

(iii) Compute P_n for $n = 0, 1, 2, 3$ and check the last sentence of (ii).

Solution.

(i) Gram–Schmidt on $\{1, x, x^2, x^3\}$ in $L^2([-1,1])$. Use $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$.

Step 0. $q_0 = 1$. Take $p_0 = 1$ (normalization by a positive constant is irrelevant).

Step 1. $q_1 = x - \frac{\langle x, 1\rangle}{\langle 1, 1\rangle} 1 = x$ (since $\int_{-1}^{1} x\,dx = 0$). Take $p_1 = x$.

Step 2.

$$q_2 = x^2 - \frac{\langle x^2, 1\rangle}{\langle 1, 1\rangle} 1 - \frac{\langle x^2, x\rangle}{\langle x, x\rangle} x = x^2 - \frac{(2/3)}{2} \cdot 1 - 0 \cdot x = x^2 - \frac{1}{3}.$$

Scale to the standard Legendre form: $p_2 = \frac{3}{2}\, q_2 = \frac{1}{2}(3x^2 - 1)$.

Step 3.

$$q_3 = x^3 - \frac{\langle x^3, 1\rangle}{\langle 1, 1\rangle} 1 - \frac{\langle x^3, x\rangle}{\langle x, x\rangle} x - \frac{\langle x^3, x^2\rangle}{\langle x^2, x^2\rangle} x^2 = x^3 - \frac{(2/5)}{(2/3)} x = x^3 - \frac{3}{5} x,$$

since $\int x^3 = 0$, $\int x^5 = 0$. Scaling: $p_3 = \frac{5}{2}\, q_3 = \frac{1}{2}(5x^3 - 3x)$.

Thus,

$$p_0 = 1, \qquad p_1 = x, \qquad p_2 = \tfrac{1}{2}(3x^2 - 1), \qquad p_3 = \tfrac{1}{2}(5x^3 - 3x).$$

(ii) Endpoint vanishing and orthogonality of $P_n(x) = \frac{d^n}{dx^n}(1-x^2)^n$. Note $(1-x)^n$ has a zero of order n at $x = 1$, hence so does $(1-x)^n(1+x)^n$; similarly at $x = -1$. Therefore $\frac{d^m}{dx^m}(1-x)^n(1+x)^n = 0$ at $x = \pm 1$ for all $m < n$.

Let $P_n(x) = \dfrac{d^n}{dx^n}(1-x^2)^n$. For $m < n$, integrate by parts n times:

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = \int_{-1}^{1} \frac{d^n}{dx^n}(1-x^2)^n\,P_m(x)\,dx = \Big[\sum_{\ell=0}^{n-1}(\cdots)\Big] + (-1)^n \int_{-1}^{1} (1-x^2)^n\,\frac{d^n}{dx^n}P_m(x)\,dx.$$

All boundary terms vanish because each boundary term involves a derivative of $(1-x^2)^n$ of order $< n$ evaluated at $x = \pm 1$, which is 0 by the first part. But $\deg P_m = m < n$, hence $\dfrac{d^n}{dx^n}P_m \equiv 0$, so the remaining integral is 0. By symmetry the same holds for $m > n$. Therefore P_n and P_m are orthogonal for $m \neq n$.

Since (i) shows that the Legendre polynomials $\{p_n\}$ are the orthogonal degree-by-degree basis for this inner product, the orthogonal family $\{P_n\}$ must satisfy $P_n = c_n\,p_n$ with nonzero constants c_n.

(iii) Explicit P_n and comparison. Compute by differentiation:

$$\begin{aligned}
P_0(x) &= (1-x^2)^0 = 1,\\
P_1(x) &= \frac{d}{dx}(1-x^2) = -2x,\\
P_2(x) &= \frac{d^2}{dx^2}(1-2x^2+x^4) = \frac{d}{dx}(-4x+4x^3) = -4+12x^2 = 4(3x^2-1),\\
P_3(x) &= \frac{d^3}{dx^3}(1-3x^2+3x^4-x^6) = \frac{d}{dx}\Big(-6+36x^2-30x^4\Big) = 72x-120x^3\\
&= -24(5x^3-3x).
\end{aligned}$$

Comparing with (i),

$$P_0 = 1 = p_0, \quad P_1 = -2\,p_1, \quad P_2 = 8\,p_2, \quad P_3 = -48\,p_3,$$

so indeed $P_n = c_n\,p_n$ with $c_n = (-1)^n\,2^n n!$, consistent with the Rodrigues formula

$$p_n(x) = \frac{1}{2^n n!}\,\frac{d^n}{dx^n}(x^2-1)^n = \frac{(-1)^n}{2^n n!}\,\frac{d^n}{dx^n}(1-x^2)^n.$$

□

Building orthogonal polynomials: Gram–Schmidt and Rodrigues (exam–ready)

Setting

Work in the polynomial space

$$\mathcal{P}_n = \{\text{polynomials of degree } \leq n\}$$

with the L^2 inner product on $[-1,1]$ (Legendre weight $w \equiv 1$):

$$\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx.$$

Goal

Orthonormalize the monomial basis $1, x, x^2, \ldots$ to get an orthogonal family $\{p_0, p_1, \ldots\}$ on $[-1, 1]$. Up to scaling, these are the **Legendre polynomials**.

Gram–Schmidt recipe on $\{1, x, x^2, \ldots\}$

For $k = 0, 1, 2, \ldots$ set

$$q_k(x) = x^k - \sum_{j=0}^{k-1} \frac{\langle x^k, p_j \rangle}{\langle p_j, p_j \rangle} p_j(x), \qquad p_k = \frac{q_k}{\|q_k\|}.$$

Because the weight is even and the interval is symmetric, you automatically get parity: p_k is even/odd for k even/odd.

First few (with the standard normalization $P_n(1) = 1$):

$$P_0 = 1, \qquad P_1 = x, \qquad P_2 = \tfrac{1}{2}(3x^2 - 1), \qquad P_3 = \tfrac{1}{2}(5x^3 - 3x).$$

Orthogonality and three–term recurrence:

$$\int_{-1}^{1} P_\ell(x) P_m(x)\, dx = 0 \quad (\ell \neq m), \qquad (n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Rodrigues formula (closed form for classical families)

For a classical weight $w(x)$ on $[a, b]$ satisfying a Pearson ODE, the orthogonal polynomials admit

$$p_n(x) = \frac{1}{\kappa_n\, w(x)} \frac{d^n}{dx^n}\Big(w(x)\, \sigma(x)^n\Big),$$

where σ, τ (from the Pearson equation) are polynomials and κ_n fixes the normalization.

Legendre (weight $w \equiv 1$ on $[-1, 1]$)

$$\boxed{P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left(x^2 - 1\right)^n} \quad = \quad \frac{(-1)^n}{2^n n!} \frac{d^n}{dx^n} \left(1 - x^2\right)^n.$$

Immediate consequences.

- Parity: $P_n(-x) = (-1)^n P_n(x)$.
- Normalization: $P_n(1) = 1$.

- Leading coefficient:

$$\mathrm{lc}(P_n) = \frac{(2n)!}{2^n (n!)^2}.$$

- Orthogonality (with the standard L^2 inner product):

$$\int_{-1}^{1} P_n(x) P_m(x)\, dx = \frac{2}{2n+1}\, \delta_{nm}.$$

Other classical families (Rodrigues forms; for reference)

- **Hermite** (weight e^{-x^2} on $\mathbb{R}$):

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\Big(e^{-x^2}\Big).$$

- **Laguerre** (weight e^{-x} on $[0, \infty)$):

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n}\Big(e^{-x} x^n\Big).$$

- **Jacobi** (weight $(1-x)^\alpha (1+x)^\beta$ on $[-1, 1]$, $\alpha, \beta > -1$):

$$P_n^{(\alpha,\beta)}(x) = \frac{(-1)^n}{2^n n!} \frac{1}{(1-x)^\alpha (1+x)^\beta} \frac{d^n}{dx^n}\Big[(1-x)^{\alpha+n} (1+x)^{\beta+n}\Big].$$

(Optional) Useful generating/ODE facts for Legendre

- Differential equation:

$$\frac{d}{dx}\Big[(1-x^2) P_n'(x)\Big] + n(n+1) P_n(x) = 0.$$

- Generating function:

$$\sum_{n=0}^{\infty} P_n(x)\, t^n = \frac{1}{\sqrt{1 - 2xt + t^2}}, \qquad |t| < 1.$$

Problem 12 — Moment conditions force $f \equiv 0$

Statement

(1) If $f \in C[0,1]$ and $\int_0^1 f(x) x^n\, dx = 0$ for all $n = 0, 1, 2, \ldots$, prove $f \equiv 0$.
(2) If only $\int_0^1 f(x) x^n\, dx = 0$ for all $n = 1, 2, \ldots$, does $f \equiv 0$ still follow?

Theory

- Polynomials are dense in $C[0,1]$ (Weierstrass).
- $\{p \in \mathcal{P} : \; p(0) = 0\}$ is dense in $\{g \in C[0,1] : \; g(0) = 0\}$.
- If $f \in C[0,1]$ and $\int fg = 0$ for all continuous g supported in a small neighborhood of $a \in (0,1]$, then $f(a) = 0$.

Solution

(1) For any polynomial p, $\int_0^1 fp = 0$. By density and continuity of $g \mapsto \int fg$, $\int_0^1 fg = 0$ for all $g \in C[0,1]$. Taking $g = f$ gives $\int_0^1 f^2 = 0$, hence $f \equiv 0$.

(2) For any polynomial p with $p(0) = 0$, $\int_0^1 fp = 0$. By density this holds for all $g \in C[0,1]$ with $g(0) = 0$. Fix $a \in (0,1]$ and choose $g \geq 0$, $g(0) = 0$, supported near a and not identically 0. If $f(a) \neq 0$, then $\int_0^1 fg \neq 0$, contradiction. Thus $f(a) = 0$ for all $a \in (0,1]$, and by continuity at 0, $f(0) = 0$. Therefore $f \equiv 0$. $\square$

Problem 13 — Orthogonality of Chebyshev polynomials

Statement

Find a positive weight w on $[-1,1]$ such that

$$\int_{-1}^{1} T_m(x) T_n(x) \, w(x) \, dx = 0 \qquad (m \neq n),$$

and prove the orthogonality.

Solution

Use $x = \cos\theta$ with $\theta \in [0,\pi]$. Then $T_k(x) = \cos(k\theta)$, $dx = -\sin\theta \, d\theta$ and $\sqrt{1-x^2} = \sin\theta$, hence

$$\int_{-1}^{1} T_m(x) T_n(x) \, \frac{dx}{\sqrt{1-x^2}} = \int_0^{\pi} \cos(m\theta)\cos(n\theta) \, d\theta = 0 \qquad (m \neq n).$$

Therefore the weight is $w(x) = (1-x^2)^{-1/2}$ and the Chebyshev polynomials are orthogonal with respect to w. $\square$

Normalization (optional)

$$\int_{-1}^{1} \frac{T_n(x)^2}{\sqrt{1-x^2}}\,dx = \begin{cases} \pi, & n = 0, \\ \frac{\pi}{2}, & n \ge 1. \end{cases}$$

Problem — From polynomial accuracy to convergence for all continuous f

For each $n \in \mathbb{N}$ let weights $A_k^{(n)} \ge 0$ and nodes $x_k^{(n)} \in [a,b]$, $k = 1, \ldots, n$. Assume that for every polynomial P,

$$\varepsilon_n(P) := \left| \int_a^b P(x)\,dx - \sum_{k=1}^{n} A_k^{(n)}\, P\Big(x_k^{(n)}\Big) \right| \xrightarrow[n\to\infty]{} 0.$$

Prove that for every $f \in C([a,b])$,

$$\varepsilon_n(f) := \left| \int_a^b f(x)\,dx - \sum_{k=1}^{n} A_k^{(n)}\, f\Big(x_k^{(n)}\Big) \right| \to 0.$$

Theory you'll use

- **Weierstrass:** Polynomials are dense in $C([a,b])$ for $\|\cdot\|_\infty$.
- **Total mass from the constant polynomial:**

$$\sum_{k=1}^{n} A_k^{(n)} = \int_a^b 1\,dx - \varepsilon_n(1) \;\longrightarrow\; b-a.$$

- **Error splitting (for any polynomial P):**

$$\varepsilon_n(f) \le \int_a^b |f-P| + \varepsilon_n(P) + \Big(\sum_{k=1}^{n} A_k^{(n)}\Big)\|f-P\|_\infty.$$

Solution

Fix $\varepsilon > 0$. By Weierstrass, choose a polynomial P with $\|f-P\|_\infty < \delta$. Then

$$\varepsilon_n(f) \le (b-a)\delta + \varepsilon_n(P) + \Big(\sum_{k=1}^{n} A_k^{(n)}\Big)\delta.$$

By hypothesis $\varepsilon_n(P) \to 0$, so pick N_1 with $\varepsilon_n(P) < \varepsilon/3$ for $n \geq N_1$.

Since $\sum_k A_k^{(n)} \to b-a$, pick N_2 such that $\sum_k A_k^{(n)} \leq b-a+1$ for $n \geq N_2$.

For $n \geq N := \max\{N_1, N_2\}$ one has

$$\varepsilon_n(f) \leq \Big(2(b-a)+1\Big)\delta + \frac{\varepsilon}{3}.$$

Choose $\delta = \varepsilon \Big/ (3(2(b-a)+1))$.

Then $\varepsilon_n(f) \leq \varepsilon$ for $n \geq N$.

Hence $\varepsilon_n(f) \to 0$ for every $f \in C([a,b])$. $\square$

Examples (what the theorem covers)

- **Midpoint/trapezoidal/Simpson families.** For each fixed polynomial P, their errors $\varepsilon_n(P) \to 0$ as the mesh is refined; hence the rules converge for every continuous f.

- **Positive Newton–Cotes with refining nodes.** If the weights stay nonnegative and the rule integrates polynomials increasingly well (in the sense above), then it converges for all continuous f.

Problem 2 — Maximal degree of exactness for an n–point rule

It is known that there exist n distinct points $\alpha_1, \ldots, \alpha_n \in [-1,1]$ and real numbers $A_1, \ldots, A_n$ such that

$$\int_{-1}^{1} p(x)\,dx = \sum_{j=1}^{n} A_j\, p(\alpha_j) \quad \text{for all } p \text{ with } \deg p \leq 2n-1$$

(Gauss–Legendre; the α_j are the zeros of P_n). Is it possible to choose n nodes and weights so that the same holds for every p with $\deg p \leq 2n$?

Theory you'll use

- **Gaussian quadrature optimality.** An n-node quadrature rule cannot be exact for all polynomials beyond degree $2n-1$ (and Gauss rules achieve this maximal degree of exactness).

- **Simple obstruction for degree $2n$.** If the rule uses nodes $\alpha_1, \ldots, \alpha_n$, consider the polynomial
$$q(x) = \prod_{j=1}^{n} (x - \alpha_j)^2.$$
Then $q \geq 0$, $q \not\equiv 0$, $\deg q = 2n$, and $q(\alpha_j) = 0$ for all j.

- **Positivity of the integral.** Since $q \geq 0$ and not identically zero,
$$\int_{-1}^{1} q(x)\, dx > 0.$$

Solution

No. Suppose the rule is exact up to degree $2n$. Let the n distinct nodes be $\alpha_1, \ldots, \alpha_n$, and define
$$q(x) = \prod_{j=1}^{n} (x - \alpha_j)^2.$$
Then $\deg q = 2n$, $q(x) \geq 0$, $q \not\equiv 0$, and $q(\alpha_j) = 0$ for each j. Exactness for degree $2n$ would give
$$\int_{-1}^{1} q(x)\, dx \;=\; \sum_{j=1}^{n} A_j\, q(\alpha_j) \;=\; 0,$$
whereas $\int_{-1}^{1} q(x)\, dx > 0$ since $q \geq 0$ and is not identically zero. This contradiction shows that no n–point rule can be exact for all polynomials of degree $\leq 2n$. Therefore Gauss–Legendre (exact to degree $2n - 1$) is optimal. □

Examples

For $n = 1$, the Gauss rule $\alpha_1 = 0$, $A_1 = 2$ is exact for degree ≤ 1 but not for x^2. For $n = 2$, the Gauss rule $\alpha_{1,2} = \pm 1/\sqrt{3}$, $A_1 = A_2 = 1$ is exact for degree ≤ 3; it fails for $q(x) = (x^2 - \frac{1}{3})^2$ (sum $= 0$ but $\int_{-1}^{1} q > 0$).

Legendre polynomials P_n: definition and core properties

Definitions

- **Rodrigues formula:**
$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n, \qquad n = 0, 1, 2, \ldots$$

- **Generating function:**

$$\sum_{n=0}^{\infty} P_n(x)\, t^n = \frac{1}{\sqrt{1-2xt+t^2}}, \qquad |t| < 1.$$

- **Three–term recurrence:**

$$(n+1)P_{n+1}(x) = (2n+1)\, x\, P_n(x) - n\, P_{n-1}(x), \quad n \geq 1,$$

with $P_0(x) = 1$, $P_1(x) = x$.

First few polynomials

$$\begin{aligned}
P_0(x) &= 1,\\
P_1(x) &= x,\\
P_2(x) &= \tfrac{1}{2}(3x^2 - 1),\\
P_3(x) &= \tfrac{1}{2}(5x^3 - 3x),\\
P_4(x) &= \tfrac{1}{8}(35x^4 - 30x^2 + 3),\\
P_5(x) &= \tfrac{1}{8}(63x^5 - 70x^3 + 15x).
\end{aligned}$$

Orthogonality and normalization

$$\int_{-1}^{1} P_m(x)\, P_n(x)\, dx = \frac{2}{2n+1}\, \delta_{mn}, \qquad P_n(1) = 1, \qquad P_n(-x) = (-1)^n P_n(x).$$

Differential equation (Sturm–Liouville)

$$\frac{d}{dx}\Big[(1-x^2)P_n'(x)\Big] + n(n+1)\, P_n(x) = 0.$$

Zeros and interlacing

All zeros of P_n lie in $(-1,1)$, are simple, and interlace with the zeros of $P_{n\pm1}$.

Gauss–Legendre quadrature

Let $x_1, \ldots, x_n$ be the (distinct) zeros of P_n. Then for polynomials of degree $\leq 2n-1$,

$$\int_{-1}^{1} p(x)\, dx = \sum_{k=1}^{n} w_k\, p(x_k), \qquad w_k = \frac{2}{\left(1 - x_k^2\right)\left[P_n'(x_k)\right]^2} \ > \ 0.$$

Useful integrals / norms

$$\int_{-1}^{1} P_n(x)^2\,dx = \frac{2}{2n+1}, \qquad \int_{-1}^{1} x\,P_n(x)\,P_m(x)\,dx = \frac{2(n+1)}{(2n+1)(2n+3)}\delta_{m,n+1}+\frac{2n}{(2n+1)(2n-1)}\delta_{m,n-1}.$$

Remarks

Legendre polynomials are the $w \equiv 1$ case of the Jacobi family $P_n^{(\alpha,\beta)}$ with $\alpha = \beta = 0$. They appear in spherical harmonics Y_ℓ^m via the associated Legendre functions.

Legendre Polynomials

Definition and Construction

The *Legendre polynomials* $\{P_n(x)\}_{n\geq 0}$ form an orthogonal sequence on $[-1,1]$ with respect to the weight $w(x) \equiv 1$. They can be obtained by applying the Gram–Schmidt process to $\{1, x, x^2, \dots\}$ in $L^2([-1,1])$, or equivalently via the *Rodrigues formula:*

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n.$$

The normalization is $P_n(1) = 1$.

Orthogonality and Normalization

For all $m, n \geq 0$,

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \frac{2}{2n+1}\,\delta_{mn}.$$

Thus, P_n are orthogonal under the standard L^2 inner product on $[-1,1]$.

Recurrence Relation

They satisfy the three-term recurrence

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x),$$

with $P_0(x) = 1$ and $P_1(x) = x$.

Differential Equation (Sturm–Liouville form)

Each P_n satisfies

$$\frac{d}{dx}\Big[(1-x^2)P_n'(x)\Big] + n(n+1)P_n(x) = 0.$$

This shows they are eigenfunctions of a self-adjoint differential operator.

Generating Function

Legendre polynomials are generated by

$$\sum_{n=0}^{\infty} P_n(x)t^n = \frac{1}{\sqrt{1-2xt+t^2}}, \qquad |t|<1.$$

Parity and Examples

They satisfy $P_n(-x) = (-1)^n P_n(x)$. The first few are

$$\begin{aligned}
P_0(x) &= 1,\\
P_1(x) &= x,\\
P_2(x) &= \tfrac{1}{2}(3x^2-1),\\
P_3(x) &= \tfrac{1}{2}(5x^3-3x),\\
P_4(x) &= \tfrac{1}{8}(35x^4-30x^2+3).
\end{aligned}$$

Zeros and Quadrature

All n zeros of P_n lie in $(-1,1)$ and are simple. If x_k are these zeros, then *Gauss–Legendre quadrature* gives exact integration for polynomials up to degree $2n-1$:

$$\int_{-1}^{1} p(x)\,dx = \sum_{k=1}^{n} w_k\, p(x_k), \qquad w_k = \frac{2}{(1-x_k^2)\,[P_n'(x_k)]^2}.$$

Orthogonality Examples

$$\int_{-1}^{1} P_0(x)P_1(x)\,dx = 0, \quad \int_{-1}^{1} P_2(x)^2\,dx = \frac{2}{5}.$$

Remarks

Legendre polynomials are a special case of the *Jacobi polynomials* $P_n^{(\alpha,\beta)}(x)$ with parameters $\alpha=\beta=0$. They play a central role in potential theory and in the expression of spherical harmonics $Y_\ell^m(\theta,\phi)$ via associated Legendre functions P_ℓ^m.

Legendre via Gram–Schmidt and the Christoffel–Darboux identity

Inner product and plan

Work in $L^2([-1,1])$ with weight $w \equiv 1$:

$$\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx.$$

Apply Gram–Schmidt to the monomials $\{1, x, x^2, \dots\}$ to obtain an orthogonal family $\{p_0, p_1, \dots\}$; after the usual normalization $P_n(1) = 1$ we get the *Legendre polynomials* P_n.

Gram–Schmidt construction

Set $q_0(x) = 1$, $p_0 = q_0/\|q_0\|$, and for $k \geq 1$ define

$$q_k(x) = x^k - \sum_{j=0}^{k-1} \frac{\langle x^k, p_j\rangle}{\langle p_j, p_j\rangle}\, p_j(x), \qquad p_k = \frac{q_k}{\|q_k\|}.$$

Because the interval and weight are even, p_k has definite parity: p_k is even for k even and odd for k odd.

First few polynomials (standard normalization $P_n(1) = 1$)

$$P_0 = 1, \qquad P_1 = x, \qquad P_2 = \tfrac{1}{2}(3x^2-1), \qquad P_3 = \tfrac{1}{2}(5x^3-3x), \qquad P_4 = \tfrac{1}{8}(35x^4-30x^2+3).$$

Orthogonality and norms

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \frac{2}{2n+1}\,\delta_{mn}.$$

Thus $\|P_n\|_{L^2}^2 = \dfrac{2}{2n+1}$.

Three–term recurrence (from Gram–Schmidt / Favard)

The multiplication operator $x : \mathrm{span}\{P_0, \dots, P_n\} \to \mathrm{span}\{P_0, \dots, P_{n+1}\}$ is symmetric, so xP_n has no $P_{n-2}, P_{n-3}, \dots$ components. Projecting xP_n onto $\{P_{n+1}, P_n, P_{n-1}\}$ and using orthogonality yields

$$(n+1)P_{n+1}(x) = (2n+1)x\,P_n(x) - n\,P_{n-1}(x), \qquad n \geq 1,$$

with $P_0 = 1$, $P_1 = x$.

Rodrigues formula and ODE (for reference)

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n, \qquad \frac{d}{dx}\Big[(1-x^2)P_n'(x)\Big] + n(n+1)P_n(x) = 0.$$

Christoffel–Darboux identity (Legendre)

For $x \neq y$,

$$\boxed{\sum_{k=0}^{n}(2k+1)\,P_k(x)\,P_k(y) = \frac{n+1}{x-y}\Big(P_{n+1}(x)P_n(y) - P_n(x)P_{n+1}(y)\Big).}$$

Taking the limit $y \to x$ gives the diagonal form

$$\boxed{\sum_{k=0}^{n}(2k+1)\,P_k(x)^2 = (n+1)\Big(P_{n+1}'(x)\,P_n(x) - P_n'(x)\,P_{n+1}(x)\Big).}$$

Proof sketch (via the three–term recurrence)

Let $S_n(x,y) = \sum_{k=0}^{n}(2k+1)P_k(x)P_k(y)$. Multiply the recurrence

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

by $P_n(y)$, and the same with $x \leftrightarrow y$; subtract. After cancellation in the telescoping sum, divide by $(x-y)$ to obtain the displayed formula.

Two useful corollaries

- **Reproducing kernel for degree $\leq n$:**

  $$K_n(x,y) = \sum_{k=0}^{n}\frac{2k+1}{2}\,P_k(x)P_k(y) = \frac{n+1}{2}\,\frac{P_{n+1}(x)P_n(y) - P_n(x)P_{n+1}(y)}{x-y}.$$

  Then for all polynomials p with $\deg p \leq n$, $p(x) = \int_{-1}^{1} K_n(x,y)\,p(y)\,dy$.

- **Partial–sum bound:**

$$\left|\sum_{k=0}^{n}(2k+1)P_k(x)P_k(y)\right| \le \frac{2(n+1)}{|x-y|} \qquad (x \neq y,\ x, y \in [-1,1]),$$

  since $|P_m| \le 1$ on $[-1,1]$.

Worked Gram–Schmidt: constructing P_2

Use $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$.

Step 1: P_0. $q_0 = 1$, $\|q_0\|^2 = \int_{-1}^{1} 1\,dx = 2$. Take $p_0 = q_0/\|q_0\|$ and then rescale to $P_0(1) = 1$:

$$P_0(x) = 1.$$

Step 2: P_1. Start with x and orthogonalize against P_0:

$$\langle x, P_0\rangle = \int_{-1}^{1} x\,dx = 0 \quad \Rightarrow \quad q_1 = x.$$

Rescale so that $P_1(1) = 1$: $P_1(x) = x$.

Step 3: P_2. Start with x^2 and subtract the P_0, P_1 components:

$$\langle x^2, P_1\rangle = \int_{-1}^{1} x^3\,dx = 0, \qquad \langle x^2, P_0\rangle = \int_{-1}^{1} x^2\,dx = \frac{2}{3}, \qquad \langle P_0, P_0\rangle = 2.$$

Thus

$$q_2(x) = x^2 - \frac{\langle x^2, P_0\rangle}{\langle P_0, P_0\rangle} P_0(x) = x^2 - \frac{1}{3}.$$

Rescale to make $P_2(1) = 1$:

$$P_2(x) = \frac{q_2(x)}{q_2(1)} = \frac{x^2 - \frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}(3x^2 - 1).$$

Weights w_k via the Christoffel–Darboux identity

Let $x_1, \ldots, x_n$ be the distinct zeros of P_n. Define the cardinal polynomials

$$\ell_k(x) = \frac{P_n(x)}{(x - x_k)\,P_n'(x_k)} \qquad (k = 1, \ldots, n),$$

so that $\ell_k(x_j) = \delta_{kj}$ and $\deg \ell_k = n - 1$.

Key identity (C–D). For $x \neq y$,

$$\sum_{m=0}^{n-1} \frac{2m+1}{2} P_m(x) P_m(y) = \frac{n}{2} \frac{P_n(x)P_{n-1}(y) - P_{n-1}(x)P_n(y)}{x-y}.$$

Set $y = x_k$; since $P_n(x_k) = 0$,

$$K_{n-1}(x, x_k) := \sum_{m=0}^{n-1} \frac{2m+1}{2} P_m(x) P_m(x_k) = \frac{n}{2} \frac{P_n(x)\, P_{n-1}(x_k)}{x - x_k}.$$

Use the derivative relation $(1-x^2)P_n'(x) = n\Big(P_{n-1}(x) - xP_n(x)\Big)$; at $x = x_k$ this gives

$$P_{n-1}(x_k) = \frac{1-x_k^2}{n}\, P_n'(x_k).$$

Hence

$$K_{n-1}(x, x_k) = \frac{1}{2}\,(1-x_k^2)\,[P_n'(x_k)]^2\, \ell_k(x).$$

Integrate both sides. By orthogonality, $\displaystyle\int_{-1}^{1} P_m(x)\, dx = 0$ for $m \geq 1$, while $\int_{-1}^{1} P_0(x)\, dx = 2$. Therefore

$$\int_{-1}^{1} K_{n-1}(x, x_k)\, dx = \sum_{m=0}^{n-1} \frac{2m+1}{2} P_m(x_k) \int_{-1}^{1} P_m(x)\, dx = \frac{2 \cdot 1}{2}\, P_0(x_k) = 1.$$

Consequently,

$$1 = \frac{1}{2}\,(1-x_k^2)\,[P_n'(x_k)]^2 \int_{-1}^{1} \ell_k(x)\, dx.$$

Weights and cardinal integrals. For Gauss–Legendre quadrature (exact for degree $\leq 2n-1$), the weights satisfy

$$w_k = \int_{-1}^{1} \ell_k(x)\, dx,$$

since for any p with $\deg p \leq 2n-1$ one has the interpolation identity $p(x) = \sum_{j=1}^{n} p(x_j)\ell_j(x) + P_n(x)r(x)$ with $\deg r \leq n-1$, and $\int P_n r = 0$ by orthogonality.

Combining with the previous line gives the standard weight formula:

$$\boxed{w_k = \frac{2}{\Big(1-x_k^2\Big)\,[P_n'(x_k)]^2}, \qquad k = 1, \ldots, n.}$$

Numerical Quadrature: setup, exactness, error, and convergence

Problem (quadrature)

Given a function f on $[a, b]$, approximate the integral

$$I(f) := \int_a^b f(x)\,dx$$

by a *quadrature rule*

$$Q_n(f) := \sum_{k=1}^{n} A_k\, f(x_k),$$

where the *nodes* $x_k \in [a, b]$ and *weights* $A_k \in \mathbb{R}$ are chosen in advance.

Degree of exactness

The *degree of exactness* (or precision) of Q_n is the largest integer $m \geq 0$ such that

$$Q_n(p) = I(p) \qquad \text{for all polynomials } p \text{ with } \deg p \leq m.$$

Examples:

- **Midpoint rule** ($n = 1$): $Q_1(f) = (b - a)\, f\Big(\frac{a+b}{2}\Big)$, exact for deg ≤ 1.
- **Trapezoidal rule** ($n = 2$): $Q_2(f) = \frac{b-a}{2}\Big(f(a) + f(b)\Big)$, exact for deg ≤ 1.
- **Simpson's rule** ($n = 3$): $Q_3(f) = \frac{b-a}{6}\Big(f(a) + 4f(\frac{a+b}{2}) + f(b)\Big)$, exact for deg ≤ 3.
- **Gauss–Legendre** (optimal): n nodes $\{x_k\}$ are the zeros of P_n; with suitable $A_k > 0$ the rule is exact for deg $\leq 2n - 1$.

Error representations

Classical remainder (smooth f). If Q_n has degree m and $f \in C^{m+1}[a, b]$, then

$$I(f) - Q_n(f) \;=\; \frac{f^{(m+1)}(\xi)}{(m+1)!} \int_a^b \omega_{m+1}(x)\,dx, \qquad \omega_{m+1}(x) := \prod_{k=1}^{n} (x - x_k),$$

for some $\xi \in (a, b)$ (exact form depends on the rule; for Newton–Cotes the exponent reflects paneling).

Peano kernel (general linear rules). If Q is linear and exact on Π_{r-1} (all polynomials of degree $\le r-1$), then for $f \in C^r[a,b]$

$$I(f)-Q(f) = \int_a^b f^{(r)}(t)\, K_r(t)\, dt, \qquad K_r(t) := \frac{1}{(r-1)!}\int_a^b (x-t)_+^{r-1}\, dx - \sum_{k=1}^n A_k \frac{(x_k-t)_+^{r-1}}{(r-1)!}.$$

Hence

$$|I(f)-Q(f)| \;\le\; \|f^{(r)}\|_\infty \int_a^b |K_r(t)|\, dt.$$

Convergence: when does $Q_n(f) \to I(f)$?

- **Polynomial density + control of weights.** Suppose for each polynomial p we have $\varepsilon_n(p) := |I(p) - Q_n(p)| \to 0$ as $n \to \infty$, the weights $A_k^{(n)} \ge 0$, and

  $$\sum_{k=1}^n A_k^{(n)} \;=\; I(1) - \varepsilon_n(1) \;\longrightarrow\; b-a.$$

  Then for every $f \in C([a,b])$,

  $$|I(f) - Q_n(f)| \;\xrightarrow[n\to\infty]{}\; 0.$$

  Idea: approximate f uniformly by a polynomial p (Weierstrass), split the error, and use positivity to bound $\sum A_k^{(n)}$.

- **Composite Newton–Cotes (refining mesh).** On a uniform partition with mesh $h \to 0$, the composite midpoint/trapezoidal/Simpson rules converge for $f \in C([a,b])$ (with classical rates if f is smoother).

- **Gaussian sequences.** Increasing n in Gauss–Legendre quadrature gives rapidly convergent approximations for analytic f (often near-exponential decay of the error).

Gaussian quadrature specifics (Legendre case)

If $x_1, \dots, x_n$ are the zeros of P_n on $[-1,1]$, then there are *unique* positive weights

$$w_k = \frac{2}{\left(1-x_k^2\right)[P_n'(x_k)]^2}$$

such that

$$\int_{-1}^1 p(x)\, dx = \sum_{k=1}^n w_k\, p(x_k) \qquad \text{for all } \deg p \le 2n-1.$$

This degree $2n-1$ is *optimal* for n nodes.

Quick comparison table

- **Midpoint / Trapezoidal / Simpson:** simple, composite versions easy; exactness ≤ 1 (mid/trap), 3 (Simpson); global error $O(h^2)$ (trap), $O(h^4)$ (Simpson) for smooth f.
- **Newton–Cotes (high order):** many nodes, can yield negative weights and instability; not recommended for large n.
- **Gauss–Legendre:** best degree for a fixed n; positive weights; excellent for smooth/analytic f.
- **Clenshaw–Curtis:** Chebyshev cosine nodes; easy to implement, FFT-accelerated; performance close to Gauss for many f.

Example: quadrature for $f(x) = \sinh x$ on $[0, 1]$

Exact integral

$$I = \int_0^1 \sinh x \, dx = \cosh(1) - 1.$$

Single-panel rules

$$\begin{aligned}
&\text{Midpoint:} && Q_{\text{mid}} = (b-a)\, f\Big(\tfrac{a+b}{2}\Big) = 1 \cdot \sinh\Big(\tfrac{1}{2}\Big), \\
&\text{Trapezoidal:} && Q_{\text{trap}} = \tfrac{1}{2}\Big(f(0) + f(1)\Big) = \tfrac{1}{2}\sinh(1), \\
&\text{Simpson:} && Q_{\text{simp}} = \tfrac{1}{6}\Big(f(0) + 4f\Big(\tfrac{1}{2}\Big) + f(1)\Big).
\end{aligned}$$

Numerically,

$$\begin{aligned}
Q_{\text{mid}} &\approx 0.5210953, \\
Q_{\text{trap}} &\approx 0.5876010, \\
Q_{\text{simp}} &\approx 0.5430699, \qquad I = \cosh(1) - 1 \approx 0.5430806.
\end{aligned}$$

Absolute errors: $|Q_{\text{mid}} - I| \approx 1.36 \cdot 10^{-3}$, $|Q_{\text{trap}} - I| \approx 5.46 \cdot 10^{-3}$, $|Q_{\text{simp}} - I| \approx 1.09 \cdot 10^{-5}$.

Composite Simpson (even N panels)

For $N = 16, 32$ panels on $[0, 1]$:

$$\begin{aligned}
N = 16: &\quad |Q_{\text{simp},\, N} - I| \approx 4.60 \times 10^{-8}, \\
N = 32: &\quad |Q_{\text{simp},\, N} - I| \approx 2.88 \times 10^{-9}.
\end{aligned}$$

Gauss–Legendre (optimal n-point rules)

Mapping the n-point Gauss–Legendre rule from $[-1,1]$ to $[0,1]$ gives

$$Q_n = \sum_{k=1}^{n} w_k \sinh(x_k), \quad x_k = \tfrac{1}{2}(1+\xi_k), \quad w_k = \tfrac{1}{2}\omega_k,$$

where (ξ_k, ω_k) are the standard nodes/weights on $[-1,1]$. Numerically,

$$\begin{aligned} n = 4 &: \ |Q_4 - I| \approx 3.0 \times 10^{-10}, \\ n = 5, 6 &: \ \text{error at machine precision.} \end{aligned}$$

Conclusion

For analytic f like $\sinh x$, Gauss–Legendre achieves very high accuracy with few nodes (degree of exactness $2n-1$), while composite Simpson converges at fourth order as the mesh refines.

Problem 3 — Chebyshev–alternation for a sparse cosine series

Problem. Let T_j be the j-th Chebyshev polynomial of the first kind and let $\{\gamma_j\}_{j\ge 1}$ be nonnegative with $\sum_{j=1}^{\infty} \gamma_j < \infty$. Show that the series

$$f(x) = \sum_{j=1}^{\infty} \gamma_j \, T_{3j}(x), \qquad x \in [-1,1],$$

defines a continuous function on $[-1,1]$. Moreover, for each $n \ge 0$ there exist points

$$-1 \le x_0 < x_1 < \cdots < x_{3^{n+1}} \le 1$$

such that, if $P_n(x) = \sum_{j=1}^{n} \gamma_j T_{3j}(x)$, then

$$f(x_k) - P_n(x_k) = (-1)^k \sum_{j=n+1}^{\infty} \gamma_j, \qquad k = 0, 1, \ldots, 3^{n+1}.$$

Theory.

(1) **Cosine representation and bound.** For all $\theta \in \mathbb{R}$,

$$T_m(\cos\theta) = \cos(m\theta),$$

hence $|T_m(x)| \le 1$ for every $x \in [-1,1]$.

(2) **Composition identity.** For positive integers a, b,

$$T_{ab}(x) = T_a\Big(T_b(x)\Big).$$

(3) **Extrema grid.** The extrema of T_M occur at

$$x_k = \cos\Big(\frac{k\pi}{M}\Big), \qquad k = 0, 1, \dots, M,$$

with values $T_M(x_k) = (-1)^k$.

(4) **Positive Chebyshev combination bound.** If $G(y) = \sum_{j\ge 1} \alpha_j T_j(y)$ with $\alpha_j \ge 0$, then for all $y \in [-1, 1]$,

$$|G(y)| \le \sum_{j\ge 1} \alpha_j,$$

with equality iff each $T_j(y)$ takes an endpoint value ± 1 and all have the same sign.

Solution. *Continuity.* Since $|T_{3j}(x)| \le 1$ and $\sum_j \gamma_j < \infty$, the M-test yields uniform convergence of $\sum_{j=1}^{\infty} \gamma_j T_{3j}$ on $[-1, 1]$. Hence f is continuous.

Alternation for the remainder. Fix $n \ge 0$ and set

$$R_n(x) = f(x) - P_n(x) = \sum_{j=n+1}^{\infty} \gamma_j T_{3j}(x), \qquad G_n(y) = \sum_{j=n+1}^{\infty} \gamma_j T_j(y).$$

Using $T_{3j} = T_j \circ T_3$, we have $R_n(x) = G_n\Big(T_3(x)\Big)$.

By (4), $|G_n(y)| \le \sum_{j>n} \gamma_j$ for all y, with equality precisely when all terms simultaneously achieve their extremal values with a common sign.

The standard equioscillation argument for positive cosine combinations (applied on the extremal grid of T_{3^n}) produces points

$$-1 \le y_0^{(n)} < y_1^{(n)} < \dots < y_{3^n}^{(n)} \le 1$$

such that

$$G_n\Big(y_k^{(n)}\Big) = (-1)^k \sum_{j>n} \gamma_j, \qquad k = 0, 1, \dots, 3^n.$$

Now use $R_n(x) = G_n\Big(T_3(x)\Big)$. For each k, let $x_{k,1} < x_{k,2} < x_{k,3}$ be the three (monotone-branch) preimages of $y_k^{(n)}$ under $T_3 : [-1, 1] \to [-1, 1]$.

Merging these ordered triples over $k = 0, \dots, 3^n$ yields an increasing sequence of $(3 \cdot 3^n) + 1 = 3^{n+1} + 1$ points

$$-1 \le x_0 < x_1 < \dots < x_{3^{n+1}} \le 1,$$

and for each such point,

$$R_n(x_\ell) = G_n\Big(T_3(x_\ell)\Big) = G_n\Big(y_{k(\ell)}^{(n)}\Big) = (-1)^{k(\ell)} \sum_{j>n} \gamma_j.$$

A direct inspection of the three monotone branches shows that along the merged list the parity alternates consecutively, hence

$$R_n(x_\ell) = (-1)^\ell \sum_{j=n+1}^{\infty} \gamma_j, \qquad \ell = 0, 1, \dots, 3^{n+1}.$$

Therefore $f(x_\ell) - P_n(x_\ell) = (-1)^\ell \sum_{j=n+1}^{\infty} \gamma_j$, as required.

Examples. (1) If $\gamma_j = 0$ for $j > m$, then for $n \geq m$ the remainder vanishes and the conclusion is trivial; for $n < m$ the same construction applies with the finite tail.

(2) For $\gamma_j = r^j$ with $0 < r < 1$, one has $G_n(y) = \sum_{j>n} (r\, T_1(y))^j$ and the above construction yields the alternation points; in particular for $n = 0$ the four points $x_0 < x_1 < x_2 < x_3$ satisfy $f(x_k) = (-1)^k \frac{r}{1-r}$.

Examples (exact formulas). *(1) Finite tail.* If $\gamma_j = 0$ for $j > m$, then for $n \geq m$ the remainder $R_n(x) = f(x) - P_n(x)$ vanishes. For $n < m$,

$$R_n(x) = \sum_{j=n+1}^{m} \gamma_j\, T_{3j}(x),$$

and at the alternation points from the theorem,

$$R_n(x_k) = (-1)^k \sum_{j=n+1}^{m} \gamma_j.$$

(2) Geometric weights $\gamma_j = r^j$ with $0 < r < 1$. Let $y := T_3(x) \in [-1, 1]$. Using the Chebyshev generating function

$$\sum_{j=1}^{\infty} r^j T_j(y) = \frac{r\, y - r^2}{1 - 2ry + r^2}, \qquad \sum_{j=1}^{n} r^j T_j(y) = \frac{r\, y - r^{n+1} T_{n+1}(y) + r^{n+2} T_n(y) - r^2 y}{1 - 2ry + r^2},$$

we obtain the exact closed forms

$$f(x) = \frac{r\, T_3(x) - r^2}{1 - 2r\, T_3(x) + r^2}, \qquad P_n(x) = \frac{r\, T_3(x) - r^{n+1} T_{n+1}(T_3(x)) + r^{n+2} T_n(T_3(x)) - r^2 T_3(x)}{1 - 2r\, T_3(x) + r^2},$$

and hence the remainder

$$R_n(x) = f(x) - P_n(x) = \frac{r^{\,n+1}\Big(T_{n+1}(T_3(x)) - r\, T_n(T_3(x))\Big)}{1 - 2r\, T_3(x) + r^2}.$$

In particular, if $T_3(x) = 1$ then

$$R_n(x) = \sum_{j=n+1}^{\infty} r^j = \frac{r^{\,n+1}}{1 - r}.$$

Therefore, for $n = 0$, the four alternation points $x_0 < x_1 < x_2 < x_3$ satisfy

$$f(x_k) = (-1)^k \frac{r}{1 - r} \qquad (k = 0, 1, 2, 3).$$

Gauss–Legendre quadrature for $\int_0^1 \sinh x\, dx$

Legendre polynomials and GL rules. The Legendre polynomials satisfy

$$P_0(t) = 1, \quad P_1(t) = t, \quad (n+1)P_{n+1}(t) = (2n+1)tP_n(t) - nP_{n-1}(t).$$

For $n = 1, 2, 3$:

$$P_1(t) = t, \quad P_2(t) = \tfrac{1}{2}(3t^2 - 1), \quad P_3(t) = \tfrac{1}{2}(5t^3 - 3t).$$

Gauss–Legendre nodes on $[-1, 1]$ are the roots t_i of $P_n(t)$ and the weights are

$$w_i = \frac{2}{(1 - t_i^2)\,[P_n'(t_i)]^2}.$$

Mapping to $[0, 1]$ with $x = \frac{t+1}{2}$ gives $x_i = \frac{t_i+1}{2}$ and $W_i = \frac{w_i}{2}$, so

$$\int_0^1 f(x)\, dx \approx \sum_{i=1}^{n} W_i f(x_i).$$

Exact value.

$$I = \int_0^1 \sinh x\, dx = \cosh(1) - 1 \approx 0.5430806348152437.$$

Case $n = 1$. $P_1(t) = t \Rightarrow t_1 = 0$, $P_1'(0) = 1$, hence $w_1 = 2$, $x_1 = \frac{1}{2}$, $W_1 = 1$.

$$Q_1 = \sinh(1/2) \approx 0.5210953054937474, \quad Q_1 - I \approx -2.1985 \times 10^{-2}.$$

Case $n = 2$. $P_2(t) = \frac{1}{2}(3t^2 - 1)$ has roots $t_{1,2} = \pm 1/\sqrt{3}$. Since $P_2'(t) = 3t$, $w_{1,2} = 1$. Mapping: $x_{1,2} = \frac{1 \pm 1/\sqrt{3}}{2}$ and $W_{1,2} = \frac{1}{2}$.

$$Q_2 = \tfrac{1}{2}\Big[\sinh\Big(\tfrac{1-1/\sqrt{3}}{2}\Big) + \sinh\Big(\tfrac{1+1/\sqrt{3}}{2}\Big)\Big] \approx 0.5429588092378294,\ Q_2 - I \approx -1.2183 \times 10^{-4}.$$

Case $n = 3$. $P_3(t) = \frac{1}{2}(5t^3 - 3t)$ has roots $t = \{0, \pm\sqrt{3/5}\}$. With $P_3'(t) = \frac{1}{2}(15t^2 - 3)$,

$$w = \left\{\tfrac{5}{9},\ \tfrac{8}{9},\ \tfrac{5}{9}\right\} \quad \text{at} \quad t = \left\{-\sqrt{\tfrac{3}{5}},\ 0,\ \sqrt{\tfrac{3}{5}}\right\}.$$

Mapping to $[0, 1]$:

$$x = \left\{\tfrac{1-\sqrt{3/5}}{2},\ \tfrac{1}{2},\ \tfrac{1+\sqrt{3/5}}{2}\right\}, \qquad W = \left\{\tfrac{5}{18},\ \tfrac{4}{9},\ \tfrac{5}{18}\right\}.$$

Thus

$$Q_3 = \tfrac{5}{18}\sinh\Big(\tfrac{1-\sqrt{3/5}}{2}\Big) + \tfrac{4}{9}\sinh\Big(\tfrac{1}{2}\Big) + \tfrac{5}{18}\sinh\Big(\tfrac{1+\sqrt{3/5}}{2}\Big) \approx 0.5430803743542271,$$

and $Q_3 - I \approx -2.6046 \times 10^{-7}$.

Problem 3: Chebyshev series with explicit γ coefficients

Setup and continuity. Let T_k denote Chebyshev polynomials of the first kind. Given nonnegative coefficients γ_j with $\sum_{j=1}^{\infty} \gamma_j < \infty$, define

$$f(x) = \sum_{j=1}^{\infty} \gamma_j \, T_{3j}(x), \qquad x \in [-1, 1].$$

Since $|T_{3j}(x)| \leq 1$ on $[-1, 1]$ and $\sum_j \gamma_j < \infty$, the Weierstrass M–test gives uniform convergence; hence f is continuous.

Partial sums, remainder, and tail mass. For $n \geq 0$ set

$$P_n(x) = \sum_{j=1}^{n} \gamma_j \, T_{3j}(x), \qquad R_n(x) = f(x) - P_n(x) = \sum_{j=n+1}^{\infty} \gamma_j \, T_{3j}(x),$$

and denote the γ-tail by

$$S_n := \sum_{j=n+1}^{\infty} \gamma_j.$$

Alternation points and formula. For each n there exist $3^{n+1}+1$ points, e.g.

$$x_k = \cos\left(\frac{k\pi}{3^{n+1}}\right), \qquad k = 0, 1, \dots, 3^{n+1},$$

such that

$$\boxed{f(x_k) - P_n(x_k) = R_n(x_k) = (-1)^k \, S_n, \qquad k = 0, 1, \dots, 3^{n+1}.}$$

Thus the remainder has constant magnitude S_n and alternates in sign on this grid.

Examples (exact). *(1) Finite tail.* If $\gamma_j = 0$ for $j > m$, then $R_n \equiv 0$ for $n \geq m$. When $n < m$,

$$R_n(x_k) = (-1)^k \sum_{j=n+1}^{m} \gamma_j.$$

(2) Geometric weights. If $\gamma_j = r^j$ with $0 < r < 1$, let $y := T_3(x)$. Using the generating function $\sum_{j \geq 1} r^j T_j(y) = \frac{ry - r^2}{1 - 2ry + r^2}$ and its truncation,

$$\sum_{j=1}^{n} r^j T_j(y) = \frac{ry - r^{n+1} T_{n+1}(y) + r^{n+2} T_n(y) - r^2 y}{1 - 2ry + r^2},$$

we obtain

$$f(x) = \frac{r \, T_3(x) - r^2}{1 - 2r \, T_3(x) + r^2}, \qquad R_n(x) = \frac{r^{n+1}\Big(T_{n+1}(T_3(x)) - r \, T_n(T_3(x))\Big)}{1 - 2r \, T_3(x) + r^2}.$$

At the alternation points,

$$R_n(x_k) = (-1)^k \, \frac{r^{n+1}}{1 - r}.$$

Applications.

(a) **Exact uniform error.** For $P_n(x) = \sum_{j\le n} \gamma_j T_{3j}(x)$ with $\gamma_j \ge 0$,

$$\|f - P_n\|_\infty = S_n := \sum_{j>n} \gamma_j,$$

and this value is attained at $x_k = \cos\!\left(\frac{k\pi}{3^{n+1}}\right)$.

(b) **Tolerance–driven truncation.** Given $\varepsilon > 0$, choose n so that $S_n \le \varepsilon$. If γ_j are unknown, estimate S_n via

$$S_n = \max_{0\le k\le 3^{n+1}} |f(x_k) - P_n(x_k)|.$$

(c) **Equioscillation certificate.** The remainder R_n alternates with constant magnitude S_n, providing a best-uniform (minimax-like) error profile for the truncation under nonnegative coefficients.

(d) **Practical diagnostics.** Sampling only at $\{x_k\}$ detects the worst-case error and guides adaptive refinement.

(e) **Closed-form analysis (geometric γ).** For $\gamma_j = r^j$, $0 < r < 1$,

$$f(x) = \frac{rT_3(x) - r^2}{1 - 2rT_3(x) + r^2}, \quad R_n(x) = \frac{r^{n+1}\Big(T_{n+1}(T_3(x)) - r\,T_n(T_3(x))\Big)}{1 - 2rT_3(x) + r^2}.$$

Approximation vs. interpolation. Given nonnegative coefficients γ_j, we approximate $f(x) = \sum_{j\ge 1} \gamma_j T_{3j}(x)$ by the partial sum $P_n(x) = \sum_{j=1}^{n} \gamma_j T_{3j}(x)$. The points $x_k = \cos\!\left(\frac{k\pi}{3^{n+1}}\right)$ are used to *certify the error*:

$$f(x_k) - P_n(x_k) = (-1)^k S_n, \qquad S_n = \sum_{j>n} \gamma_j = \|f - P_n\|_\infty.$$

They are *not* interpolation nodes. Interpolation would require constructing a polynomial Q such that $Q(y_i) = f(y_i)$ at prescribed nodes y_i.

From f to γ: Chebyshev projection. Write $x = \cos\theta$ and expand

$$f(x) = \sum_{m=0}^{\infty} a_m T_m(x), \quad a_m = \frac{2}{\pi}\int_0^\pi f(\cos\theta)\cos(m\theta)\,d\theta \ (m \ge 1), \quad a_0 = \frac{1}{\pi}\int_0^\pi f(\cos\theta)\,d\theta.$$

Set $\gamma_j := a_{3j}$ so that

$$f(x) = \sum_{j\ge 1} \gamma_j T_{3j}(x) + a_0 T_0(x) \quad \text{(with } a_m = 0 \text{ for } m \not\equiv 0 \pmod 3 \text{ if the expansion is exact).}$$

Given nonnegative γ_j, the n-term truncation $P_n(x) = \sum_{j=1}^{n} \gamma_j T_{3j}(x)$ has uniform error

$$\|f - P_n\|_\infty = S_n := \sum_{j>n} \gamma_j,$$

attained at $x_k = \cos\!\left(\frac{k\pi}{3^{n+1}}\right)$, where $f(x_k) - P_n(x_k) = (-1)^k S_n$.

Practical procedure: from f to γ_j and error evaluation

Step 1 — Start with a known function $f(x)$ on $[-1,1]$. For example,

$$f(x) = e^x, \quad f(x) = \sinh x, \quad \text{or any continuous function you can evaluate.}$$

Step 2 — Expand f in Chebyshev polynomials. Use the cosine change of variable $x = \cos\theta$. The Chebyshev expansion is

$$f(x) = \sum_{m=0}^{\infty} a_m T_m(x),$$

where the coefficients are given by

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(\cos\theta)\, d\theta, \qquad a_m = \frac{2}{\pi}\int_0^{\pi} f(\cos\theta)\cos(m\theta)\, d\theta, \quad (m \geq 1).$$

In practice, these integrals are evaluated efficiently using a *Discrete Cosine Transform (DCT).*

Step 3 — Extract the coefficients that correspond to indices $3j$.

$$\gamma_j := a_{3j}, \qquad j = 1, 2, \ldots$$

Now you have the nonnegative (if f satisfies the theorem's assumption) sequence γ_j.

Step 4 — Form the truncated approximation.

$$P_n(x) = \sum_{j=1}^{n} \gamma_j\, T_{3j}(x).$$

This P_n is your n-term Chebyshev approximation to f.

Step 5 — Compute the tail (expected uniform error bound).

$$S_n = \sum_{j=n+1}^{\infty} \gamma_j.$$

In practice, S_n is approximated by summing until γ_j become negligible.

Step 6 — Evaluate at the special alternation points.

$$x_k = \cos\left(\frac{k\pi}{3^{n+1}}\right), \qquad k = 0, 1, \ldots, 3^{n+1}.$$

At these points, the error has the explicit alternating form

$$f(x_k) - P_n(x_k) = (-1)^k\, S_n,$$

so that $\|f - P_n\|_\infty = S_n$.

Summary.

Symbol	Meaning
$f(x)$	Given continuous function on $[-1, 1]$
$T_m(x)$	Chebyshev polynomial of degree m
a_m	Chebyshev coefficients of f
$\gamma_j = a_{3j}$	Extracted subsequence (used in this theorem)
$P_n(x) = \sum_{j=1}^{n} \gamma_j T_{3j}(x)$	n-term approximation
$R_n(x) = f(x) - P_n(x)$	Remainder (error)
$S_n = \sum_{j>n} \gamma_j$	Uniform error magnitude
$x_k = \cos \frac{k\pi}{3^{n+1}}$	Alternation points of R_n
$f(x_k) - P_n(x_k) = (-1)^k S_n$	Alternating error pattern

Problem 4 — Arbitrarily slow polynomial approximation

Statement. For each $f \in C([-1, 1])$, let

$$E_n(f) = \inf_{p \in \mathcal{P}_n} \sup_{x \in [-1,1]} |f(x) - p(x)|$$

be the distance from f to the space $\mathcal{P}_n$ of polynomials of degree at most n. By the Weierstrass theorem, $E_n(f) \to 0$ for every $f \in C([-1, 1])$. Using the result of Question 3, show that the convergence $E_n(f) \to 0$ can be *arbitrarily slow*: for any decreasing sequence $\{\delta_n\}_{n \geq 1}$ of nonnegative numbers with $\delta_n \to 0$, there exists $f \in C([-1, 1])$ such that

$$E_n(f) \geq \delta_n, \qquad n = 1, 2, \dots.$$

Theory. The Weierstrass approximation theorem ensures $E_n(f) \to 0$, but gives no rate. From Problem 3, any series

$$f(x) = \sum_{j \geq 1} \gamma_j T_{3j}(x), \qquad \gamma_j \geq 0, \ \sum_j \gamma_j < \infty,$$

defines a continuous function satisfying

$$\|f - P_n\|_\infty = \sum_{j>n} \gamma_j =: S_n.$$

Hence, by prescribing the tail S_n, we can fully control the uniform error of the polynomial truncation.

Solution. Given a decreasing sequence $\delta_n \downarrow 0$, define coefficients

$$\gamma_j = \delta_{j-1} - \delta_j, \qquad (\delta_0 := \sum_{j\geq 1} \gamma_j < \infty).$$

Then $\gamma_j \geq 0$ and

$$S_n = \sum_{j>n} \gamma_j = \delta_n.$$

Define

$$f(x) = \sum_{j\geq 1} \gamma_j T_{3^j}(x).$$

By Problem 3, this series converges uniformly on $[-1,1]$, so $f \in C([-1,1])$. Its partial sums $P_n(x) = \sum_{j\leq n} \gamma_j T_{3^j}(x)$ satisfy

$$\|f - P_n\|_\infty = S_n = \delta_n,$$

thus

$$E_n(f) \geq \|f - P_n\|_\infty = \delta_n, \qquad n = 1, 2, \dots.$$

Therefore, $E_n(f) \to 0$ but as slowly as the prescribed sequence δ_n.

Examples.

(1) **Linear rate.** For $\delta_n = \frac{1}{n+1}$, one has $\gamma_j = \frac{1}{j} - \frac{1}{j+1} = \frac{1}{j(j+1)}$ and

$$f(x) = \sum_{j\geq 1} \frac{1}{j(j+1)} T_{3^j}(x), \qquad E_n(f) \geq \frac{1}{n+1}.$$

(2) **Logarithmic rate.** For $\delta_n = \frac{1}{\log(n+2)}$, $\gamma_j = \frac{1}{\log(j+1)} - \frac{1}{\log(j+2)}$ and $E_n(f) \geq 1/\log(n+2)$.

(3) **Exponential rate.** For $\delta_n = r^n$ with $0 < r < 1$, $\gamma_j = r^{j-1}(1-r)$, giving

$$f(x) = \sum_{j\geq 1} (1-r) r^{j-1} T_{3^j}(x), \quad E_n(f) \geq r^n.$$

Conclusion. By suitable choice of γ_j, one can make the polynomial approximation of f converge to f at any desired (arbitrarily slow) rate.

Problem 5 — Hat functions and piecewise–linear interpolation

Problem. For $n, r \in \mathbb{Z}$ and $n \geq 1$, define $\Delta_{n,r} : [-1,1] \to \mathbb{R}$ by

$$\Delta_{n,r}(x) = \max\{0,\ 1 - n|x - rn^{-1}|\}.$$

Sketch $\Delta_{n,r}$. For $f : [-1, 1] \to \mathbb{R}$, show that

$$f_n(x) = \sum_{m=-n}^{n} f\left(\frac{m}{n}\right) \Delta_{n,m}(x)$$

is piecewise linear with $f_n(r/n) = f(r/n)$. Show that if f is continuous, then $\|f_n - f\|_\infty \to 0$ as $n \to \infty$.

Theory. Each $\Delta_{n,r}$ is a tent function supported on $[(r-1)/n, (r+1)/n]$ with value 1 at $x = r/n$ and slopes $\pm n$. On each cell $[k/n, (k+1)/n]$, exactly two hats are nonzero, $\Delta_{n,k}$ and $\Delta_{n,k+1}$, and they satisfy $\Delta_{n,k}(x) + \Delta_{n,k+1}(x) = 1$. If f is continuous on $[-1, 1]$, then its modulus of continuity $\omega_f(\delta) = \sup_{|x-y|\le\delta} |f(x) - f(y)|$ obeys $\omega_f(\delta) \to 0$ as $\delta \downarrow 0$.

Sketch / explicit form of $\Delta_{n,r}$.

$$\Delta_{n,r}(x) = \begin{cases} n\Big(x - \frac{r-1}{n}\Big), & x \in \Big[\frac{r-1}{n}, \frac{r}{n}\Big], \\ n\Big(\frac{r+1}{n} - x\Big), & x \in \Big[\frac{r}{n}, \frac{r+1}{n}\Big], \\ 0, & \text{otherwise.} \end{cases}$$

Piecewise linearity and interpolation. If $x \in [k/n, (k+1)/n]$, then

$$\Delta_{n,k}(x) = \frac{k+1}{n} - x, \qquad \Delta_{n,k+1}(x) = x - \frac{k}{n},$$

and all other $\Delta_{n,m}(x)$ vanish. Hence

$$f_n(x) = f\left(\frac{k}{n}\right)\left(\frac{k+1}{n} - x\right) + f\left(\frac{k+1}{n}\right)\left(x - \frac{k}{n}\right),$$

which is linear in x on that interval. In particular, $f_n(r/n) = f(r/n)$ for each r.

Uniform convergence for continuous f. For $x \in [k/n, (k+1)/n]$ write $x = \lambda\frac{k}{n} + (1-\lambda)\frac{k+1}{n}$ with $\lambda = \frac{k+1}{n} - x \in [0, 1]$. Then

$$f_n(x) = \lambda f\left(\frac{k}{n}\right) + (1-\lambda) f\left(\frac{k+1}{n}\right),$$

so

$$|f(x) - f_n(x)| \le \lambda\left|f(x) - f\left(\frac{k}{n}\right)\right| + (1-\lambda)\left|f(x) - f\left(\frac{k+1}{n}\right)\right| \le \omega_f\left(\frac{1}{n}\right).$$

Therefore $\|f - f_n\|_\infty \le \omega_f(1/n) \to 0$ as $n \to \infty$.

Examples.

(1) $f(x) = x^2$: f_n is the polygonal line through $\{(m/n, (m/n)^2)\}$; $\|f - f_n\|_\infty = \Theta(n^{-2})$.

(2) $f(x) = |x|$: convergence is uniform with $\|f - f_n\|_\infty = O(n^{-1})$ (Lipschitz 1).

(3) $f = \operatorname{sign}(x)$ (discontinuous): f_n does not converge uniformly to f; uniform convergence requires continuity.

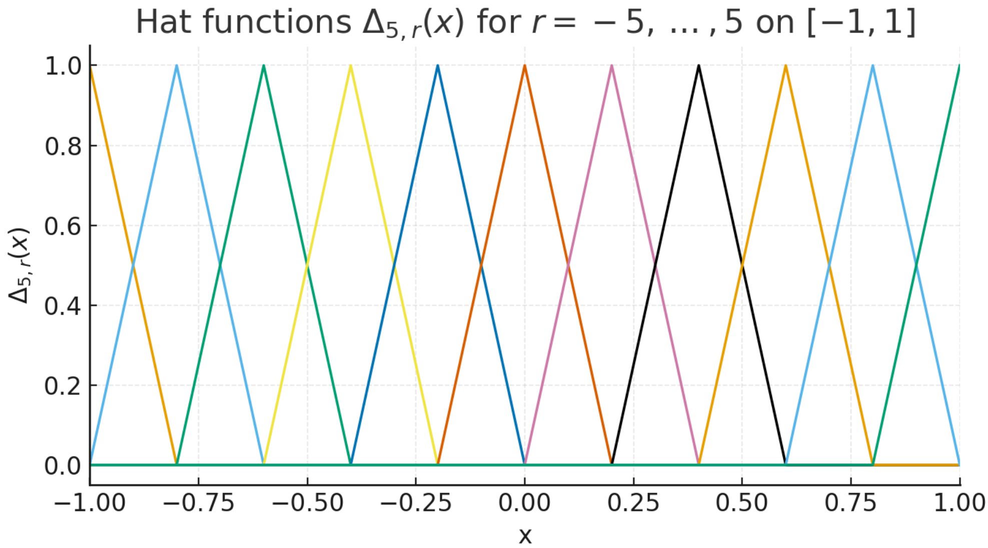

Figure 1: Hat functions $\Delta_{5,r}(x)$ for $r = -5, \ldots, 5$ on $[-1, 1]$.

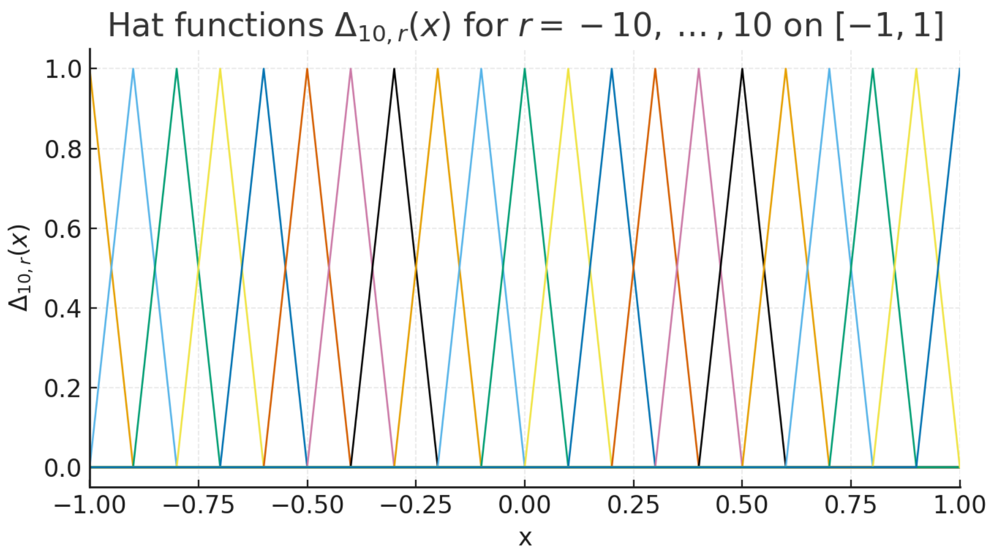

Figure 2: Denser fan of hats $\Delta_{10,r}(x)$ for $r = -10, \ldots, 10$.

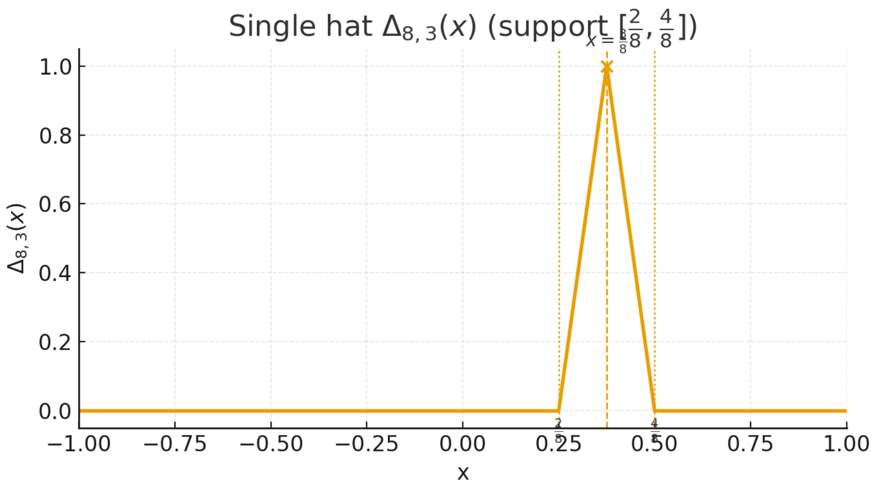

Figure 3: Single hat $\Delta_{8,3}(x)$ with peak at $x = \frac{3}{8}$ and support $\left[\frac{2}{8}, \frac{4}{8}\right]$.

Problem 6 — A Schauder basis for $C([-1,1])$ from hat functions

Statement. Using the result of Question 5, prove that there exists a sequence $\{\phi_n\}_{n\geq 0} \subset C([-1,1])$ such that for every $f \in C([-1,1])$ there exists a *unique* series $\sum_{n=0}^{\infty} a_n\phi_n$ which converges uniformly to f.

Theory. We recall the hat (tent) framework from Question 5 and collect the identities used in the proof.

(a) **Hat functions on a uniform grid.** For $n \in \mathbb{N}$ and $r \in \mathbb{Z}$ define

$$\Delta_{n,r}(x) = \max\{0,\ 1 - n|x - r/n|\}, \qquad x \in [-1,1].$$

Then $\Delta_{n,r}$ is supported on $[(r-1)/n, (r+1)/n]$, satisfies $\Delta_{n,r}(r/n) = 1$, is linear with slopes $\pm n$ on its support, and for each $x \in [k/n, (k+1)/n]$ one has

$$\Delta_{n,k}(x) + \Delta_{n,k+1}(x) = 1, \qquad \Delta_{n,m}(x) = 0 \text{ for } m \notin \{k, k+1\}.$$

Consequently, for any $f \in C([-1,1])$ the piecewise–linear interpolant

$$f_n(x) = \sum_{m=-n}^{n} f\left(\frac{m}{n}\right)\Delta_{n,m}(x)$$

is linear on each $[k/n, (k+1)/n]$, interpolates the data $f_n(m/n) = f(m/n)$, and satisfies the uniform error bound

$$\|f - f_n\|_\infty \leq \omega_f\left(\frac{1}{n}\right),$$

where $\omega_f(\delta) = \sup_{|x-y|\leq\delta} |f(x) - f(y)|$ is the modulus of continuity.

(b) **Dyadic refinement and midpoint hats (Faber–Schauder atoms).** On $[0,1]$ (with $u = \frac{x+1}{2}$) define the dyadic midpoint hats

$$h_{k,m}(u) = \max\Big\{0,\ 1 - 2^{k+1}\Big|u - \frac{2m-1}{2^{k+1}}\Big|\Big\}, \qquad k \geq 0,\ m = 1, \ldots, 2^k.$$

Each $h_{k,m}$ is the tent of height 1 with support $\big[(m-1)/2^k, (m+1)/2^k\big]$ and peak at $(2m-1)/2^{k+1}$. Transport to $[-1,1]$ via $\phi_{k,m}(x) = h_{k,m}\big(\frac{x+1}{2}\big)$ and add two low-order functions $\phi_0(x) = 1$, $\phi_1(x) = \frac{x+1}{2}$.

(c) **Telescoping of piecewise–linear interpolants.** Let $g(u) = f(2u-1)$ and let g_{2^K} be the piecewise–linear interpolant on the dyadic grid $\{j/2^K\}_{j=0}^{2^K}$. Then

$$g_{2^K}(u) = g_{2^0}(u) + \sum_{k=0}^{K-1} \big(g_{2^{k+1}}(u) - g_{2^k}(u)\big),$$

and each difference admits the representation

$$g_{2^{k+1}}(u) - g_{2^k}(u) = \sum_{m=1}^{2^k} d_{k,m}\, h_{k,m}(u),$$

with *detail coefficients* (midpoint second differences)

$$d_{k,m} = g\Big(\frac{2m-1}{2^{k+1}}\Big) - \frac{1}{2}\Big[g\Big(\frac{m-1}{2^k}\Big) + g\Big(\frac{m}{2^k}\Big)\Big].$$

Hence, for $K \geq 1$,

$$g_{2^K}(u) = a_0 + a_1 u + \sum_{k=0}^{K-1} \sum_{m=1}^{2^k} d_{k,m}\, h_{k,m}(u), \qquad a_0 = g(0), \quad a_1 = g(1) - g(0).$$

Pulling back to $[-1,1]$ yields

$$f_{2^K}(x) = a_0\, \phi_0(x) + a_1\, \phi_1(x) + \sum_{k=0}^{K-1} \sum_{m=1}^{2^k} d_{k,m}\, \phi_{k,m}(x).$$

(d) **Uniform convergence and uniqueness.** By the bound in (a), $f_{2^K} \to f$ uniformly as $K \to \infty$, hence

$$f(x) = a_0\, \phi_0(x) + a_1\, \phi_1(x) + \sum_{k=0}^{\infty} \sum_{m=1}^{2^k} d_{k,m}\, \phi_{k,m}(x)$$

is a uniformly convergent series in $\{\phi_n\} \subset C([-1,1])$. Uniqueness of the coefficients follows from the triangular structure with respect to levels: if $\sum a_n \phi_n \equiv 0$, then $a_0 = a_1 = 0$ (evaluation at the endpoints), and at each level k the functions $\{\phi_{k,m}\}_m$ have disjoint supports and are the only basis elements nonzero at their midpoints; thus all detail coefficients vanish by induction.

Construction (Faber–Schauder system). Work on $[0,1]$ with $u \in [0,1]$ and define

$$\psi_0(u) = 1, \quad \psi_1(u) = u, \qquad h_{k,m}(u) = \max\left\{0,\ 1 - 2^{k+1}\left|u - \frac{2m-1}{2^{k+1}}\right|\right\}$$

for $k \geq 0$ and $m = 1, \ldots, 2^k$. Transport to $[-1,1]$ by $u = (x+1)/2$:

$$\phi_0(x) = 1, \qquad \phi_1(x) = \tfrac{x+1}{2}, \qquad \phi_{k,m}(x) = h_{k,m}\left(\tfrac{x+1}{2}\right).$$

Enumerate $\{\phi_n\}$ as $(\phi_0, \phi_1, \phi_{0,1}, \phi_{1,1}, \phi_{1,2}, \ldots)$.

Coefficients and partial sums. Given $f \in C([-1,1])$, set $g(u) = f(2u-1)$. Define

$$a_0 = g(0), \qquad a_1 = g(1) - g(0),$$

and for $k \geq 0$, $m = 1, \ldots, 2^k$,

$$d_{k,m} = g\left(\frac{2m-1}{2^{k+1}}\right) - \frac{1}{2}\left[g\left(\frac{m-1}{2^k}\right) + g\left(\frac{m}{2^k}\right)\right].$$

Let g_{2^K} be the piecewise–linear interpolant on the dyadic grid $\{j/2^K\}$. Then the standard decomposition (telescoping across levels) gives

$$g_{2^K}(u) = a_0 + a_1 u + \sum_{k=0}^{K-1} \sum_{m=1}^{2^k} d_{k,m}\, h_{k,m}(u).$$

Hence, on $[-1,1]$,

$$f_{2^K}(x) = a_0\, \phi_0(x) + a_1\, \phi_1(x) + \sum_{k=0}^{K-1} \sum_{m=1}^{2^k} d_{k,m}\, \phi_{k,m}(x).$$

Uniform convergence. By Question 5, $f_{2^K} \to f$ uniformly on $[-1,1]$. Letting $K \to \infty$ yields the uniformly convergent expansion

$$f(x) = a_0\, \phi_0(x) + a_1\, \phi_1(x) + \sum_{k=0}^{\infty} \sum_{m=1}^{2^k} d_{k,m}\, \phi_{k,m}(x),$$

i.e. a series $\sum_{n\geq 0} a_n \phi_n$ converging uniformly to f.

Uniqueness. Suppose $\sum_{n\geq 0} a_n \phi_n \equiv 0$. Evaluating at $x = -1$ and $x = 1$ gives $a_0 = 0$ and $a_1 = 0$. Proceeding by levels, at level k the functions $\{\phi_{k,m}\}_m$ have disjoint supports and each is the only basis function nonzero at its own midpoint; hence each coefficient must be zero. By induction, all coefficients vanish, so the representation is unique.

Examples.

(1) $f(x) = ax + b$: all detail coefficients $d_{k,m}$ vanish; only the low–order terms ϕ_0, ϕ_1 remain.

(2) $f(x) = x^2$: $d_{k,m} = \frac{1}{4}\, 2^{-2k}$ for all m, giving a rapidly convergent sum of midpoint hats across levels.

Conclusion. The system $\{\phi_n\}$ is a Schauder basis for $C([-1,1])$ (the Faber–Schauder basis): every continuous function admits a unique uniformly convergent series in these functions.

Problem 7 — Powers preserve uniform convergence

Statement. Let $f_n : [0,1] \to \mathbb{R}$ satisfy $f_n \to f$ uniformly on $[0,1]$, and suppose f is bounded. Show that for any integer $m \geq 1$, the functions $g_n(t) = f_n(t)^m$ converge uniformly on $[0,1]$ to $g(t) = f(t)^m$.

Theory. For all $a, b \in \mathbb{R}$ and $m \geq 1$,

$$a^m - b^m = (a-b) \sum_{k=0}^{m-1} a^{m-1-k} b^k,$$

hence if $|a|, |b| \leq M$ then $|a^m - b^m| \leq m M^{m-1} |a - b|$. Uniform convergence $f_n \to f$ implies $\|f_n - f\|_\infty \to 0$.

Proof. Let $M = \|f\|_\infty < \infty$. Since $f_n \to f$ uniformly, there exists N with $\|f_n - f\|_\infty \leq 1$ for all $n \geq N$. Then for $n \geq N$,

$$\|f_n\|_\infty \leq \|f\|_\infty + \|f_n - f\|_\infty \leq M + 1 =: M'.$$

For any $t \in [0,1]$,

$$|f_n(t)^m - f(t)^m| \leq |f_n(t) - f(t)| \sum_{k=0}^{m-1} |f_n(t)|^{m-1-k} |f(t)|^k \leq m(M')^{m-1}\, |f_n(t) - f(t)|.$$

Taking sup over t yields

$$\|f_n^m - f^m\|_\infty \leq m(M')^{m-1}\, \|f_n - f\|_\infty \xrightarrow[n\to\infty]{} 0.$$

Therefore $f_n^m \to f^m$ uniformly on $[0,1]$. $\square$

Examples.

(1) $f_n(t) = t + \frac{1}{n}$, $f(t) = t$, $m = 3$: $\|f_n^3 - f^3\|_\infty \le 12/n \to 0$.

(2) $f_n(t) = \sin(nt)/n$, $f \equiv 0$: for any $m \ge 1$, $\|f_n^m - f^m\|_\infty = \|f_n\|_\infty^m \le (1/n)^m \to 0$.

Problem 8 — Polynomial (vs. rational) approximation on a punctured disc

paragraphProblem (exact). Let $B_r(z)$ denote the open disc of radius r about $z \in \mathbb{C}$ and set

$$U \;=\; B_2(1) \setminus B_1(0).$$

Suppose f is holomorphic on U.

(i) Prove that there exists a sequence of *rational functions* which converges to f uniformly on compact subsets of U.

(ii) Must there be a sequence of *polynomials* which converges to f uniformly on U?

(iii) If, in addition, f is holomorphic on some open set containing $\overline{U}$, must there be a sequence of *polynomials* which converges to f uniformly on U?

Theory. We will use the following results.

(a) **Runge's theorem (rational form).** Let $K \subset \mathbb{C}$ be compact and let $E = \widehat{\mathbb{C}} \setminus K$. If E has finitely many components, then every function holomorphic on a neighborhood of K can be uniformly approximated on K by rational functions whose poles lie in a prescribed finite set containing at least one point from each bounded component of E. If E is connected and contains ∞, the approximants can be chosen to be polynomials.

(b) **Mergelyan's theorem (polynomial form).** If $K \subset \mathbb{C}$ is compact with connected complement and f is continuous on K and holomorphic in $\operatorname{int} K$, then there exists a sequence of polynomials converging uniformly to f on K.

(c) **Cauchy integral test for non-approximability.** If $p_n \to g$ uniformly on a simple closed curve Γ, then $\int_\Gamma p_n(z)\,dz \to \int_\Gamma g(z)\,dz$. For every polynomial p, $\int_\Gamma p(z)\,dz = 0$.

(d) **Geometry of U.** The complement of U in the Riemann sphere has two components: the unbounded $\mathbb{C} \setminus B_2(1)$ (containing ∞) and the bounded $\overline{B_1(0)}$ (containing 0). Thus Runge's theorem yields rational approximation on compacts of U with all poles placed in the bounded component (e.g. at 0), but not, in general, polynomial approximation on all of U unless f extends holomorphically across the hole so that the relevant compact has connected complement.

Setting. Let $B_r(z)$ be the open disc of radius r about $z \in \mathbb{C}$ and $U = B_2(1) \setminus B_1(0)$. Assume f is holomorphic on U.

(i) Local uniform approximation. The complement $\widehat{\mathbb{C}} \setminus U$ has two components: the unbounded $\mathbb{C} \setminus B_2(1)$ and the bounded $\overline{B_1(0)}$. By Runge's theorem, for every compact $K \Subset U$ there exist rational functions r_n with all poles in $\overline{B_1(0)}$ (indeed at 0) such that $r_n \to f$ uniformly on K. Thus f is locally uniformly approximable by rational functions with poles at 0. *(If the statement asks for polynomials, this is false in general; see (ii).)*

(ii) Uniform approximation on U by polynomials? *No.* Consider $f(z) = 1/z \in O(U)$. Suppose polynomials p_n satisfy $p_n \to f$ uniformly on U. Let $\Gamma = \{\, |z| = r \,\}$ with $1 < r < 2$ and $\Gamma \subset U$. Then

$$\int_\Gamma p_n(z)\, dz \longrightarrow \int_\Gamma \frac{1}{z}\, dz = 2\pi i,$$

whereas for all polynomials p_n we have $\int_\Gamma p_n(z)\, dz = 0$ by Cauchy's theorem, a contradiction. Hence, in general, no such polynomial sequence exists.

(iii) If f extends holomorphically across $\overline{U}$. Assume f is holomorphic on some open set $W \supset \overline{U}$. Then f is holomorphic on a neighborhood of the compact set $K = \overline{B_2(1)}$, whose complement is connected and contains ∞. By Runge's theorem (or Mergelyan's theorem), there are polynomials p_n with $p_n \to f$ uniformly on K; in particular, $p_n \to f$ uniformly on U. Therefore under this additional assumption the answer is *yes*.

Problem 9 — Polynomial approximation of $1/z$ on a semicircle

Statement. Construct polynomials p_n converging uniformly to $1/z$ on $K = \{z \in \mathbb{C} : |z| = 1,\ \Re z \geq 0\}$.

Theory. By Mergelyan's theorem, if K is compact with connected complement and $f \in C(K)$ is holomorphic on $\operatorname{int} K$, then there exist polynomials P_n with $P_n \to f$ uniformly on K. Our K is a closed Jordan arc, $\operatorname{int} K = \varnothing$, and $\widehat{\mathbb{C}} \setminus K$ is connected.

Solution. The function $f(z) = 1/z$ is continuous on K. Since K has connected complement and empty interior, Mergelyan applies; therefore there exist polynomials p_n with $\sup_{z \in K} |p_n(z) - 1/z| \to 0$.

Example. One may build p_n via Faber polynomials of the arc or via discrete least–squares fits on K; both converge uniformly by Mergelyan.

Problem 10 — Characterization on a Runge domain

Statement. Let $U \subset \mathbb{C}$ be bounded and assume $\mathbb{C} \setminus U$ is connected. Show that $f : U \to \mathbb{C}$ is holomorphic iff for every compact $K \subset U$ and $\varepsilon > 0$ there exists a polynomial P with $\sup_K |f - P| < \varepsilon$.

Theory. If $\mathbb{C} \setminus U$ is connected, U is a Runge domain; by the Runge–Oka–Weil theorem, holomorphic functions on U can be uniformly approximated on compacts by polynomials. Uniform limits of holomorphic functions on open sets are holomorphic.

Proof. ($\Rightarrow$) For $f \in \mathcal{O}(U)$ and $K \Subset U$, Runge–Oka–Weil yields a polynomial P with $\sup_K |f - P| < \varepsilon$. ($\Leftarrow$) If for each $K \Subset U$ there are polynomials P_n with $P_n \to f$ uniformly on K, then by Weierstrass f is holomorphic on K°, hence on U.

Problem 11 — A pointwise–vanishing polynomial sequence outside 0

Statement. Does there exist polynomials p_n with $p_n(0) = 1$ and $p_n(z) \to 0$ for all $z \neq 0$?

Construction & Proof. Enumerate a dense set $\{w_j\}_{j \geq 1} \subset \mathbb{Q}(i) \setminus \{0\}$ and choose integers $N_j \uparrow \infty$. Define partial products

$$p_n(z) = \prod_{j=1}^{n} \left(1 - \tfrac{z}{w_j}\right)^{N_j}.$$

Then $p_n(0) = 1$. For fixed $z \neq 0$ there are infinitely many j with $|1 - z/w_j| \leq \frac{1}{2}$; hence $|p_n(z)| \leq \prod_{j \leq n,\ |1-z/w_j| \leq 1/2} 2^{-N_j} \to 0$. Therefore $p_n(z) \to 0$ for each $z \neq 0$.

Remark. Choosing $N_j = j$ suffices; taking $N_j = 2^j$ accelerates decay.

Problem 12 — Extending a uniform polynomial limit from an annulus

Statement. Let $A = \{z : \frac{1}{2} \le |z| \le 1\}$ and $f : A \to \mathbb{C}$ be continuous, holomorphic on $\{\frac{1}{2} < |z| < 1\}$. If polynomials $p_n \to f$ uniformly on A, prove that there is $g \in C(\{|z| \le 1\})$, holomorphic on $|z| < 1$, with $g = f$ on A.

Proof. Fix $\varepsilon > 0$. From uniform convergence on A, choose N so that $\sup_A |p_n - p_m| < \varepsilon$ for $n, m \ge N$. By the maximum modulus principle on the annulus, $\sup_{\frac{1}{2} \le |z| \le 1} |p_n - p_m| < \varepsilon$. On the disk $|z| \le \frac{1}{2}$,

$$\sup_{|z| \le \frac{1}{2}} |p_n - p_m| \le \sup_{|z| = \frac{1}{2}} |p_n - p_m| < \varepsilon.$$

Therefore $\sup_{|z| \le 1} |p_n - p_m| < \varepsilon$, so $\{p_n\}$ is uniformly Cauchy on $|z| \le 1$, hence converges uniformly to some $g \in C(\{|z| \le 1\})$, holomorphic on $|z| < 1$. Uniform convergence on A gives $g = f$ on A.

Definition (Jordan curve). A subset $J \subset \mathbb{R}^2$ is a *Jordan curve* if there exists a continuous injective map $\gamma : \mathbb{S}^1 \to \mathbb{R}^2$ such that $J = \gamma(\mathbb{S}^1)$. Equivalently, J is a simple closed curve (continuous, no self-intersections).

Related notions. A *Jordan arc* is the image of a homeomorphism $\eta : [a, b] \to \mathbb{R}^2$. A *Jordan domain* is a bounded open set $D \subset \mathbb{R}^2$ whose boundary ∂D is a Jordan curve (typically oriented positively).

Jordan curve theorem. Every Jordan curve J divides $\mathbb{R}^2$ into exactly two connected components, an interior and an exterior, with J as their common boundary.

Mergelyan's Theorem (Polynomial Approximation)

Statement. Let $K \subset \mathbb{C}$ be compact with **connected complement** $\mathbb{C} \setminus K$. If f is continuous on K and holomorphic in the interior K°, then there exists a sequence of **polynomials** p_n with

$$\sup_{z\in K} |f(z) - p_n(z)| \longrightarrow 0.$$

- If $K^\circ = \varnothing$ (for example, K is a Jordan curve or arc), the condition "holomorphic in the interior" is vacuous; every continuous f on K is uniformly approximable by polynomials.
- If $\mathbb{C} \setminus K$ is not connected (i.e., K has holes), polynomials are in general *not* dense, but **rational functions with poles outside** K are dense (Runge–Oka–Weil).

Useful Variants and Corollaries.

- **Runge–Oka–Weil (rational form).** For any compact K and any finite set E meeting every bounded component of $\mathbb{C} \setminus K$, functions holomorphic near K can be uniformly approximated on K by rational functions with all poles in E.
- **Mergelyan on Jordan domains.** If $K = \overline{\Omega}$ with Ω a bounded Jordan domain, then every $f \in A(\Omega)$ (continuous on $\overline{\Omega}$, holomorphic on Ω) is a uniform limit of polynomials.
- **Maximum modulus transfer.** Uniform convergence on K of polynomials implies uniform convergence on compact subsets of K° of their derivatives (by Cauchy estimates).

When It Fails (and What to Use Instead).

- On an *annulus* $K = \{r \le |z| \le R\}$, the complement $\mathbb{C} \setminus K$ has two components, so polynomials are not generally dense; but rational functions with poles at 0 (or ∞) are dense (Runge).
- If f *extends holomorphically across the holes*, one can enlarge K to a compact with connected complement and then apply Mergelyan.

Tiny Proof Sketch (ideas only).

(a) Consider the uniform algebra $P(K)$ (uniform closure of polynomials on K).

(b) If $\mathbb{C} \setminus K$ is connected, K is *polynomially convex*; by deep results (Mergelyan 1951), any $f \in A(K)$ can be uniformly approximated in $P(K)$.

(c) Tools behind the proof include Runge's theorem, peak functions, and the theory of uniform algebras.

Historical Note. The theorem was proved by S. N. Mergelyan in 1951 and generalizes earlier approximation results of Weierstrass (on intervals) and Runge (on rational approximation). It represents the final step in characterizing all compact sets K in $\mathbb{C}$ for which polynomials are dense in $A(K)$.

Problem 1 — Irrationality of $\sqrt{3}+\sqrt{5}$ and e^2

Statement. Prove that $\sqrt{3}+\sqrt{5}$ and e^2 are irrational.

Solution. (1) If $\sqrt{3}+\sqrt{5}=r\in\mathbb{Q}$, then $8+2\sqrt{15}=r^2\in\mathbb{Q}$, so $\sqrt{15}\in\mathbb{Q}$, a contradiction. Hence $\sqrt{3}+\sqrt{5}\notin\mathbb{Q}$.

(2) Write $e^2=\sum_{k=0}^{\infty}\frac{2^k}{k!}$. Suppose $e^2=\frac{p}{q}$ with $p,q\in\mathbb{N}$. Let $N\geq\max\{q,5\}$ and set

$$A_N := N!\sum_{k=0}^{N}\frac{2^k}{k!}\in\mathbb{Z},\qquad T_N := N!\sum_{k=N+1}^{\infty}\frac{2^k}{k!}.$$

Then $N!\,e^2=A_N+T_N$ and $N!\,e^2=\frac{pN!}{q}\in\mathbb{Z}$, so $T_N\in\mathbb{Z}$. But for $N\geq 5$,

$$0<T_N<\frac{2^{N+1}}{N+1}\cdot\frac{1}{1-\frac{2}{N+2}}<1,$$

a contradiction. Hence $e^2\notin\mathbb{Q}$.

Problem 2 — Algebraic zeros and coefficients

Statement. (i) Show that a real number α is algebraic iff it is a zero of a polynomial with rational coefficients. (ii) Is it true that if all the roots of a polynomial are algebraic numbers, then the polynomial must have rational coefficients?

Solution. (i) If $\sum_{k=0}^{m}a_k\alpha^k=0$ with $a_k\in\mathbb{Z}$, then certainly α is a root of a polynomial with rational coefficients. Conversely, if $\sum_{k=0}^{m}q_k\alpha^k=0$ with $q_k\in\mathbb{Q}$, multiply by a common denominator to obtain an integer–coefficient polynomial vanishing at α.

(ii) *False.* Example: $p(x) = x - \sqrt{2}$ has the algebraic root $\sqrt{2}$ but p has nonrational coefficient $\sqrt{2}$. In general, if all roots are algebraic, the coefficients are algebraic numbers (by Vieta), not necessarily rational.

Problem 3 — Uncountably many transcendental numbers

Statement. Let $b_n \in \{1,2\}$ and define

$$x = \sum_{n=0}^{\infty} \frac{b_n}{10^{n!}}.$$

Show that this gives another proof that the set of transcendental numbers is uncountable.

Solution. Let p_N/q_N be the Nth partial sum with $q_N = 10^{N!}$. Then

$$0 < \left| x - \frac{p_N}{q_N} \right| < \frac{1}{10^{(N+1)!}} < \frac{1}{q_N^m} \quad \text{for every fixed } m \text{ and all sufficiently large } N.$$

Thus x is a Liouville number and hence transcendental. The map $\{1,2\}^{\mathbb{N}} \ni (b_n) \mapsto x$ is injective, so there are $2^{\aleph_0}$ many such numbers.

Problem 4 — Step–function limits by polynomials

Statement. (i) Show there exists a sequence of polynomials P_n such that

$$P_n(z) \to \begin{cases} 1, & |z| \le 1,\ \Re z \ge 0, \\ 0, & |z| \le 1,\ \Re z < 0, \end{cases} \qquad (n \to \infty).$$

(ii) Show there exists a sequence of polynomials Q_n such that

$$Q_n(z) \to \begin{cases} 1, & \Re z \ge 0, \\ 0, & \Re z < 0, \end{cases} \qquad (n \to \infty).$$

Theory. If $\Omega \subset \mathbb{C}$ is open and $\mathbb{C} \setminus \Omega$ is connected (a Runge domain), then any $f \in \mathcal{O}(\Omega)$ can be uniformly approximated on each compact $K \Subset \Omega$ by polynomials. A diagonal argument over an exhaustion of Ω yields pointwise convergence.

Proof. (i) Let $\Omega_+ = \{z : |z| < 1,\ \Re z > 0\}$ and $\Omega_- = \{z : |z| < 1,\ \Re z < 0\}$, and set $\Omega = \Omega_+ \cup \Omega_-$. Define $f \equiv 1$ on Ω_+ and $f \equiv 0$ on Ω_-. Then $f \in \mathcal{O}(\Omega)$ and $\mathbb{C} \setminus \Omega$ is connected; hence by Runge there are polynomials approximating f uniformly on each compact $K \Subset \Omega$. Choosing a diagonal sequence gives polynomials P_n with the stated pointwise limits on the two halves of the closed unit disk.

(ii) Let $\Omega = \{\Re z > 0\} \cup \{\Re z < 0\} = \mathbb{C} \setminus i\mathbb{R}$ and define $f \equiv 1$ on the right half–plane and $f \equiv 0$ on the left. Again $\mathbb{C} \setminus \Omega$ is connected, so Runge yields polynomials Q_n converging to f on compacta; the diagonal argument gives the claimed pointwise limits.

Vieta's Formulas and Liouville's Theorem

Vieta's Formulas. Let

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = a_n \prod_{j=1}^{n} (x - r_j),$$

with roots $r_1, \ldots, r_n$ (counted with multiplicity). Denote the elementary symmetric polynomials

$$e_k(r_1, \ldots, r_n) = \sum_{1 \le j_1 < \cdots < j_k \le n} r_{j_1} \cdots r_{j_k}, \qquad e_0 := 1.$$

Then for $k = 0, 1, \ldots, n$,

$$a_{n-k} = (-1)^k \, a_n \, e_k(r_1, \ldots, r_n).$$

In particular, for a monic polynomial ($a_n = 1$),

$$\sum_{j=1}^{n} r_j = -a_{n-1}, \quad \sum_{i<j} r_i r_j = a_{n-2}, \ \ldots, \ \prod_{j=1}^{n} r_j = (-1)^n a_0.$$

Hence each coefficient is a symmetric polynomial in the roots. If all r_j are algebraic, then every coefficient a_k is algebraic (though not necessarily rational).

Example. If $r_1 = \sqrt{2}$, $r_2 = \sqrt{3}$, then

$$P(x) = (x - \sqrt{2})(x - \sqrt{3}) = x^2 - (\sqrt{2} + \sqrt{3})x + \sqrt{6},$$

so by Vieta $a_1 = -(r_1 + r_2)$ and $a_0 = r_1 r_2$.

Liouville's Theorem (Classical). If α is algebraic of degree $d \ge 2$, then there exists $C(\alpha) > 0$ such that

$$\left| \alpha - \frac{p}{q} \right| > \frac{C(\alpha)}{q^d} \qquad \text{for all } p/q \in \mathbb{Q},\ q \ge 1.$$

Corollary (Liouville numbers). A real number x is a Liouville number if for every $m \in \mathbb{N}$ there exist infinitely many p/q with $0 < |x - \frac{p}{q}| < q^{-m}$. Every Liouville number is transcendental.

Example (Liouville constant).

$$L = \sum_{n=1}^{\infty} 10^{-n!}.$$

Let $p_N/10^{N!}$ be the Nth partial sum. Then

$$0 < \left| L - \frac{p_N}{10^{N!}} \right| < 10^{-(N+1)!} < \left(10^{N!}\right)^{-m} \quad \text{for any fixed } m \text{ and } N \text{ large,}$$

hence L is Liouville and therefore transcendental.

Context (Roth's Theorem). If α is irrational algebraic, then for every $\varepsilon > 0$,

$$\left| \alpha - \frac{p}{q} \right| > \frac{c(\alpha, \varepsilon)}{q^{2+\varepsilon}}$$

for all but finitely many rationals p/q. This is much stronger than Liouville's q^{-d} bound.

Problem 5 — Continued Fractions for $100/37$ and $\sqrt{7}$

Statement. Find the simple continued fraction expansions for $100/37$ and $\sqrt{7}$.

Solution. **(a)** $100/37$**.**

$100 = 2{\cdot}37{+}26, \quad 37 = 1{\cdot}26{+}11, \quad 26 = 2{\cdot}11{+}4, \quad 11 = 2{\cdot}4{+}3, \quad 4 = 1{\cdot}3{+}1, \quad 3 = 3{\cdot}1.$

Hence

$$\frac{100}{37} = [2; 1, 2, 2, 1, 3].$$

(b) $\sqrt{7}$**.** With $a_0 = \lfloor \sqrt{7} \rfloor = 2$ and the standard recurrences $m_{k+1} = d_k a_k - m_k,\ d_{k+1} = (7 - m_{k+1}^2)/d_k,\ a_{k+1} = \lfloor (a_0 + m_{k+1})/d_{k+1} \rfloor$, one obtains

$$\sqrt{7} = [2; \overline{1, 1, 1, 4}].$$

Problem 6 — Best Denominator < 10 Approximating $71/49$

Statement. Which rational with denominator less than 10 best approximates $71/49$?

Solution. Let $x = 71/49 \approx 1.4489796$. For each $q = 1, \ldots, 9$ choose $p = \text{round}(qx)$. The smallest error occurs at $q = 9$, $p = 13$:

$$\left|\frac{13}{9} - x\right| \approx 0.00458,$$

which beats all other denominators < 10. Therefore the best is $\boxed{13/9}$.

Problem 7 — Alternating Continued Fraction with $a, b > 0$

Statement. Let

$$x = \cfrac{1}{a + \cfrac{1}{b + \cfrac{1}{a + \cfrac{1}{b + \cdots}}}}, \qquad a, b \in \mathbb{N}.$$

(a) Show that x solves $ax^2 + abx - b = 0$.

(b) For $a = b = 1$, prove $x = \frac{-1+\sqrt{5}}{2}$ and that the convergents p_n/q_n satisfy $p_n = F_n$, $q_n = F_{n+1}$ where (F_n) are Fibonacci numbers, $F_0 = 0$, $F_1 = 1$.

(c) Show Cassini's identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^{n+1}$.

(d) Prove $F_{2n+1} = F_n^2 + F_{n+1}^2$ and $F_{2n} = F_n(F_{n-1} + F_{n+1})$. With $x_n = F_{n+1}/F_n$ express x_{2n} as a function of x_n and deduce that $y_k = x_{2^k}$ converges very rapidly to $\phi = (1+\sqrt{5})/2$.

Proof. (a) Since the pattern repeats after the inner b, we have $x = \cfrac{1}{a + \cfrac{1}{b+x}}$. Thus

$$\frac{1}{x} = a + \frac{1}{b+x} \implies (b+x)\left(\frac{1}{x} - a\right) = 1 \implies ax^2 + abx - b = 0.$$

(b) With $a = b = 1$, the quadratic gives $x^2 + x - 1 = 0$ and $x = \frac{-1+\sqrt{5}}{2}$. The SCF is $[0; 1, 1, 1, \ldots]$. Convergent recurrences $p_{-1} = 1, p_0 = 0,\ p_n = p_{n-1} + p_{n-2}$ and $q_{-1} = 0, q_0 = 1,\ q_n = q_{n-1} + q_{n-2}$ yield $p_n = F_n$, $q_n = F_{n+1}$.

(c) Cassini: $F_{n+1}F_{n-1} - F_n^2 = (-1)^{n+1}$, proved by induction or Binet's formula.

(d) Identities (by induction or Binet):

$$F_{2n+1} = F_n^2 + F_{n+1}^2, \qquad F_{2n} = F_n(F_{n-1} + F_{n+1}).$$

Therefore, with $x_n = F_{n+1}/F_n$,

$$x_{2n} = \frac{F_{2n+1}}{F_{2n}} = \frac{F_n^2 + F_{n+1}^2}{F_n(F_{n-1} + F_{n+1})} = \frac{1 + x_n^2}{2x_n - 1} =: T(x_n).$$

Since $\phi^2 = \phi + 1$, we have $T(\phi) = \phi$, and

$$T'(x) = \frac{2(x^2 - x - 1)}{(2x - 1)^2}, \qquad T'(\phi) = 0,$$

so the iteration $y_{k+1} = T(y_k)$ has quadratic convergence to ϕ.

Problem 8 — A CF for $\tanh y$ and irrationality of e

Statement. From the continued fraction for $\tan z$ (valid for $z \in \mathbb{C}$, $|z| \le 1$), obtain a continued fraction for $\tanh y$ ($y \in \mathbb{R}$) and deduce

$$\frac{e+1}{e-1} = 2 + \cfrac{1}{6 + \cfrac{1}{10 + \cfrac{1}{14 + \cfrac{1}{18 + \cdots}}}}.$$

Explain why this gives another proof that e is irrational.

Theory. Euler's CF:

$$\tan z = \cfrac{z}{1 - \cfrac{z^2}{3 - \cfrac{z^2}{5 - \cfrac{z^2}{7 - \cdots}}}} \qquad (|z| \le 1).$$

Since $\tan(iy) = i\,\tanh y$, we obtain CFs for $\tanh y$ and $\coth y = 1/\tanh y$.

Solution. Substitute $z = iy$ and simplify:

$$\tanh y = \cfrac{y}{1 + \cfrac{y^2}{3 + \cfrac{y^2}{5 + \cfrac{y^2}{7 + \cdots}}}}, \qquad \coth y = \frac{1}{y} + \cfrac{y}{3 + \cfrac{y^2}{5 + \cfrac{y^2}{7 + \cdots}}}.$$

With $y = 1$,

$$\coth 1 = \frac{e+1}{e-1} = 2 + \cfrac{1}{6 + \cfrac{1}{10 + \cfrac{1}{14 + \cdots}}}.$$

The CF is infinite with integer partial denominators, so its value is irrational; hence $\coth 1$ is irrational. Solving $r = \frac{e+1}{e-1}$ for e gives $e = \frac{1+r}{1-r}$, so r rational would force e rational—contradiction.

Problem 9 — From $f(nx) \to 0$ to $f(t) \to 0$

Statement. Let $f : [1, \infty) \to \mathbb{R}$ be continuous and suppose $f(nx) \to 0$ as $n \to \infty$ for each fixed $x \geq 1$. Show $f(t) \to 0$ as $t \to \infty$.

Proof. Fix $\varepsilon > 0$ and set $Q_k = \{x \geq 1 : \ |f(nx)| \leq \varepsilon \ \forall n \geq k\}$. Each $x \mapsto f(nx)$ is continuous, hence $Q_k = \bigcap_{n \geq k}\{x : \ |f(nx)| \leq \varepsilon\}$ is closed. By the assumption $[1, \infty) = \bigcup_{k \geq 1} Q_k$. By Baire, some Q_{k_0} contains an interval $[a, b] \subset [1, \infty)$. For $t \geq T := ab/(b-a)$, choose an integer $n \in [t/b, \, t/a]$ (nonempty). Then $x := t/n \in [a, b] \subset Q_{k_0}$ and with $n \geq k_0$ we have $|f(t)| = |f(nx)| \leq \varepsilon$. Thus $f(t) \to 0$ as $t \to \infty$.

Problem 10 — Limsup of Open Dense Sets

Statement. Let (A_j) be subsets of $[0, 1]$ such that for each $N \geq 1$, the union $\bigcup_{j \geq N} A_j$ is open and dense in $[0, 1]$. Show that

$$S = \{x \in [0, 1] : x \in A_j \text{ for infinitely many } j\}$$

is dense. Must S be open? Must $\bigcap_{j=1}^{\infty} A_j \neq \varnothing$?

Solution. Note $S = \limsup A_j = \bigcap_{N \geq 1} \left(\bigcup_{j \geq N} A_j \right)$, an intersection of open dense sets. By Baire, S is dense (indeed a dense G_δ). In general S need not be open. Moreover, $\bigcap_{j \geq 1} A_j$ can be empty (e.g. $A_j = [0, 1] \setminus \{q_j\}$ removing a different rational each j).

Problem 11 — Removing $\mathbb{Q}$

Statement. If G is open dense in $\mathbb{R}$, show $G \setminus \mathbb{Q}$ is dense. If only G is uncountable and dense, must $G \setminus \mathbb{Q}$ be dense?

Solution. For any nonempty open interval U, $G \cap U$ is a nonempty open interval; removing the countable set $\mathbb{Q}$ leaves it nonempty. Hence $G \setminus \mathbb{Q}$ is dense. If G is merely uncountable and dense, not necessarily: take $G = \mathbb{Q} \cup C$ with C a Cantor set. Then G is dense and uncountable, but $G \setminus \mathbb{Q} = C$ is nowhere dense.

Problem 12 — A Dense Set with Super–Polynomial Approximations

Statement. Show there is a dense set of reals x such that for each $n \in \mathbb{N}$ there exist integers p, q with $q \geq 2$ and $0 < \left|x - \frac{p}{q}\right| < \frac{1}{q^n}$.

Construction & Proof. Fix an interval (α, β). Choose $r = \frac{u}{10^M} \in (\alpha, \beta)$ and N large with $\sum_{k \geq N} 10^{-k!} < \beta - \alpha$. Define

$$x = r + \sum_{k=N}^{\infty} 10^{-k!} \in (\alpha, \beta).$$

For any $m \in \mathbb{N}$, taking $q = 10^{K!}$ with $K \geq \max\{N, m\}$ and p the truncation numerator,

$$0 < \left|x - \frac{p}{q}\right| < 10^{-(K+1)!} < \frac{1}{q^m}.$$

Thus every interval contains such x; the set is dense.

Liouville numbers and Problem 12.

Definition 0.1 (Liouville number)**.** A real number x is a *Liouville number* if for every $n \in \mathbb{N}$ there exist infinitely many rationals p/q $(q \geq 2)$ such that

$$0 < \left|x - \frac{p}{q}\right| < \frac{1}{q^n}.$$

Proposition 0.2 (Problem 12 $\Rightarrow$ Liouville)**.** *Let*

$$x \;=\; r + \sum_{k=N}^{\infty} 10^{-k!}, \qquad r = \frac{u}{10^M} \in \mathbb{Q},$$

as constructed in Problem 12. Then x is a Liouville number (hence transcendental).

Proof. Fix $n \in \mathbb{N}$ and let $K \geq \max\{N, n\}$. Define the truncation

$$\frac{p_K}{q_K} = r + \sum_{k=N}^{K} 10^{-k!}, \qquad q_K = 10^{K!} \in \mathbb{N}.$$

Then

$$0 < \left| x - \frac{p_K}{q_K} \right| = \sum_{j=K+1}^{\infty} 10^{-j!} < 10^{-(K+1)!} < \frac{1}{(10^{K!})^n} = \frac{1}{q_K^n}.$$

Since this holds for every $K \geq \max\{N, n\}$, we obtain *infinitely many* such rationals for each fixed n. Hence x is a Liouville number. $\square$

Corollary 0.3. *The set $\mathcal{L}$ of Liouville numbers is dense in $\mathbb{R}$; in particular, the set produced in Problem 12 is a dense subset of $\mathcal{L}$.*

Remark (Topological/measure size of $\mathcal{L}$). One has

$$\mathcal{L} = \bigcap_{n=1}^{\infty} \bigcup_{q \geq 2} \bigcup_{p \in \mathbb{Z}} \left(\tfrac{p}{q} - q^{-n}, \, \tfrac{p}{q} + q^{-n} \right),$$

so $\mathcal{L}$ is a dense G_δ (comeager) subset of $\mathbb{R}$, but it has Lebesgue measure 0. Moreover, by Liouville's theorem (and Roth's refinement), no irrational algebraic number lies in $\mathcal{L}$.